David L. Nelson and Michael M. Cox

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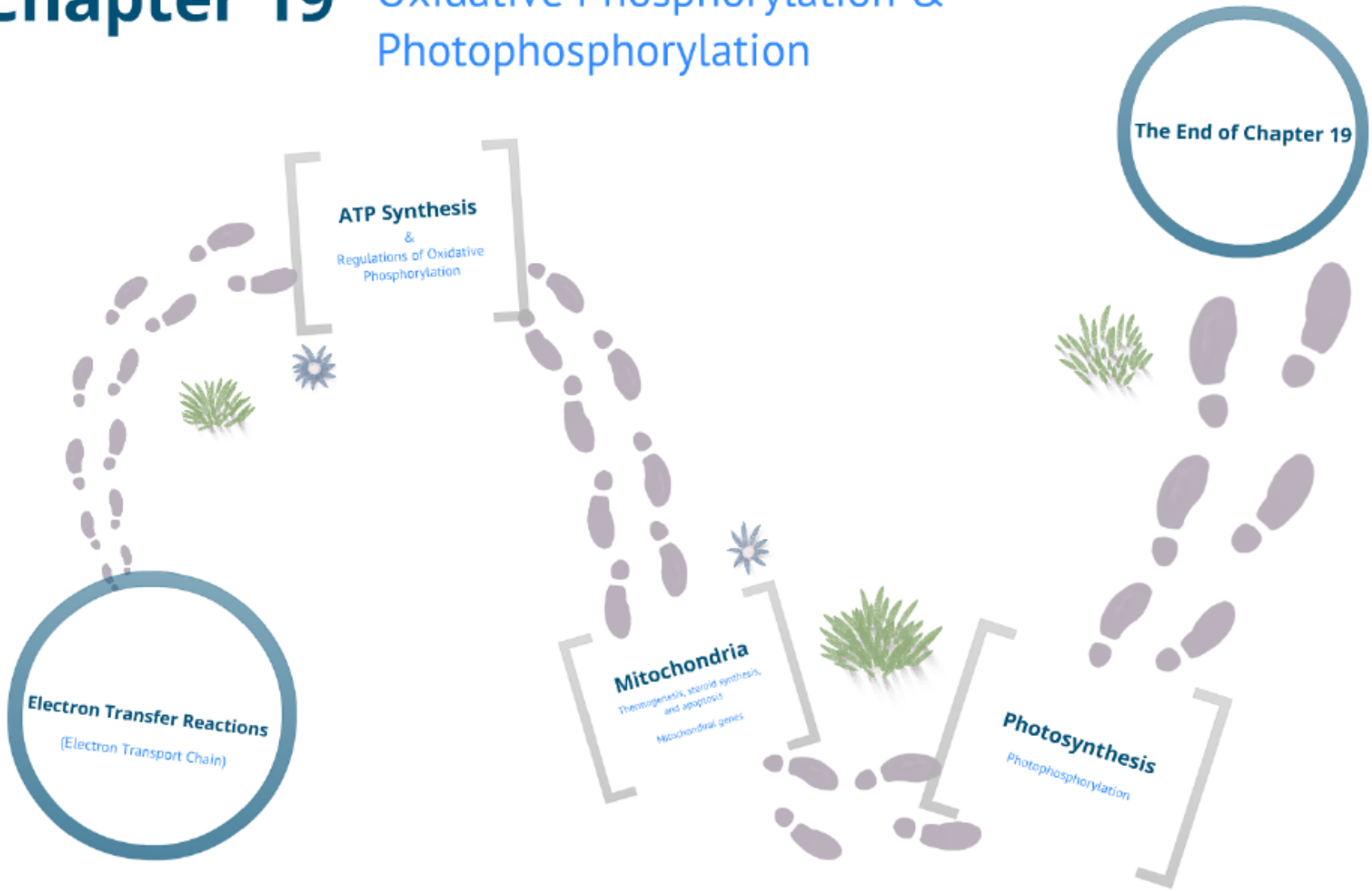
CHAPTER 19
Oxidative Phosphorylation and Photophosphorylation

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“어떤 일을 하는 방법에는 여러 가지가 있고, 그중 하나가 문제를 일으킬 수 있다면 누군가는 꼭 그 방법을 사용한다.” … (머피의 법칙)

1. 어떤 일도 겉으로 보이는 것만큼 간단하진 않다.
2. 모든 임무의 완성 주기는 당신이 예측한 시간보다 길다.
3. 어떤 일이 잘못될 가능성이 있으면, 굉장한 확률로 그 일은 잘못된다.
4. 당신이 잘못될 가능성을 예감한다면, 반드시 그 일은 잘못된다….

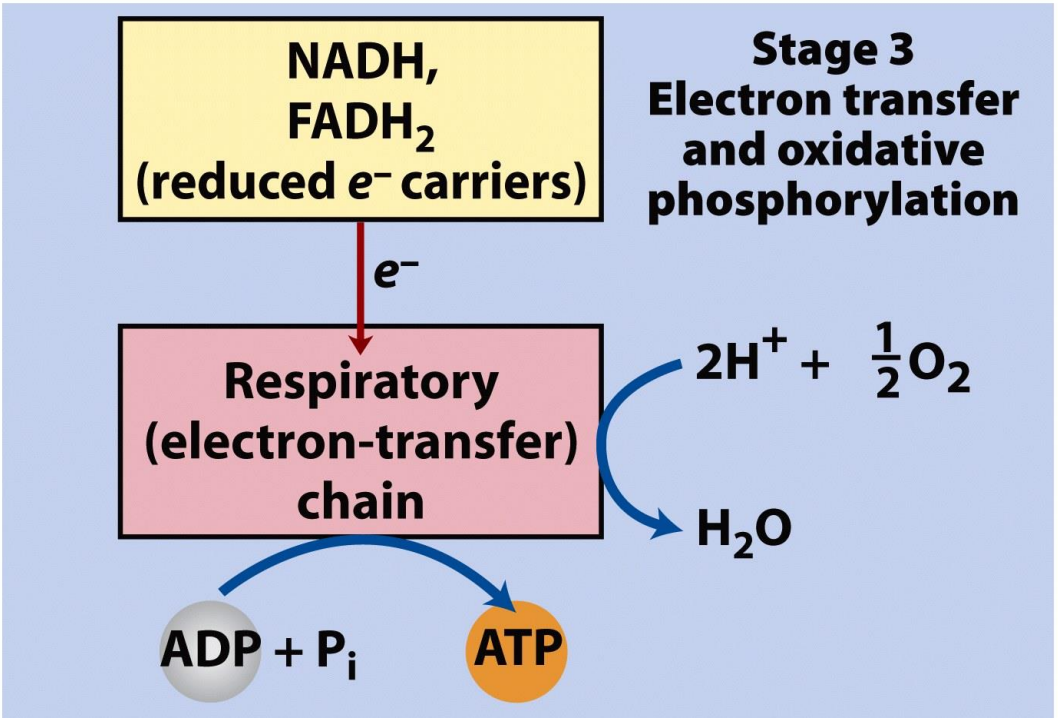
Chapter 19 Oxidative Phosphorylation & Photophosphorylation



19.1 Electron-transfer reactions in mitochondria

Oxidative phosphorylation

- Electrons from the reduced cofactors NADH and $FADH_2$ are passed to proteins in the respiratory chain
- In eukaryotes, oxygen is the ultimate electron acceptor for these electrons
- Energy of oxidation is used to produce ATP



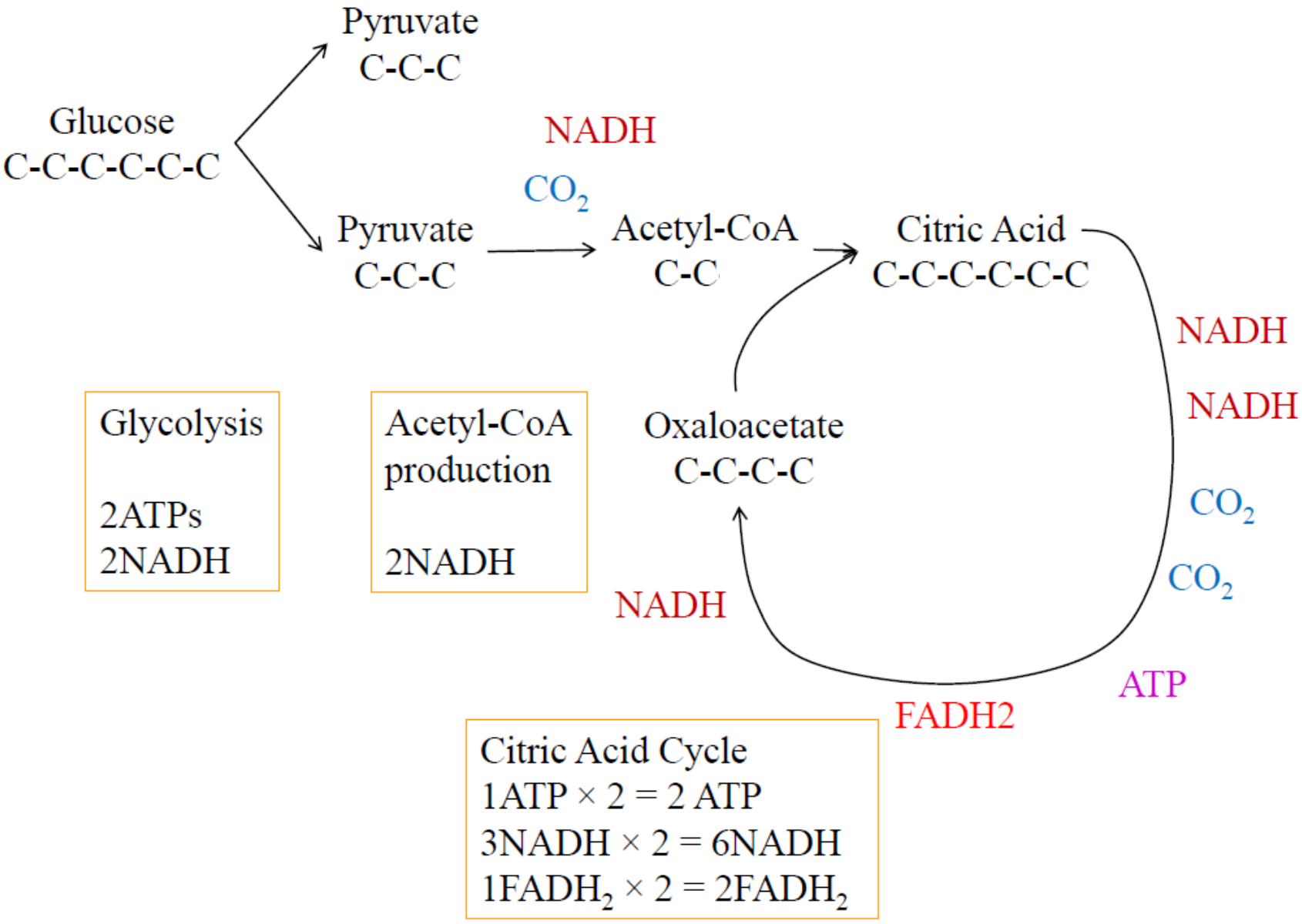

Pyruvate
C-C-C
Glucose
C-C-C-C-C-C
NADH
CO_2
Pyruvate
C-C-C
Acetyl-CoA
C-C
Citric Acid
C-C-C-C-C-C
NADH
NADH
CO_2
CO_2
ATP
FADH2
NADH
Oxaloacetate
C-C-C-C
Glycolysis
2ATPs
2NADH
Acetyl-CoA
production
2NADH
Citric Acid Cycle
1ATP × 2 = 2 ATP
3NADH × 2 = 6NADH
1$FADH_2$ × 2 = 2$FADH_2$

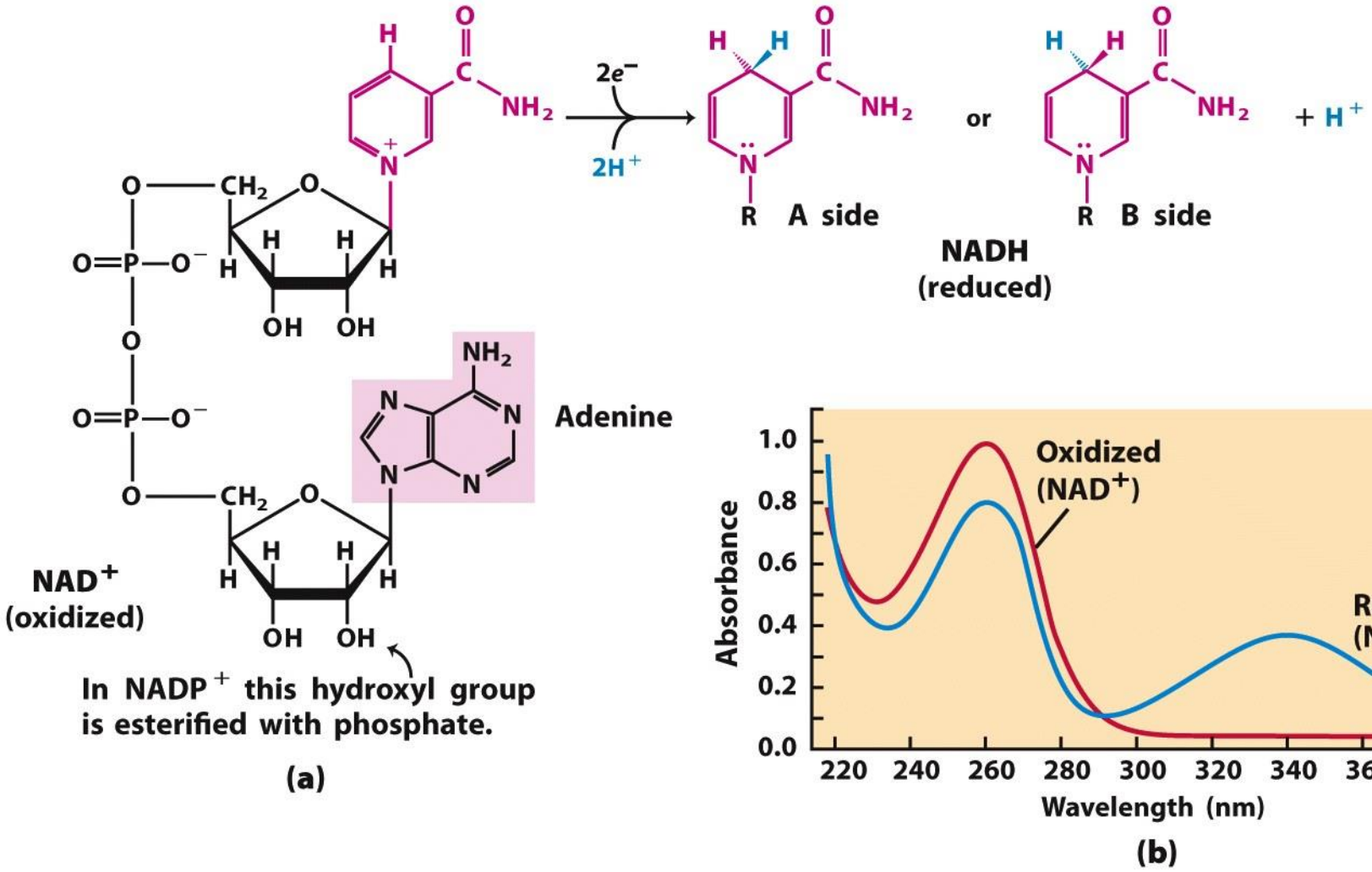
TABLE 19–5 ATP Yield from Complete Oxidation of Glucose

Process	Direct product	Final ATP
Glycolysis	2 NADH (cytosolic)	3 or 5*
	2 ATP	2
Pyruvate oxidation (two per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl-CoA oxidation in citric acid cycle (two per glucose)	6 NADH (mitochondrial matrix)	15
	2 $FADH_2$	3
	2 ATP or 2 GTP	2
Total yield per glucose		30 or 32

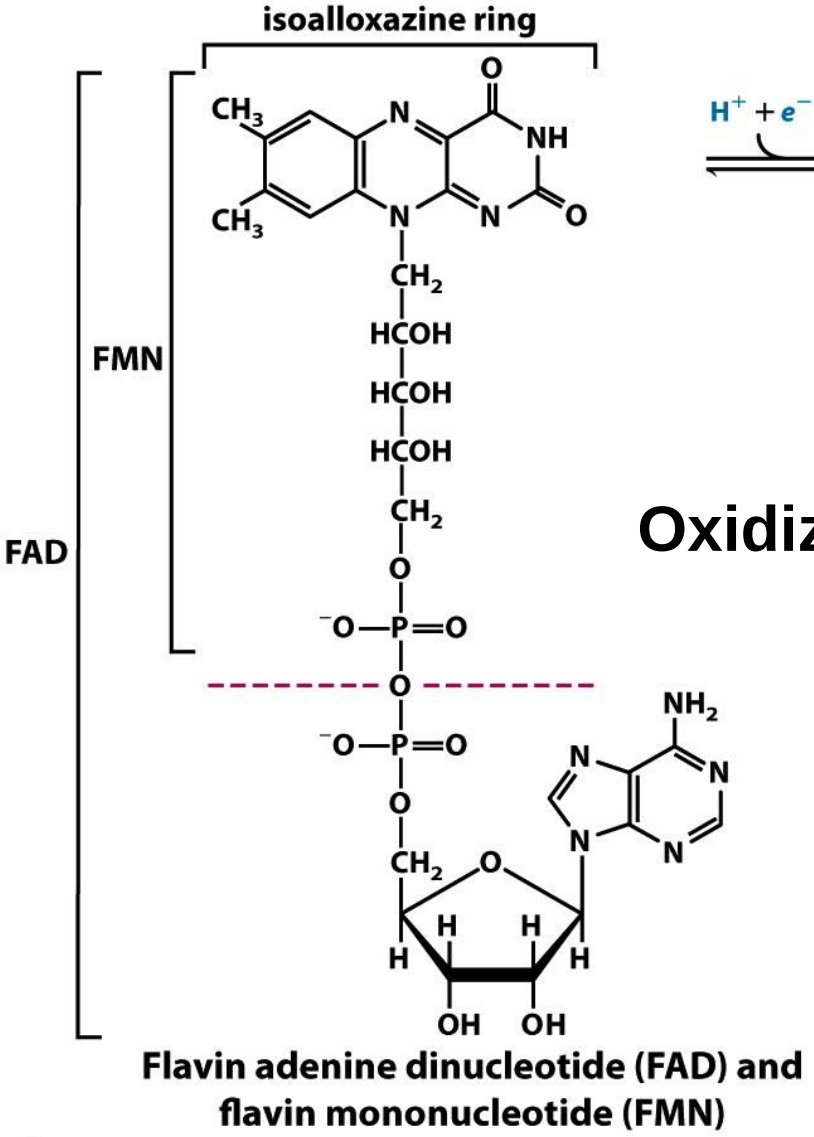
*The number depends on which shuttle system transfers reducing equivalents into the mitochondrion.

Table 19-5
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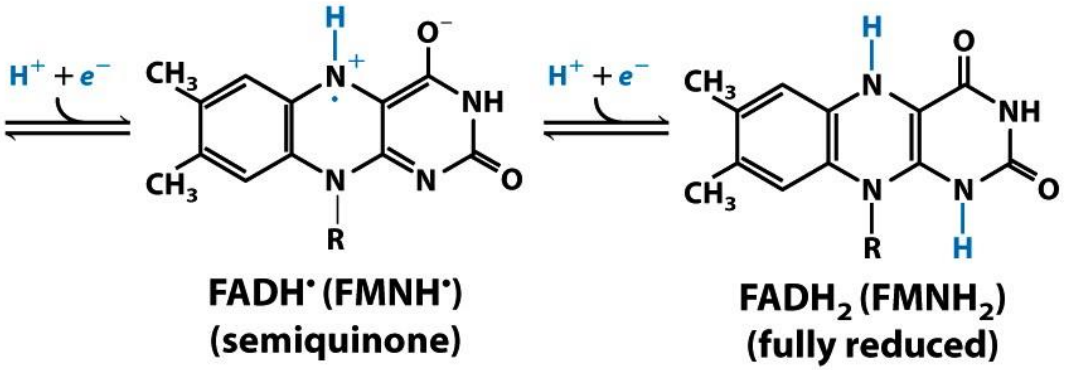
(1) Electrons are funneled to universal electron acceptors



(1) Electrons are funneled to universal electron acceptors



Flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN)



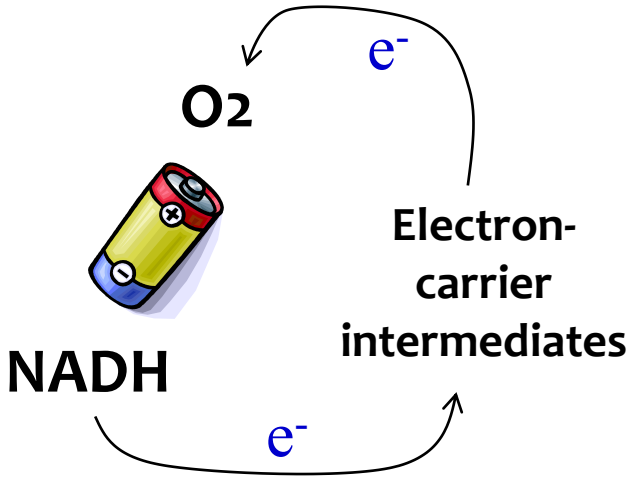
FADH• (FMNH•) (semiquinone)

$FADH_2$ ($FMNH_2$) (fully reduced)

Oxidized and reduced FAD and FMN

The standard reduction potential ($E'°$) of FAD or FMN depends on the protein associated

Stage 3 of the Cellular Respiration
: electron transfer chain



1 glucose → 10 NADHs + 2 $FADH_2$s

$NADH \rightarrow NAD^+ + H^+ + 2e^-$

High E

Low E

Energy → H^+ gradient

CoQ

Energy → H^+ gradient

Cyt bc

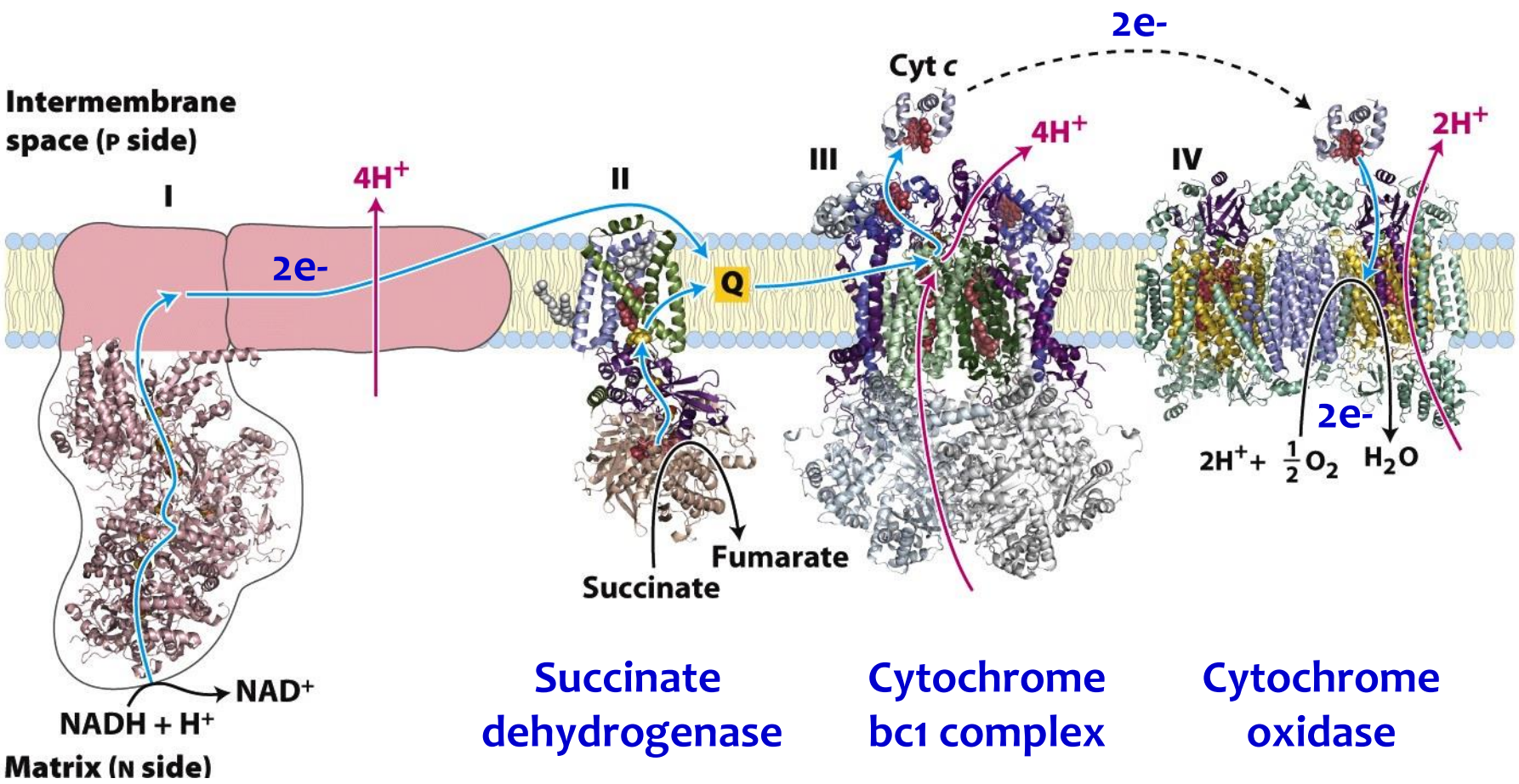
Energy → H^+ gradient

$2e^- + 2\ H^+ + ½\ O_2 \rightarrow H_2O$

ATP synthesis

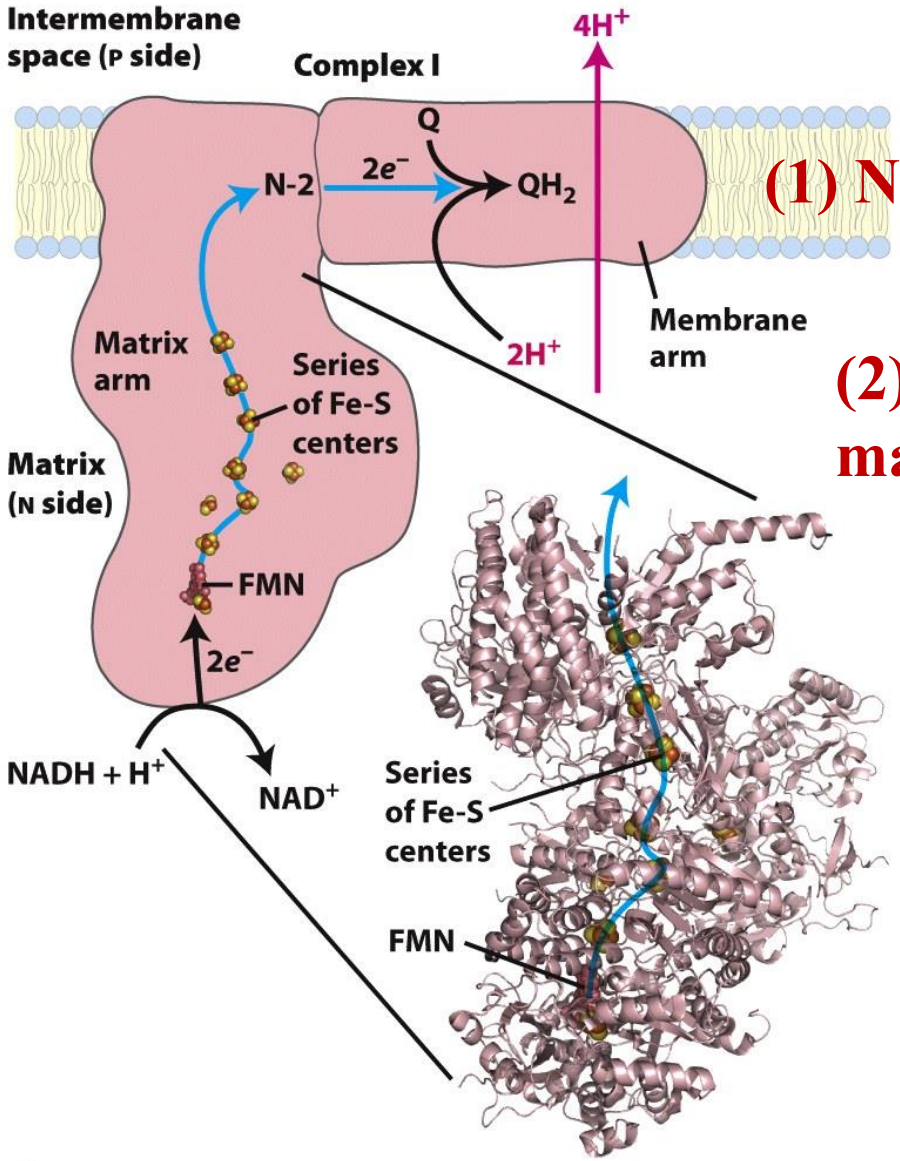
https://youtu.be/xbJ0nbzt5Kw

Overview of the four complexes of the respiratory chain



Chap19_ETC.mp4

NADH:ubiquinone oxidoreductase (Complex I). NADH dehydrogenase



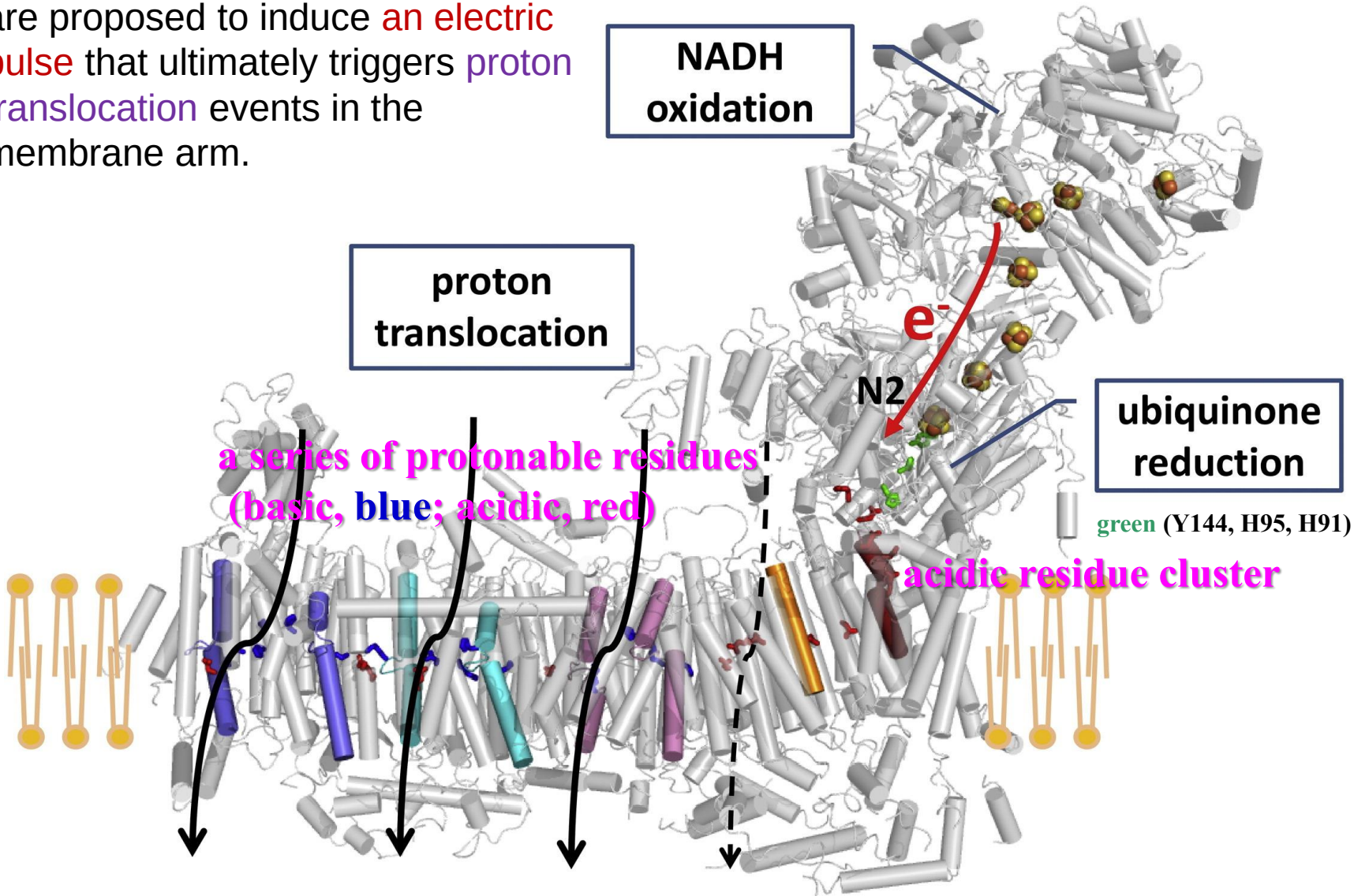
(1) $NADH + H^+ + Q \rightarrow NAD^+ + QH_2$

(2) Transfer of 4 protons from matrix to intermembrane space

→ Proton pump by the energy of electron transfer

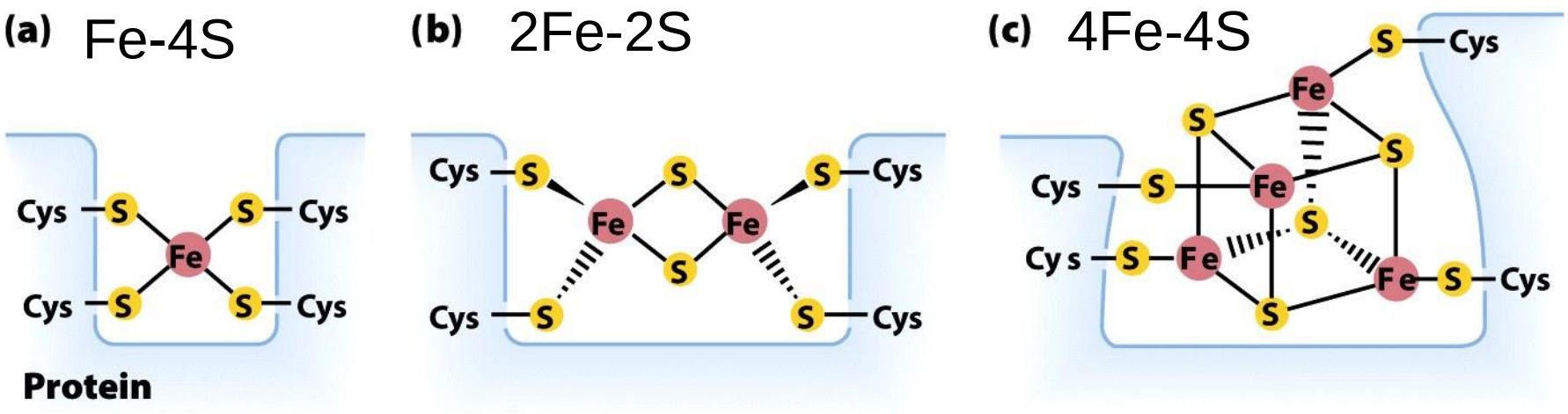
$NADH + 5H^+_N + Q \rightarrow NAD^+ + QH_2 + 4H^+_p$

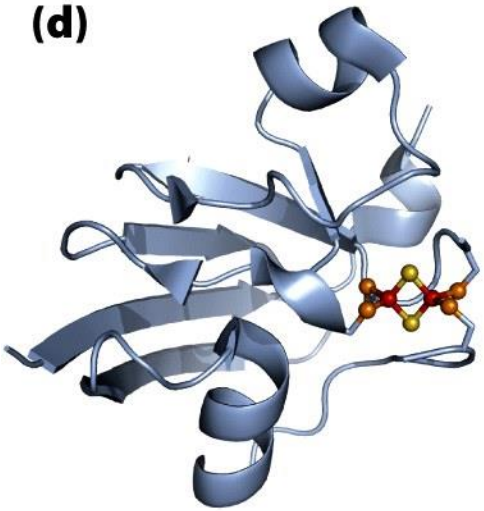
Conformational changes linked to the redox chemistry of ubiquinone are proposed to induce an electric pulse that ultimately triggers proton translocation events in the membrane arm.



Biochimica et Biophysica Acta (2016) 1857, 902

The Fe-S centers of iron-sulfur proteins





- Found in several proteins of electron transport chain, including NADH:ubiquinone oxidoreductase
- Transfers one electron at a time

Electrons pass thru a series of memb-bound carriers

Coenzyme Q or Ubiquinone

- Ubiquinone is a lipid-soluble conjugated dicarbonyl compound that readily accepts electrons
- Upon accepting two electrons, it picks up two protons to give an alcohol, ubiquinol
- Ubiquinol can freely diffuse in the membrane, carrying electrons with protons from one side of the membrane to another side

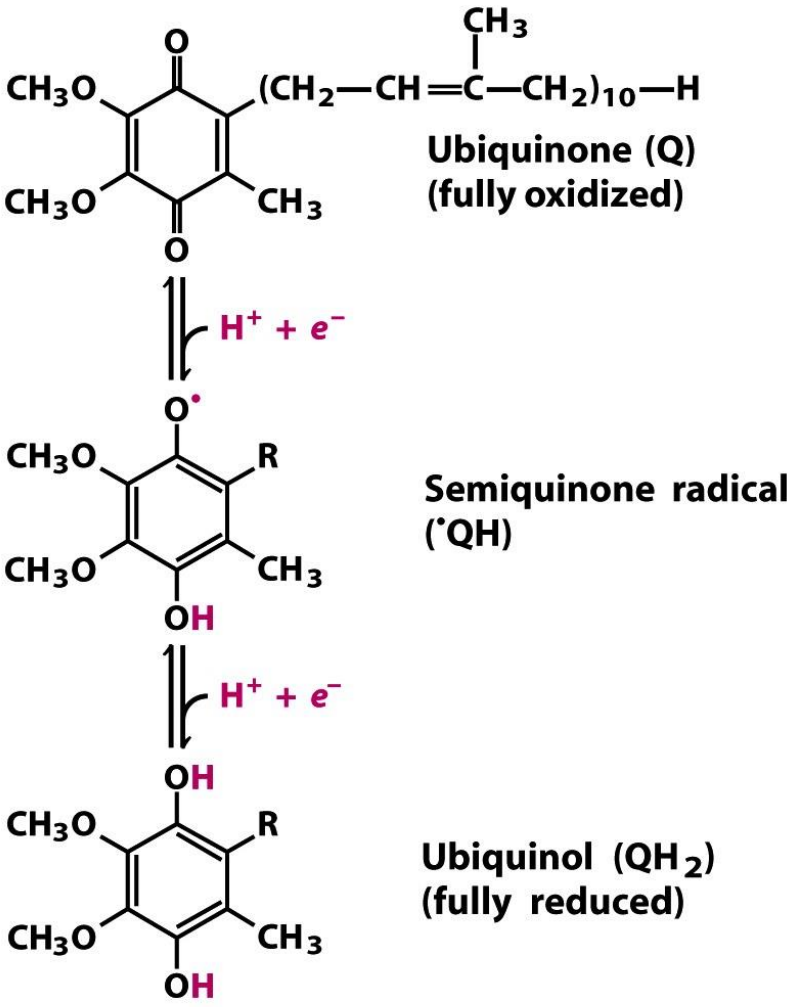
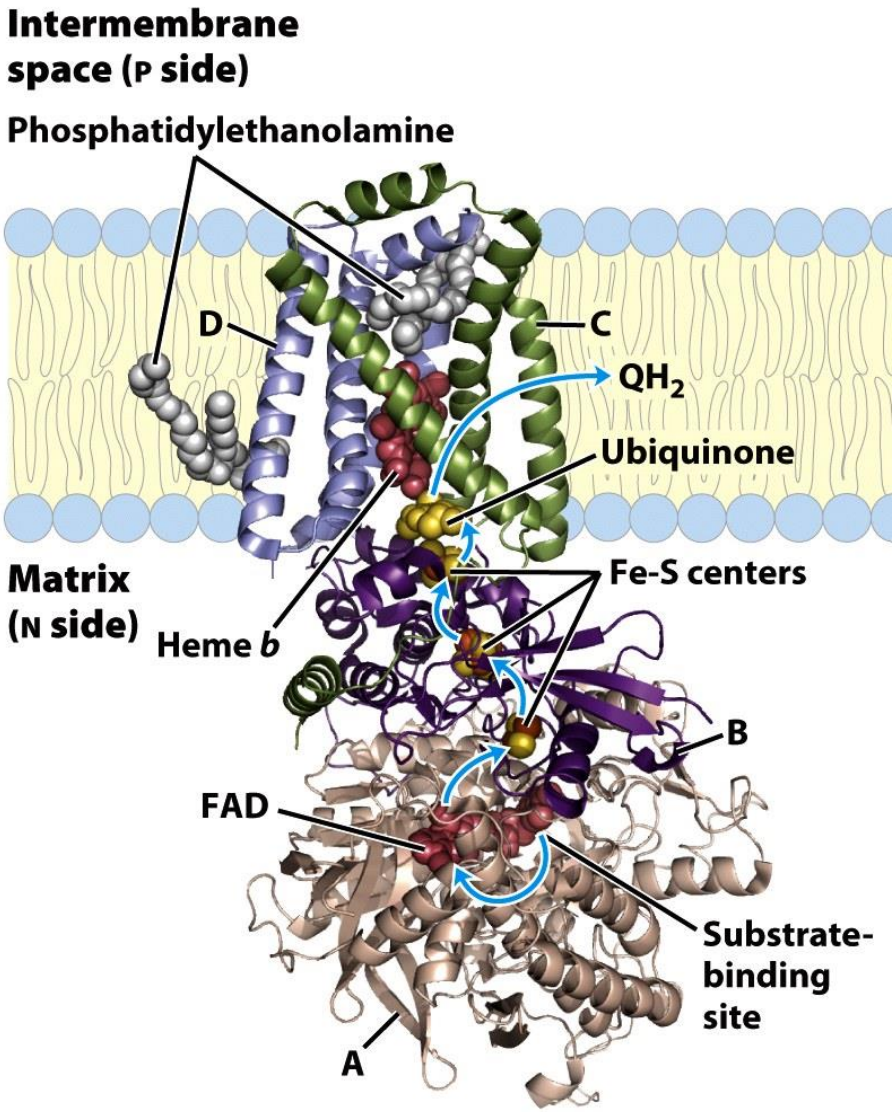


Figure 19-2
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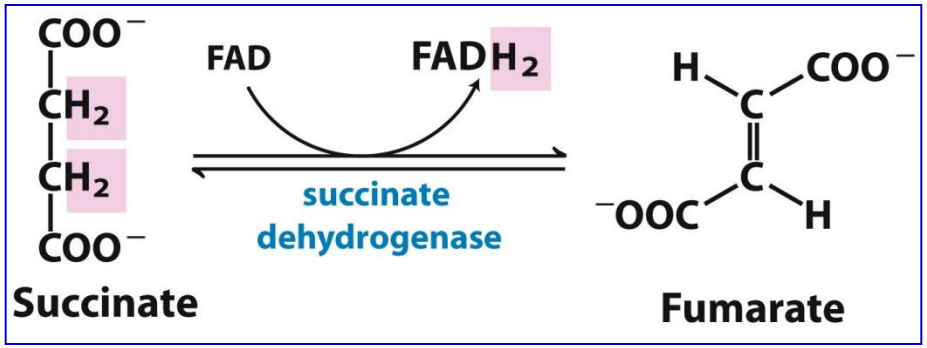
Complex II (succinate dehydrogenase)



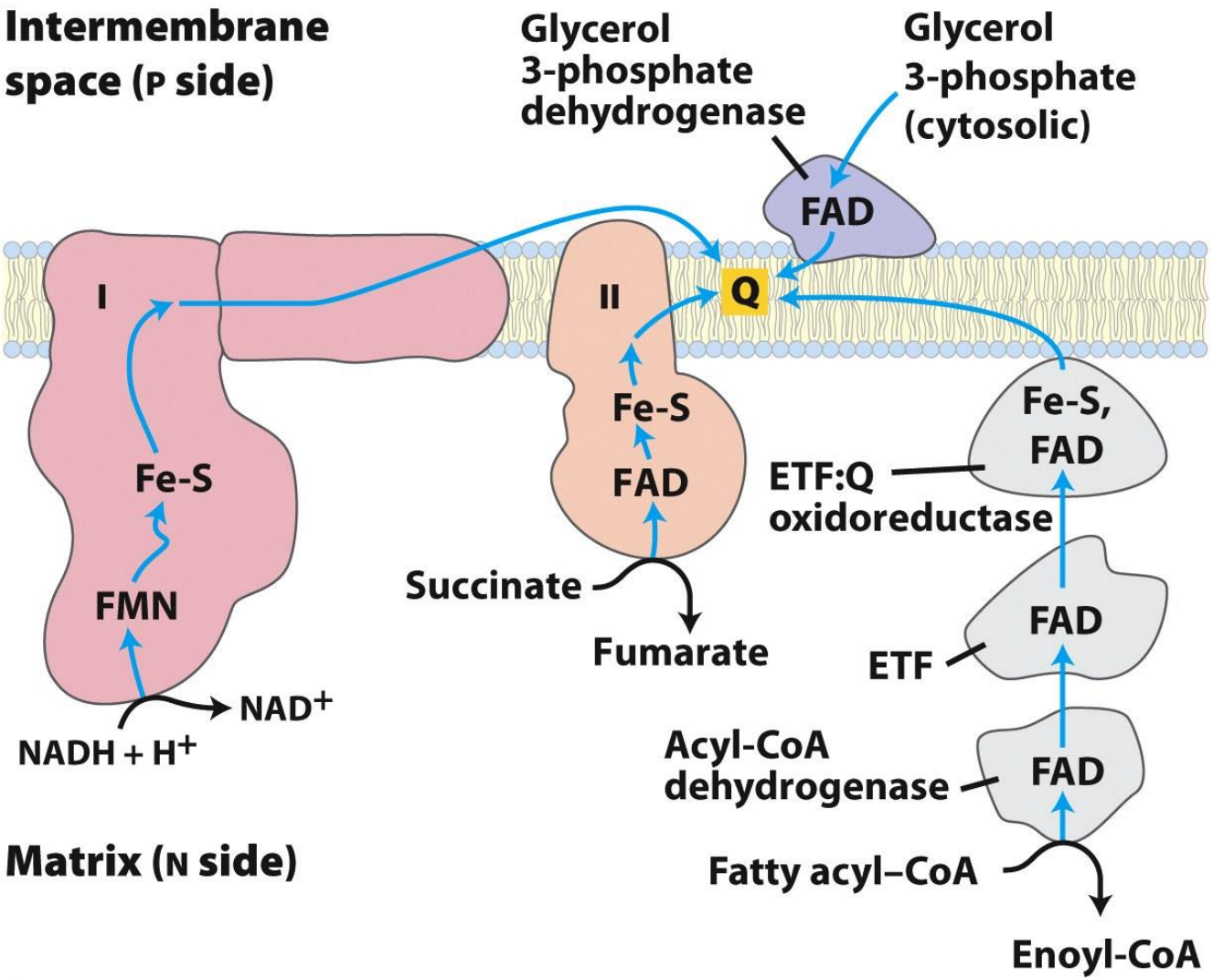
Electron transfer from succinate to ubiquinone (CoQ)

Succinate → FAD → 3 Fe-S centers →Q binding site

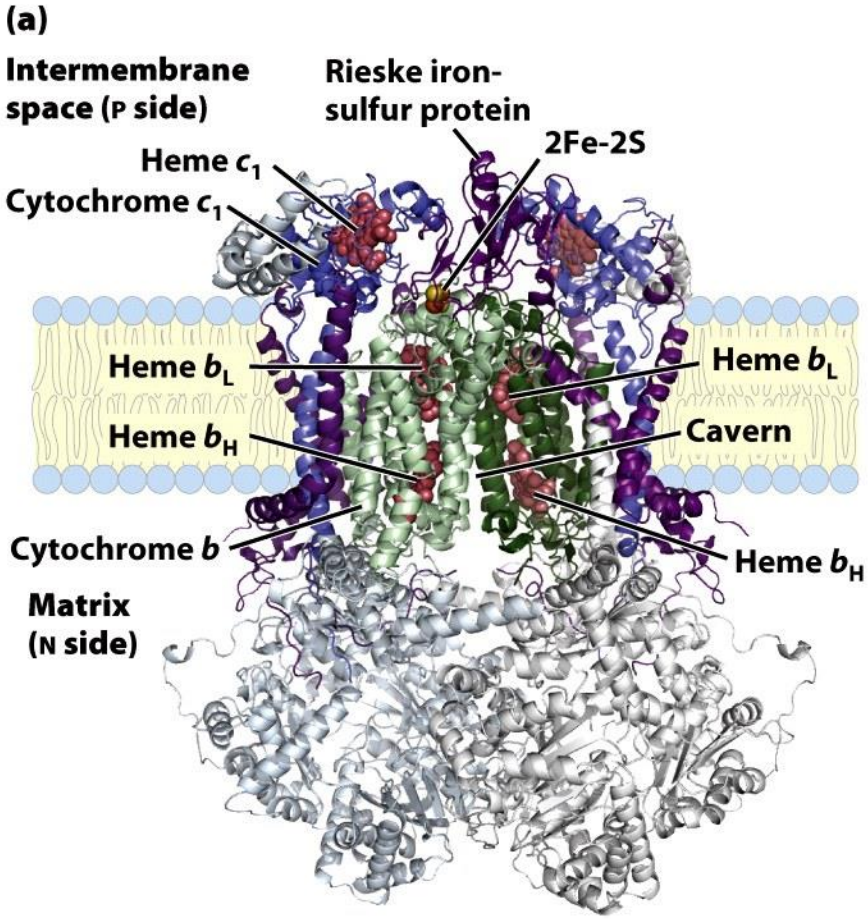
Complex II is not in the direct path of e^- transfer, but to prevent leak out of ROS (H_2O_2) and superoxide radical (O_2^-) from succinate

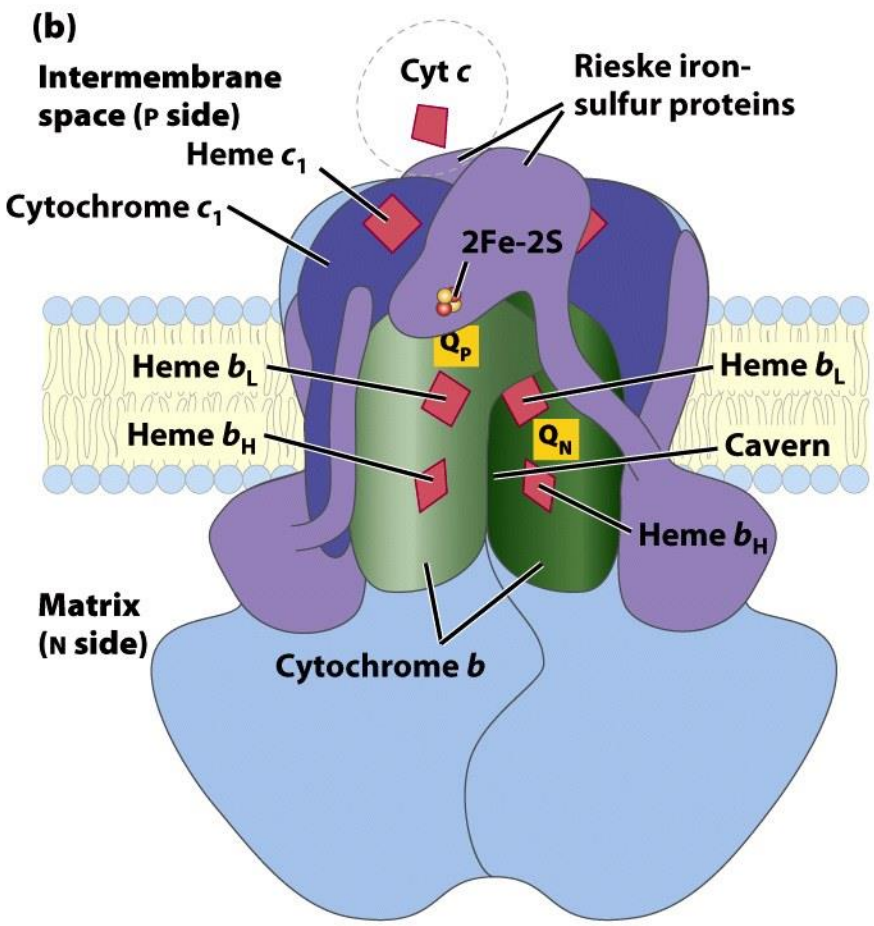


Path of electrons from NADH, succinate, fatty acyl–CoA, and glycerol 3-phosphate to ubiquinone



Cytochrome bc_1 complex (Complex III)

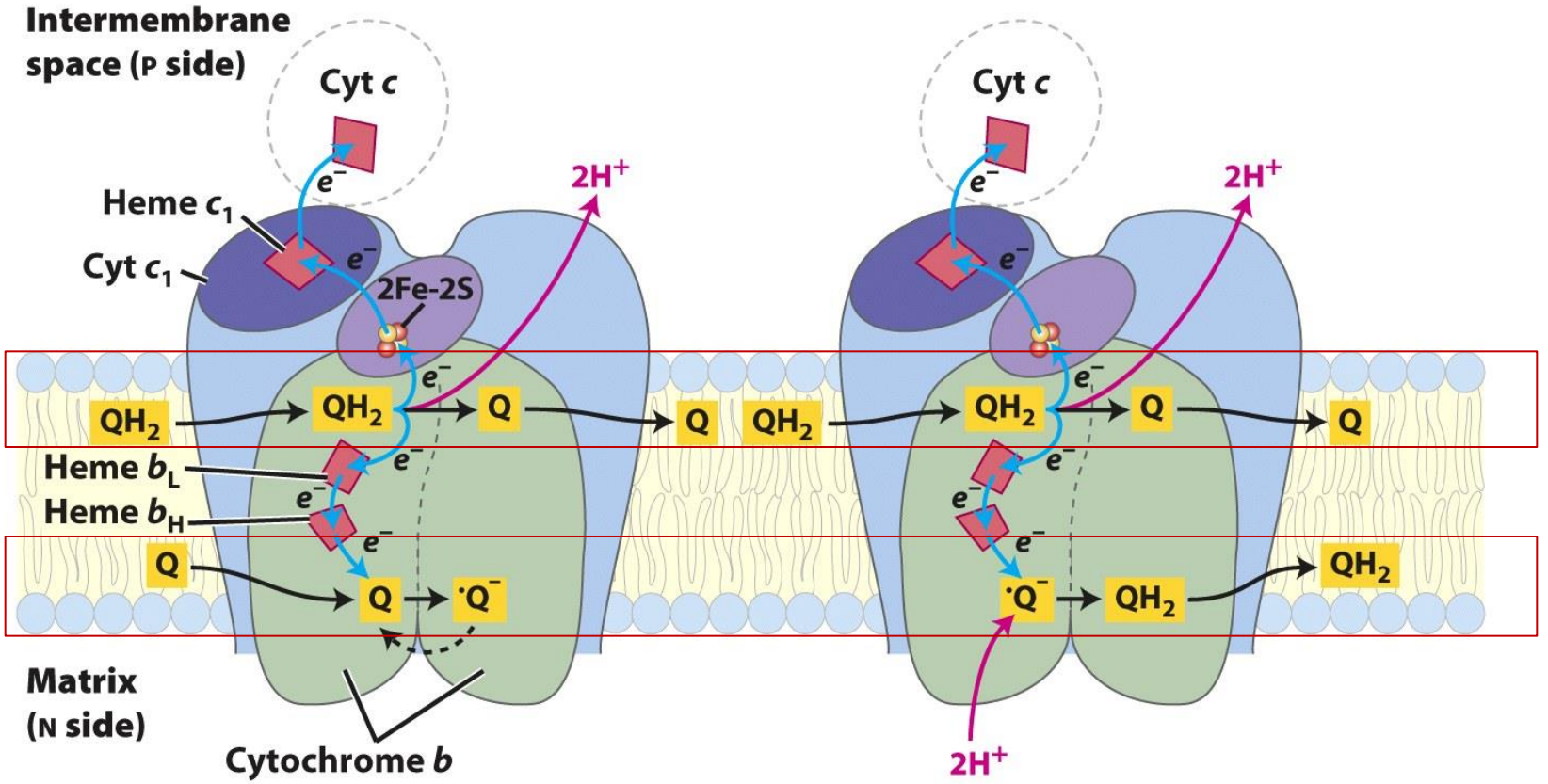




Q_N : matrix side of ubiquinone binding site

Q_P : ubiquinone binding site at intermembrane space

The Q cycle, shown in two stages



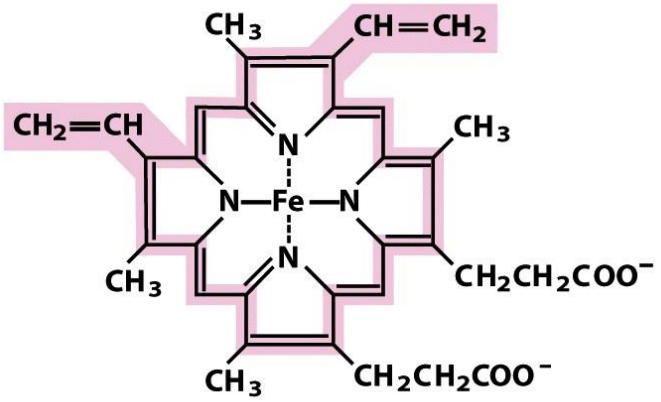
$QH_2 + Q + \text{cyt } c_1 \text{ (oxidized)} \longrightarrow \cdot Q^- + Q + 2H_P^+ + \text{cyt } c_1 \text{ (reduced)}$

$QH_2 + \cdot Q^- + 2H_N^+ + \text{cyt } c_1 \text{ (oxidized)} \longrightarrow QH_2 + 2H_P^+ + Q + \text{cyt } c_1 \text{ (reduced)}$

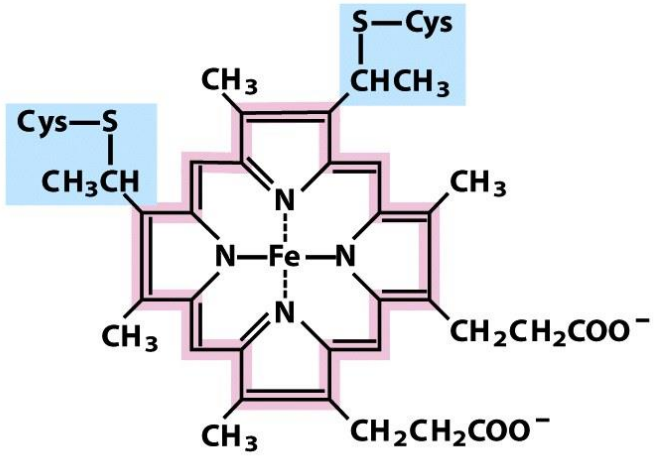
Net equation: $QH_2 + 2 \text{ cyt } c_1 \text{ (oxidized)} + 2H_N^+ \longrightarrow Q + 2 \text{ cyt } c_1 \text{ (reduced)} + 4H_P^+$

Electrons pass thru a series of memb-bound carriers

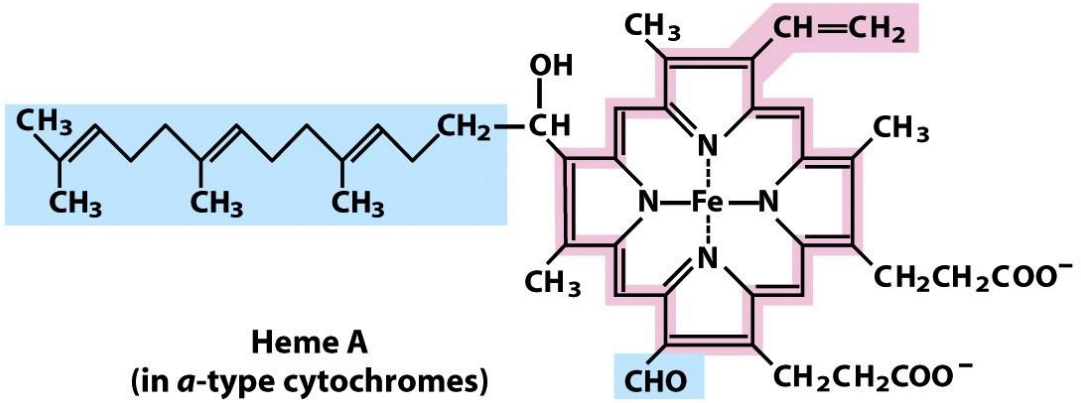
Cytochromes a, b, and c



Iron protoporphyrin IX
(in *b*-type cytochromes)



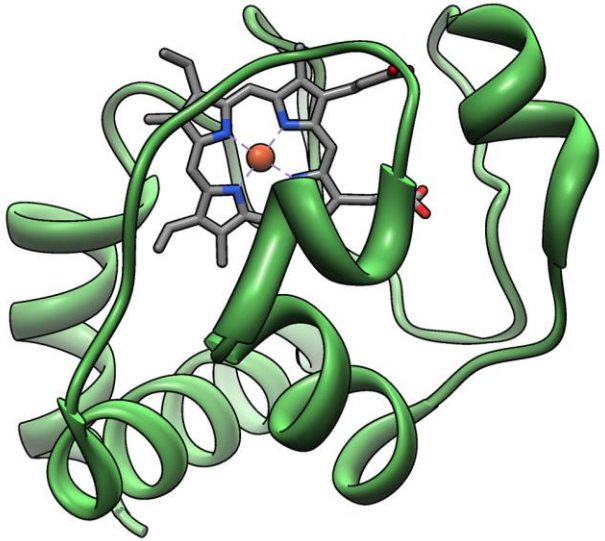
Heme C
(in *c*-type cytochromes)



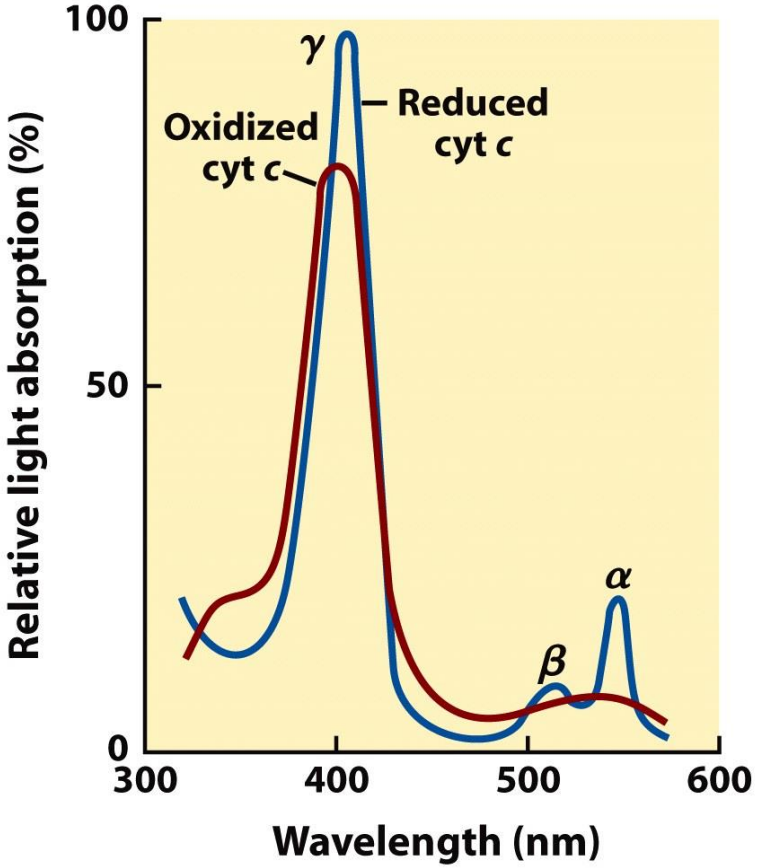
Heme A
(in *a*-type cytochromes)

Prosthetic groups of cytochromes (4 porphyrins)

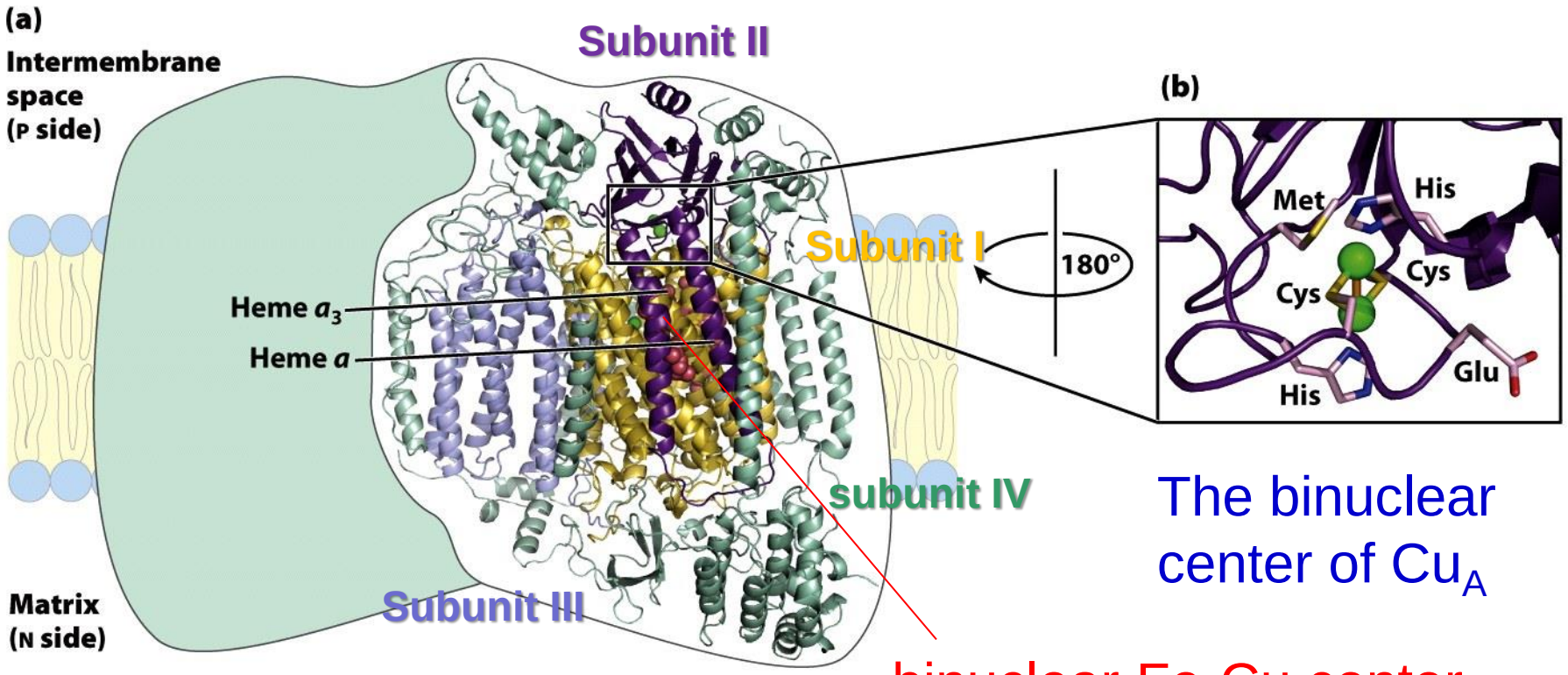
Absorption spectra of cytochrome *c* (cyt *c*) in its oxidized (red) and reduced (blue) forms.



cytochrome *c*

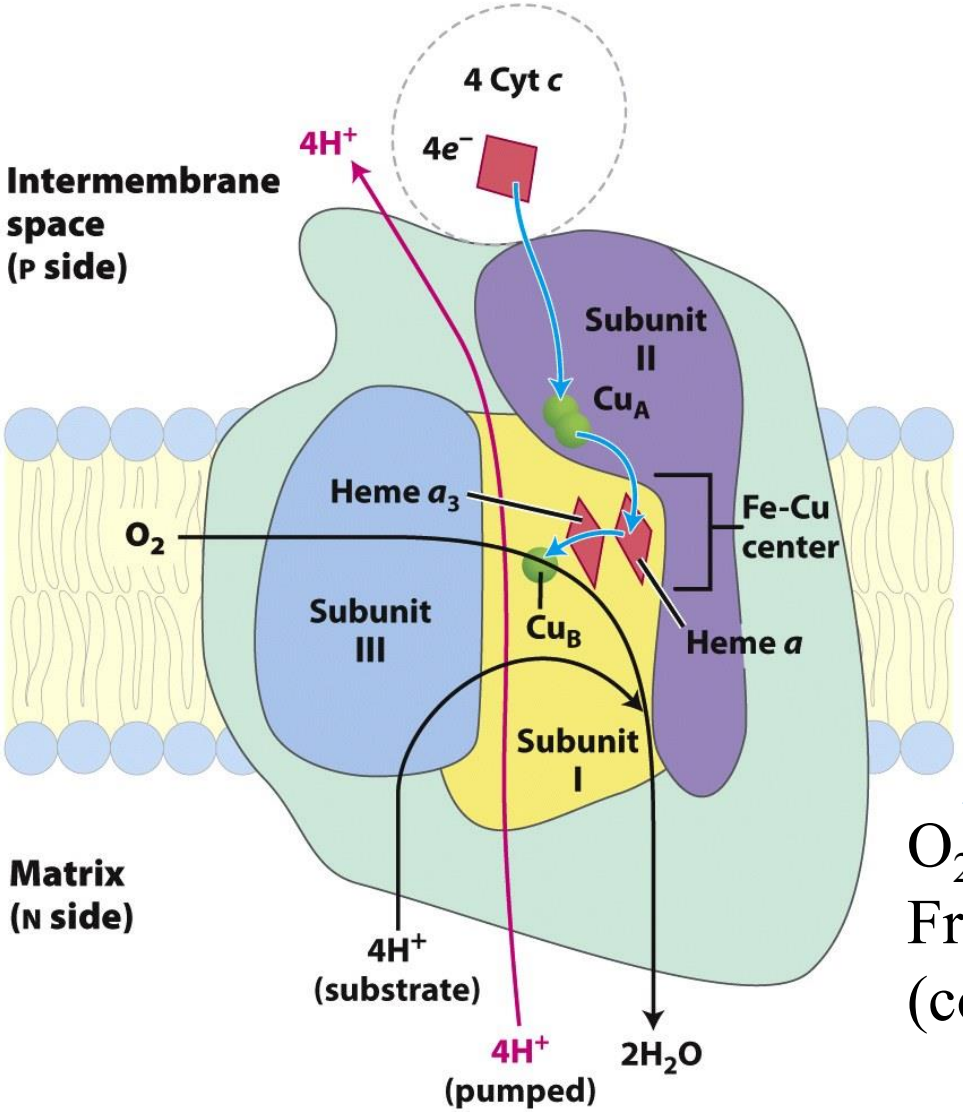


Cytochrome oxidase (Complex IV)



The complex IV from bovine mitochondria has 13 subunits, but only four core proteins are shown here. **Subunit I** has Heme *a*, a_3, and Cu_B(binuclear center). **Subunit II** contains two Cu ions complexed with Cys residues. **Subunit III** is essential for rapid proton movement through subunit II.

Path of electrons through Complex IV



Electron:
cytochrome c
→ binuclear center CuA
→ Heme a
→ Fe-Cu center

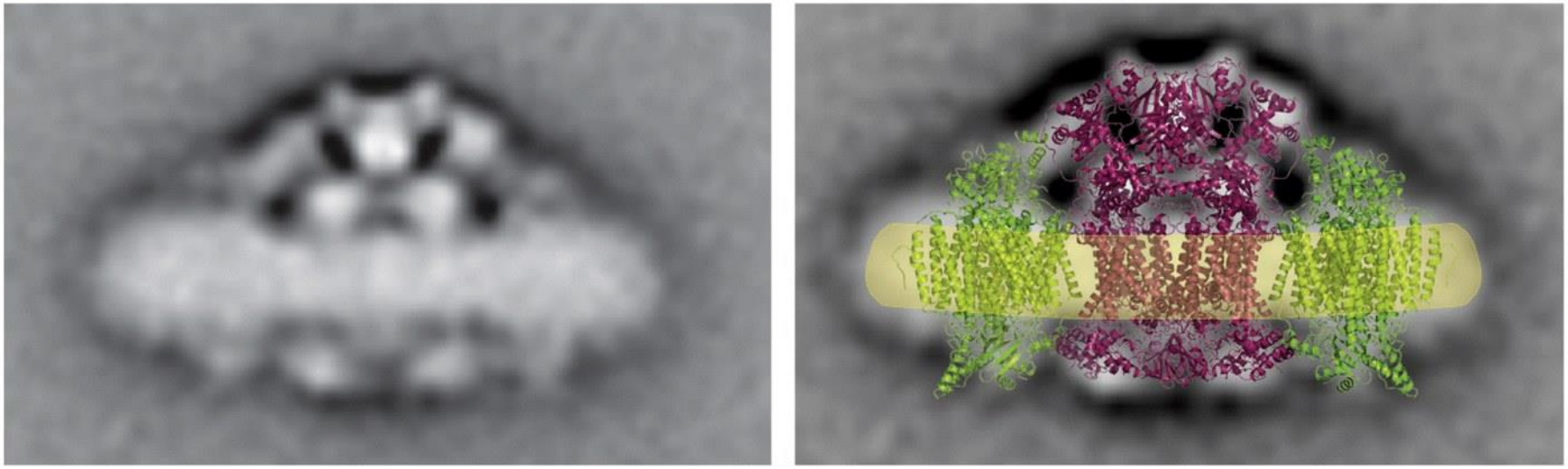
$O_2 \xrightarrow{2e^-} O_2^{2-} \xrightarrow{2e^-} 2H_2O$

From $4e^- \rightarrow 4H^+_p$ pumped

(corresponding to : $2e^- \rightarrow 2H^+_p$)

$4\ \text{Cyt c (with } 4e^-) + 8H^+_N + O_2 \rightarrow 4\ \text{Cyt c (oxidized)} + 4H^+_P + 2H_2O$

A putative respirasome composed of Complexes III and IV

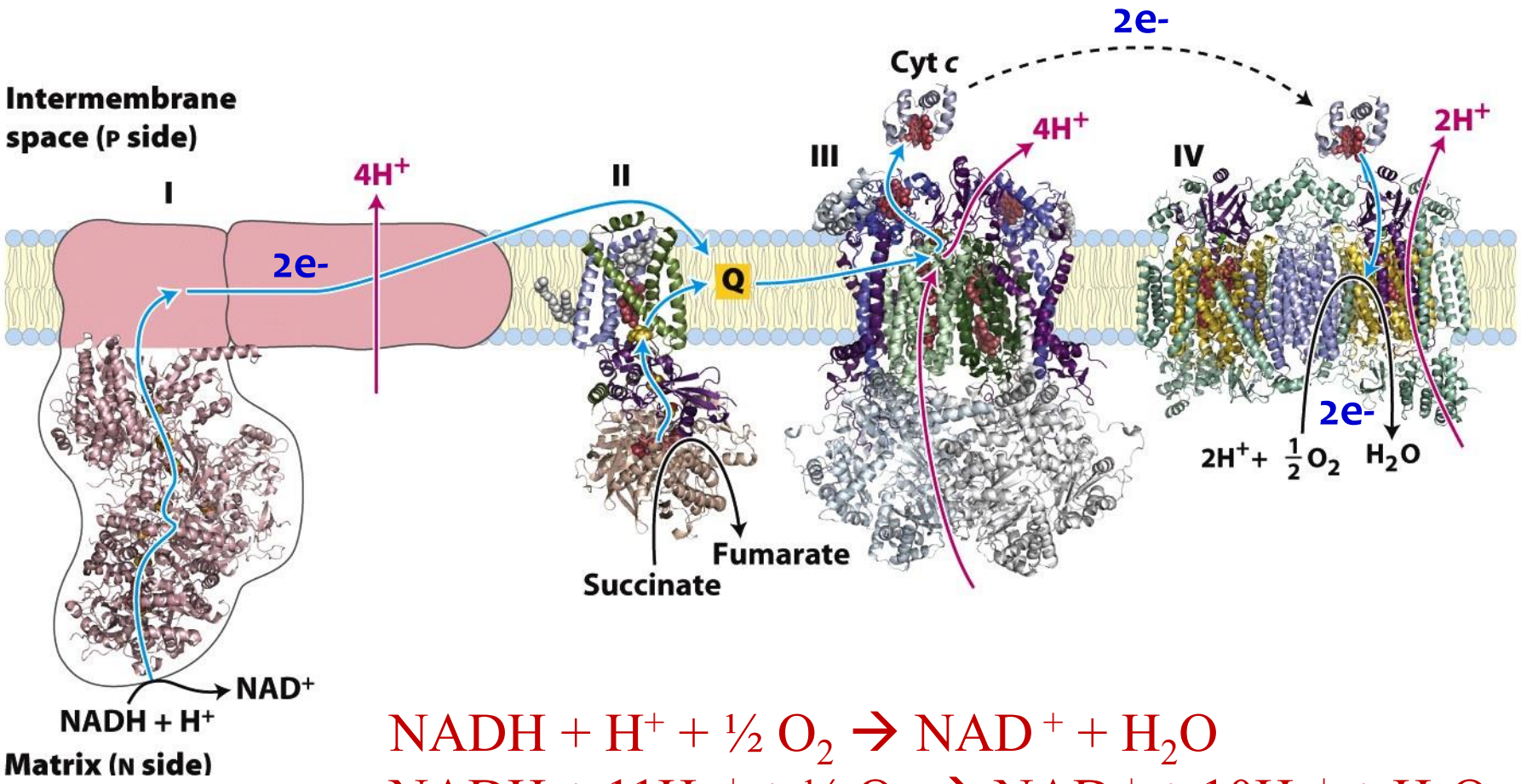


(a) (b)

Figure 19-15
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Complex I and III, or complex III and IV are associated

Summary of the flow of electrons and protons through the four complexes of the respiratory chain

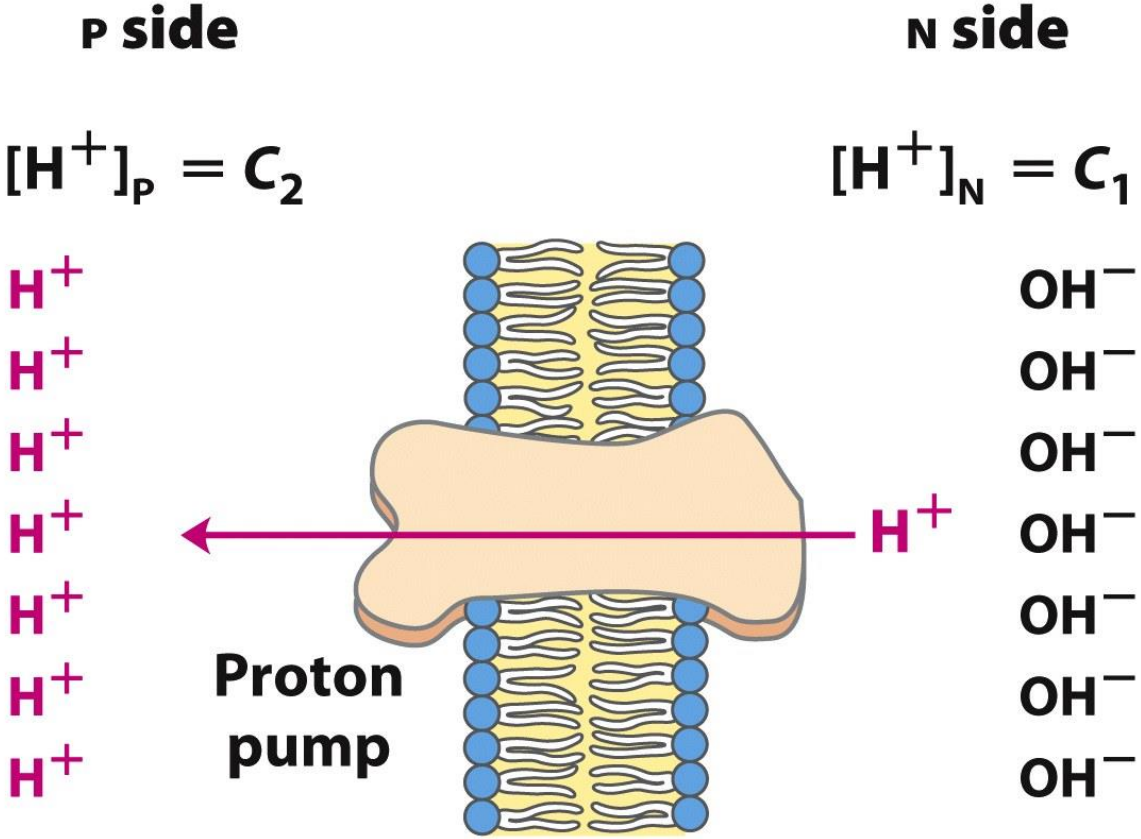


$NADH + H^+ + ½\ O_2 \rightarrow NAD^+ + H_2O$

$NADH + 11H_N^+ + ½\ O_2 \rightarrow NAD^+ + 10H_P^+ + H_2O$

$\Delta G'^\circ = -n\mathcal{F}\Delta E'^\circ = -220$ kJ/mol (of NADH)

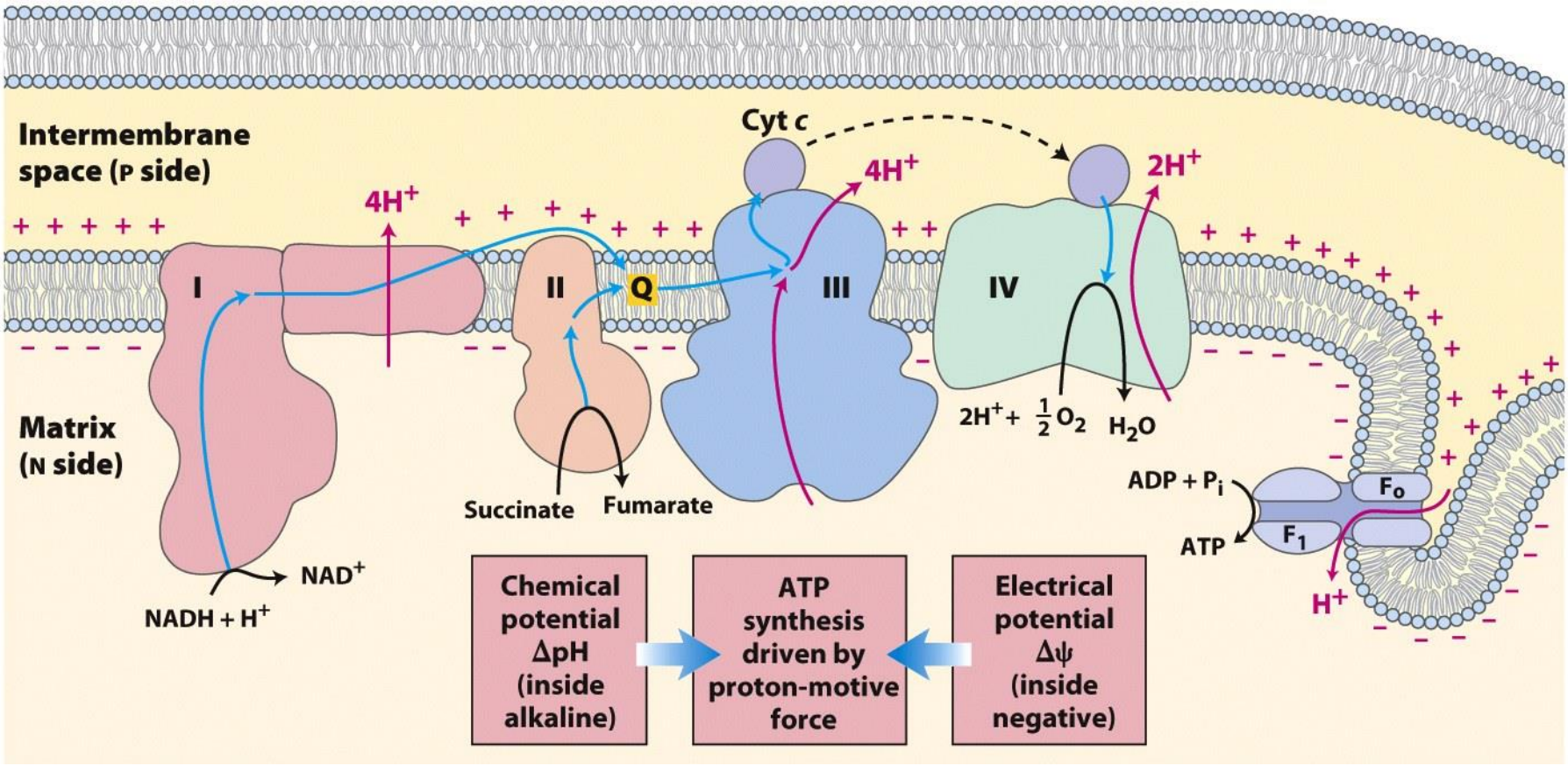
Proton-motive force



$$\Delta G = RT \ln (C_2/C_1) + Z\mathcal{F}\Delta\psi$$

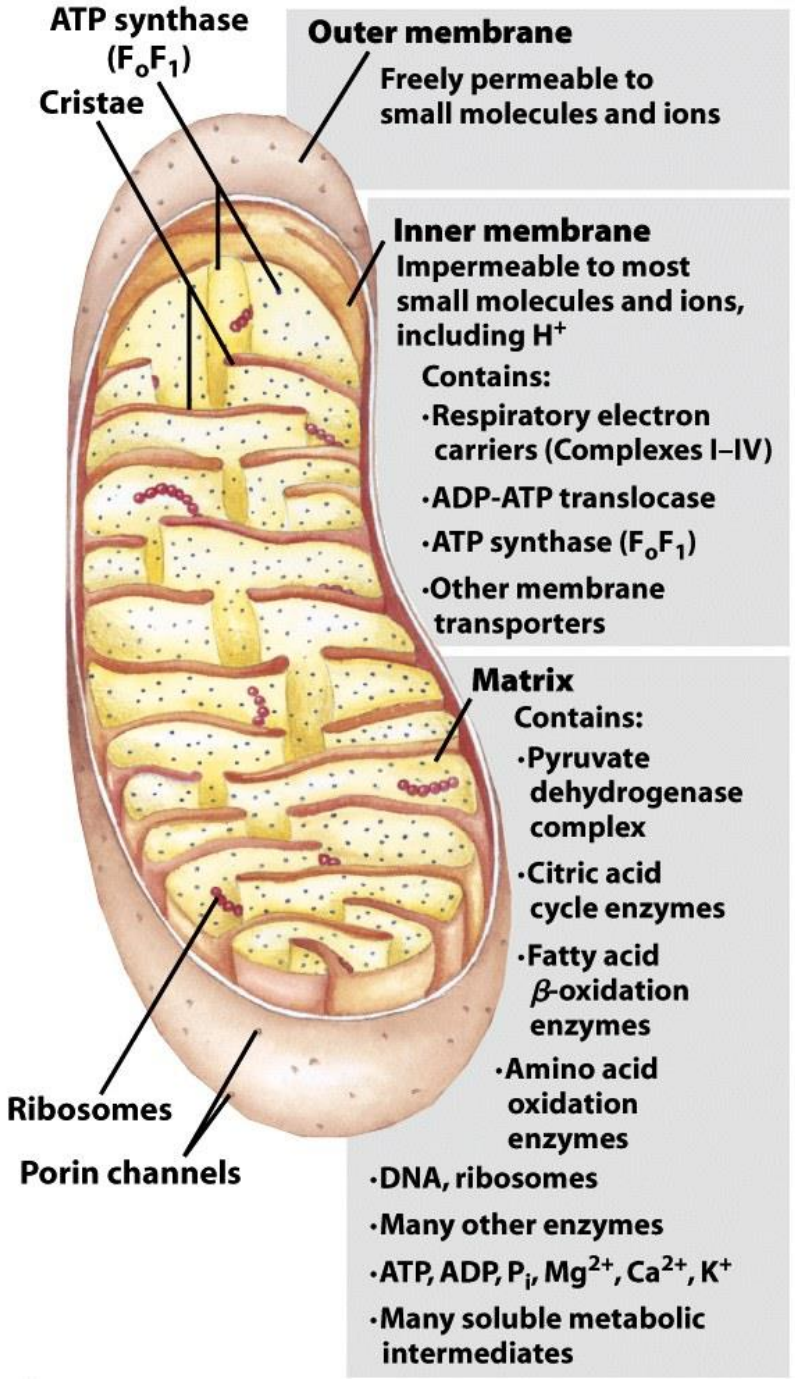
$$= 2.3RT\,\Delta\text{pH} + \mathcal{F}\Delta\psi$$

Chemiosmotic model



Protons flow passively back into the matrix → associated with ATP synthesis

A mitochondrion

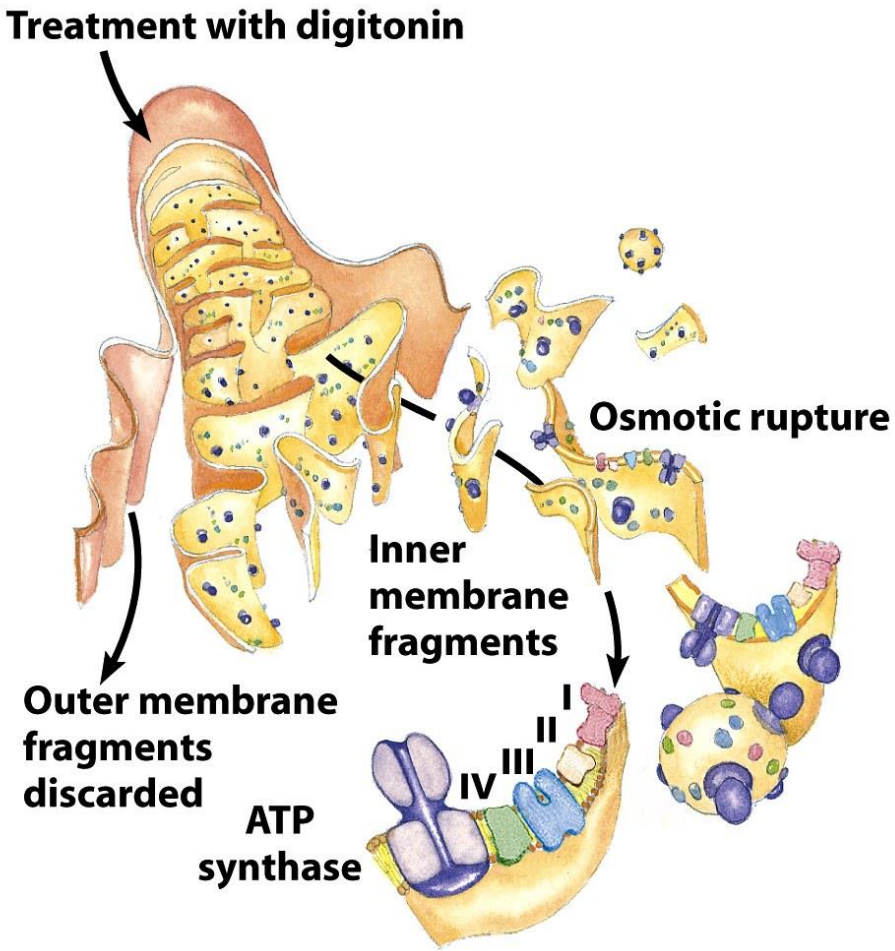


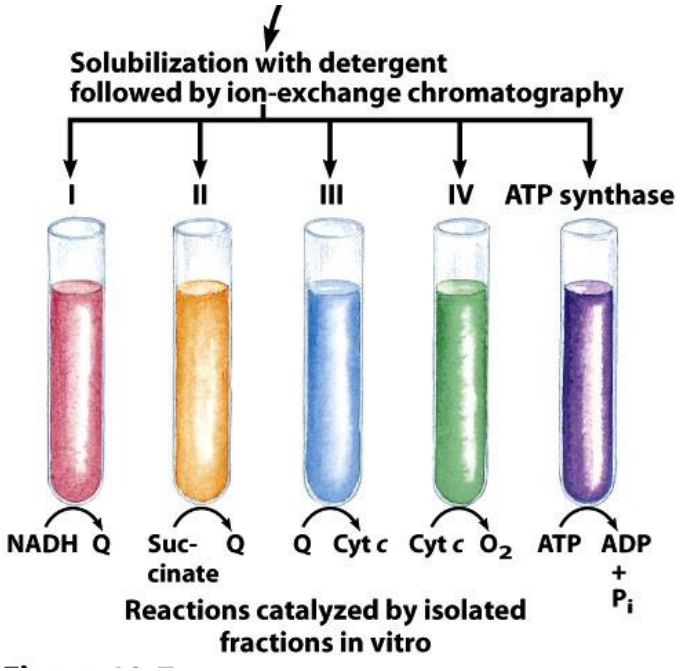
Components of the **respiratory chain** and **ATP synthesis**

All the pathways of **fuel oxidation** except glycolysis

How do you study the components of electron transfer complexes ?

Separation of functional complexes of the respiratory chain





NADH → Q → Cyt b → Cyt c_1 → Cyt c → Cyt $(a+a_3)$ → O_2

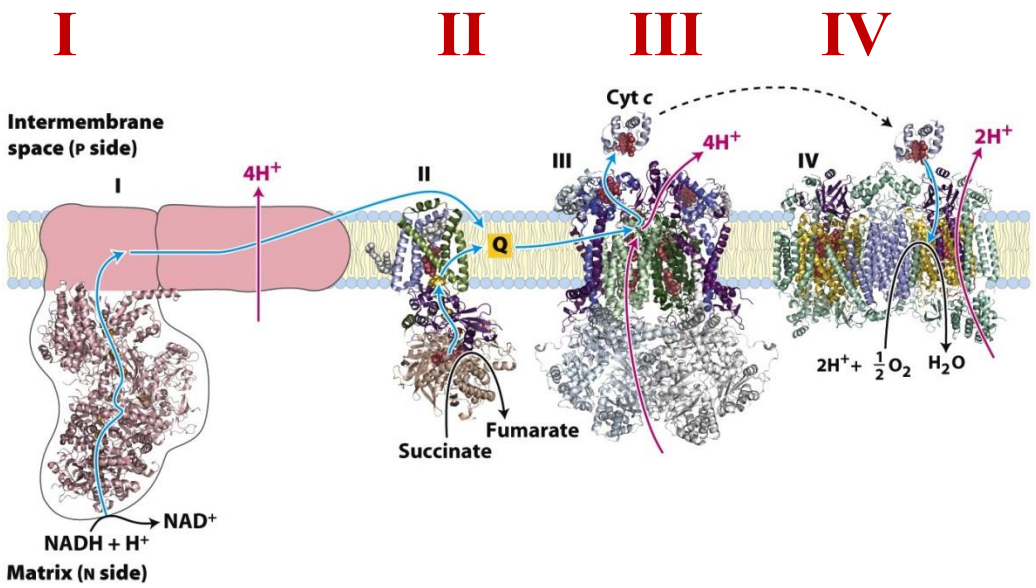
The Protein Components of the Mitochondrial Electron-Transfer Chain

TABLE 19–3 The Protein Components of the Mitochondrial Electron-Transfer Chain

Enzyme complex/protein	Mass (kDa)	Number of subunits*	Prosthetic group(s)
I NADH dehydrogenase	850	43 (14)	FMN, Fe-S
II Succinate dehydrogenase	140	4	FAD, Fe-S
III Ubiquinone:cytochrome *c* oxidoreductase	250	11	Hemes, Fe-S
Cytochrome *c*†	13	1	Heme
IV Cytochrome oxidase	160	13 (3–4)	Hemes; Cu_A, Cu_B

*Numbers of subunits in the bacterial equivalents in parentheses.

†Cytochrome *c* is not part of an enzyme complex; it moves between Complexes III and IV as a freely soluble protein.



How do you know the order?

Method for determining the sequence of electron carriers (1) : ordering by the reduction potentials

TABLE 19–2 Standard Reduction Potentials of Respiratory Chain and Related Electron Carriers

Redox reaction (half-reaction)	E'° (V)
$2H^+ + 2e^- \longrightarrow H_2$	−0.414
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
NADH dehydrogenase (FMN) + $2H^+ + 2e^- \longrightarrow$ NADH dehydrogenase ($FMNH_2$)	−0.30
Ubiquinone + $2H^+ + 2e^- \longrightarrow$ ubiquinol	0.045
Cytochrome *b* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	0.077
Cytochrome c_t (Fe^{3+}) + $e^- \longrightarrow$ cytochrome c_t (Fe^{2+})	0.22
Cytochrome *c* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *c* (Fe^{2+})	0.254
Cytochrome *a* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	0.29
Cytochrome a_3 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.8166

NADH → Q → Cyt b → Cyt c_1 → Cyt c → Cyt (a+a_3) → O_2

Method for determining the sequence of electron carriers (2) : measuring the rate of oxidation

Reducing the carriers by providing electron source
→ Introduce O_2 and identify the rate of oxidation
→ The carrier nearest O_2 gives up its electron first

NADH → Q → Cyt b → Cyt c_1 → Cyt c → Cyt $(a+a_3)$ → O_2

Method for determining the sequence of electron carriers (3): using inhibitors (with measuring the degree of oxidation of each carrier)

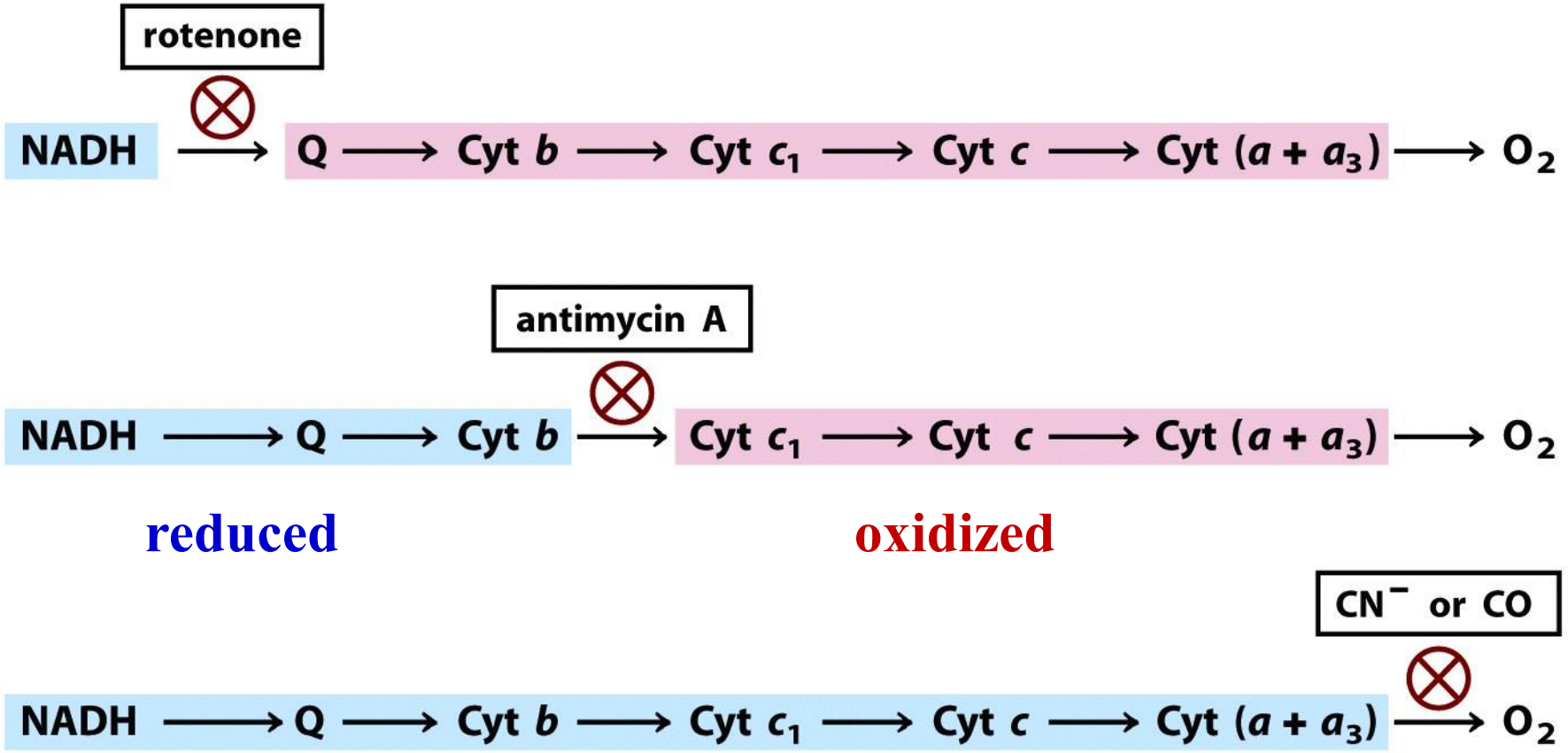


Figure 19-6
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TABLE 19–4 Agents That Interfere with Oxidative Phosphorylation or Photophosphorylation

Type of interference	Compound*	Target/mode of action
Inhibition of electron transfer	Cyanide, Carbon monoxide	Inhibit cytochrome oxidase
	Antimycin A	Blocks electron transfer from cytochrome *b* to cytochrome c_1
	Myxothiazol, Rotenone, Amytal, Piericidin A	Prevent electron transfer from Fe-S center to ubiquinone
	DCMU	Competes with Q_B for binding site in PSII
Inhibition of ATP synthase	Aurovertin	Inhibits F_1
	Oligomycin, Venturicidin	Inhibit F_o and CF_o
	DCCD	Blocks proton flow through F_o and CF_o
Uncoupling of phosphorylation from electron transfer	FCCP, DNP	Hydrophobic proton carriers
	Valinomycin	K^+ ionophore
	Thermogenin	In brown adipose tissue, forms proton-conducting pores in inner mitochondrial membrane
Inhibition of ATP-ADP exchange	Atractyloside	Inhibits adenine nucleotide translocase

*DCMU is 3-(3,4-dichlorophenyl)-1,1-dimethylurea; DCCD, dicyclohexylcarbodiimide; FCCP, cyanide-*p*-trifluoromethoxyphenylhydrazone; DNP, 2,4-dinitrophenol.

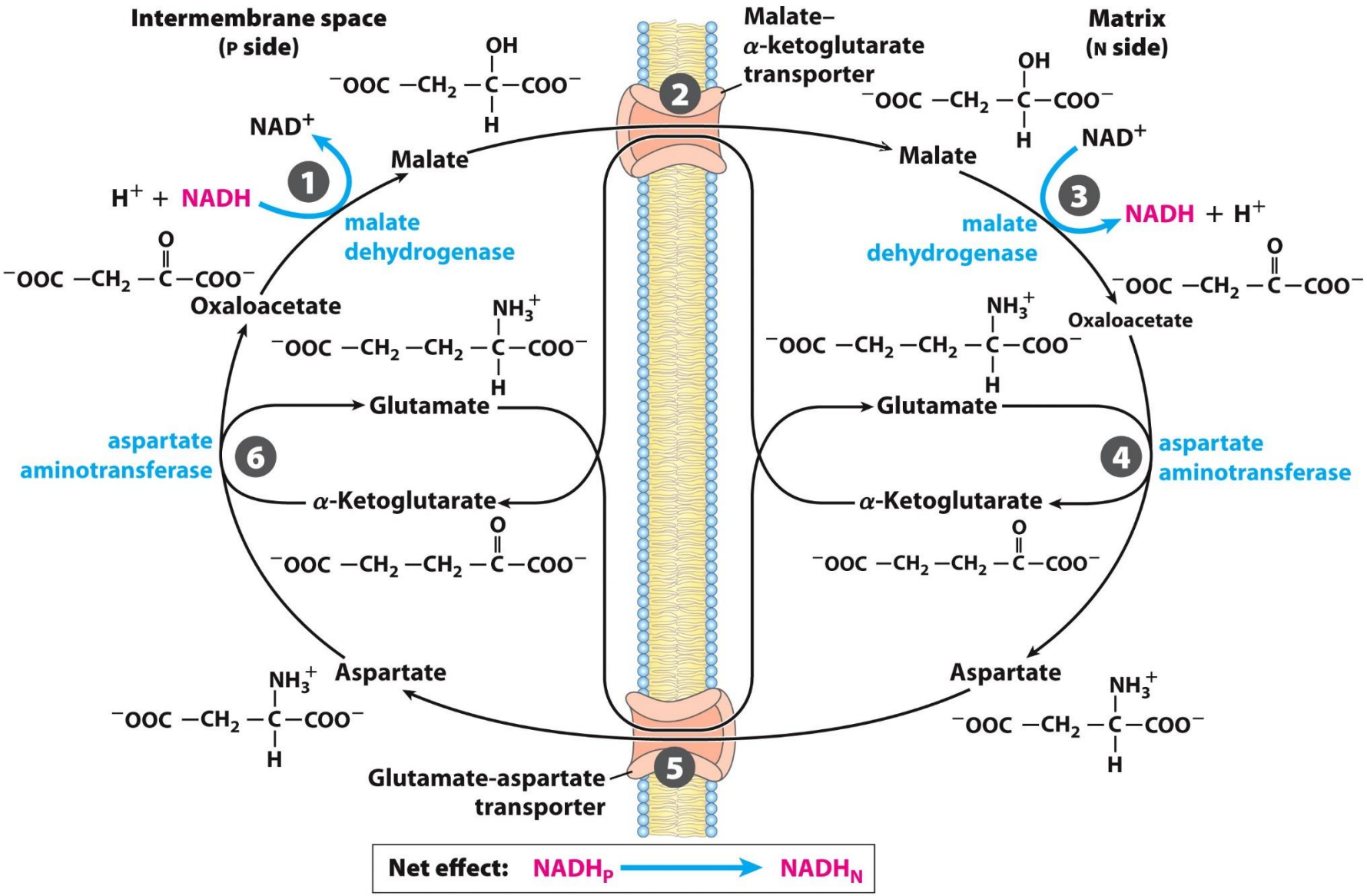
Table 19-4

Lehninger Principles of Biochemistry, Fifth Edition

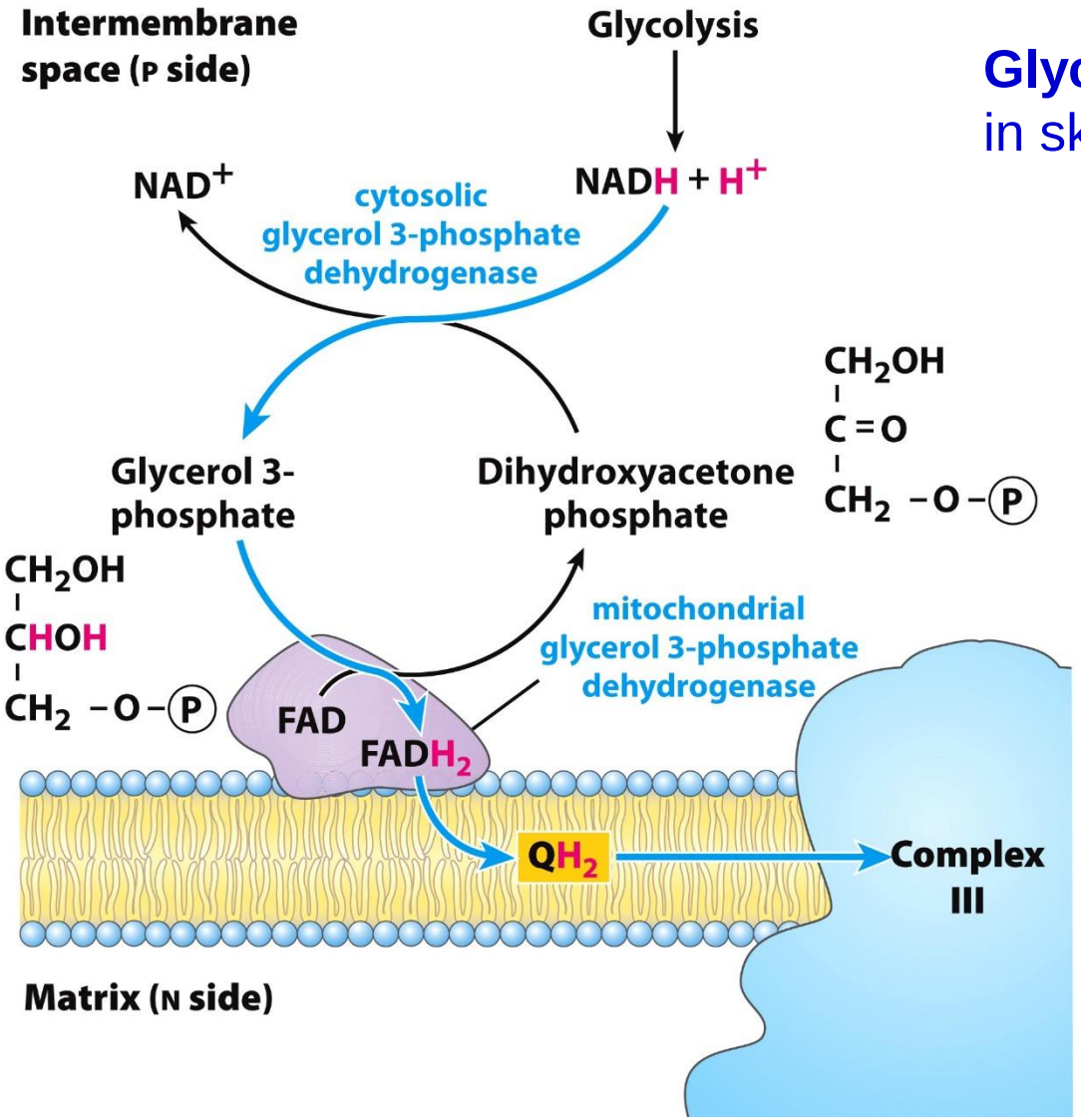
© 2008 W. H. Freeman and Company

How cytosolic NADH is transferred into mitochondria?

Malate-aspartate shuttle for transporting reducing equivalents from cytosolic NADH into the mitochondrial matrix



Alternative way of moving reducing equivalents from the cytosol



Glycerol 3-phosphate shuttle
in skeletal muscle and the brain

Synthesize 1.5 ATP / $2e^-$

Summary

1. Electrons are funneled to universal electron acceptors (NAD^+, $NADP^+$, FMN or FAD)
2. Electrons pass thru a series of membrane-bound carriers
3. Electron carriers function in multienzyme complexes (complex I ~ IV)
4. Mitochondrial complexes may associate in respirasomes
5. The energy of electron transfer is efficiently conserved in a proton gradient
6. Reactive oxygen species are generated during oxidative phosphorylation
7. Plant mitochondria have alternative mechanisms for oxidizing NADH

소진증후군 burnout syndrome

불타버린 연료와 같이 극도의 피로감을 느끼며 무기력해지는 것을 의미

개인 차원의 문제, 집단의 시스템 문제, 사회적 분위기 등이 복합적으로 작용

Today in this class ...



ATP synthesis

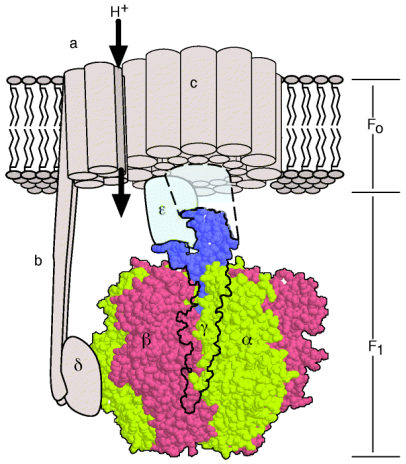


Regulation of oxidative phosphorylation

Mitochondria genes and functions

Oxidative phosphorylation

19.2 ATP synthesis

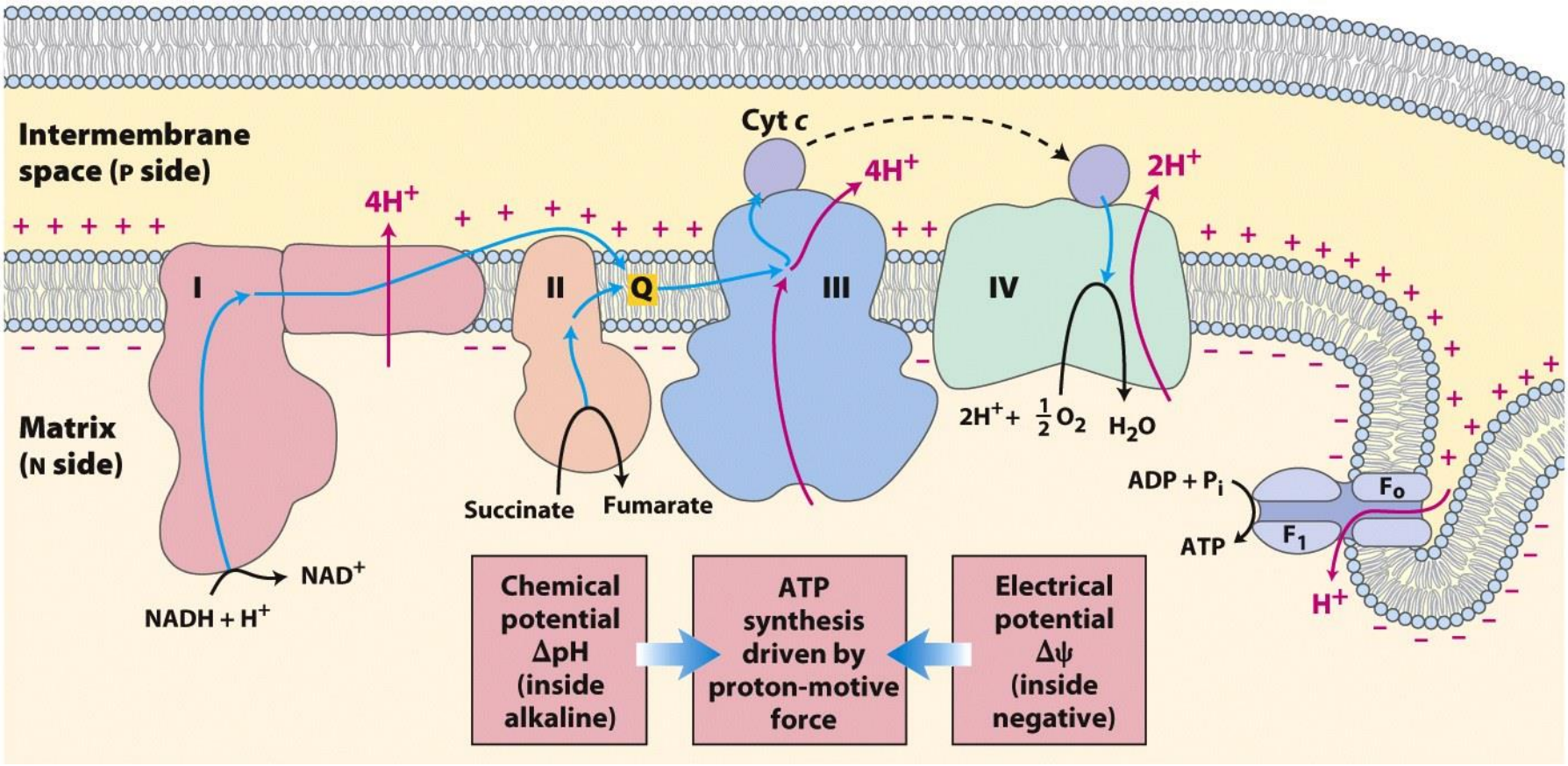




Chap19_ATPsynthesis.mp4

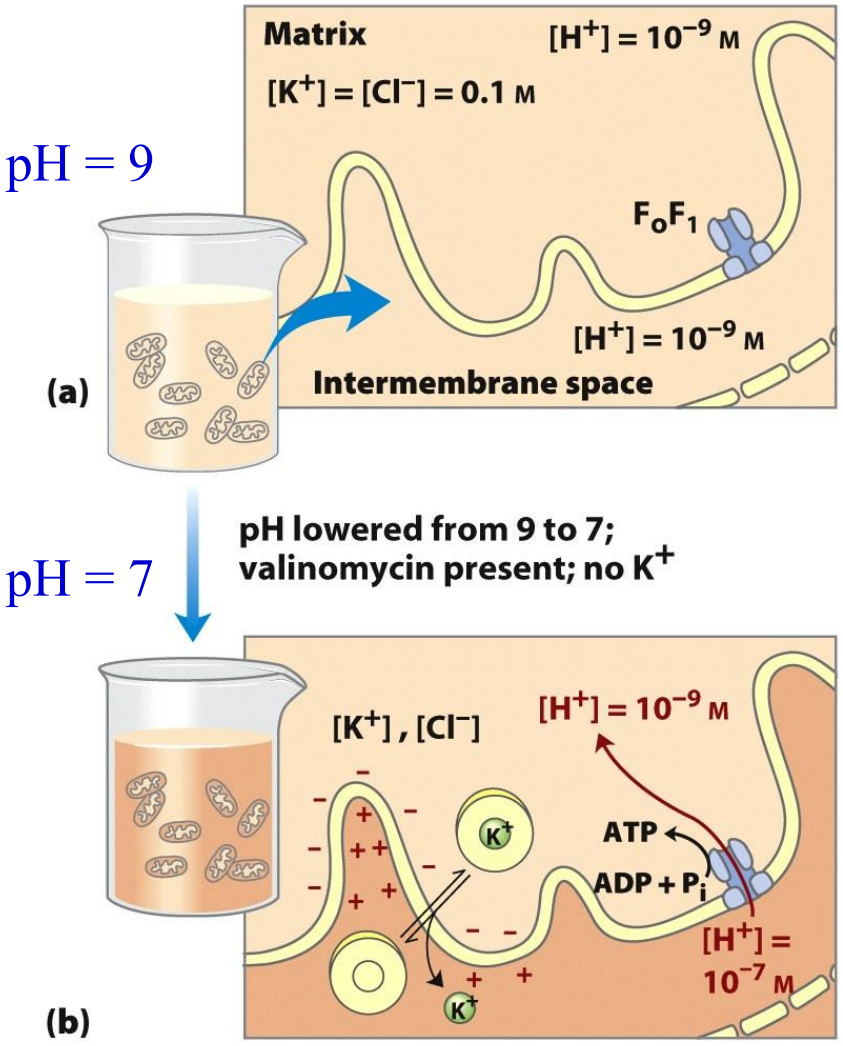
Chemiosmosis

Chemiosmotic model



Protons flow passively back into the matrix → associated with ATP synthesis

An artificially created proton gradient → driving ATP synthesis



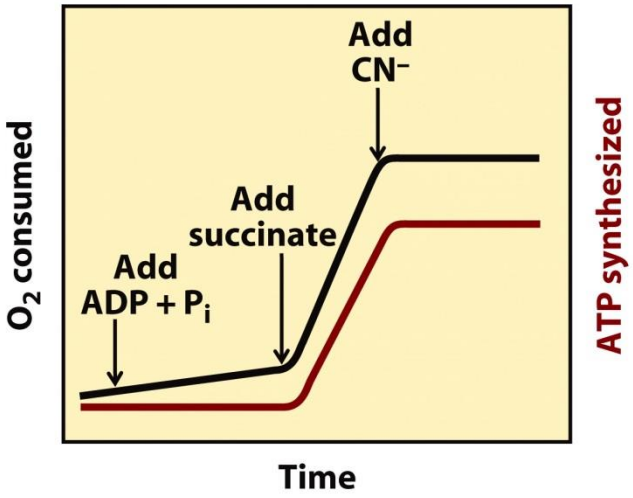
Proton gradient &
Outward flow of K^+
→ ATP synthesis

: proton-motive force suffices to drive ATP synthesis

“ Coupling” : neither reaction occurs without the other

Inhibitors of electron transfer → block ATP synthesis

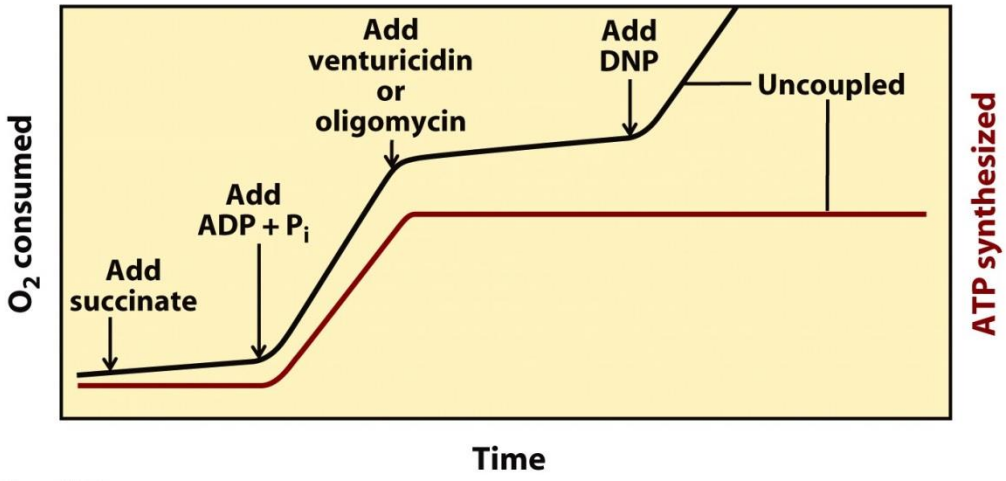
Inhibitors of ATP synthesis → block electron transfer



substrate

Providing electrons

Inhibition of electron transfer



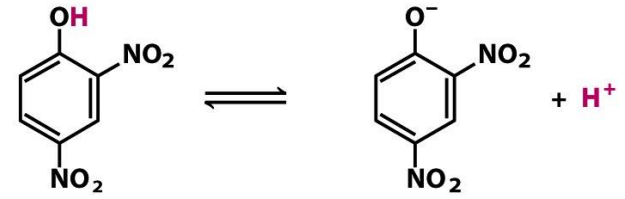
Providing electrons

substrate

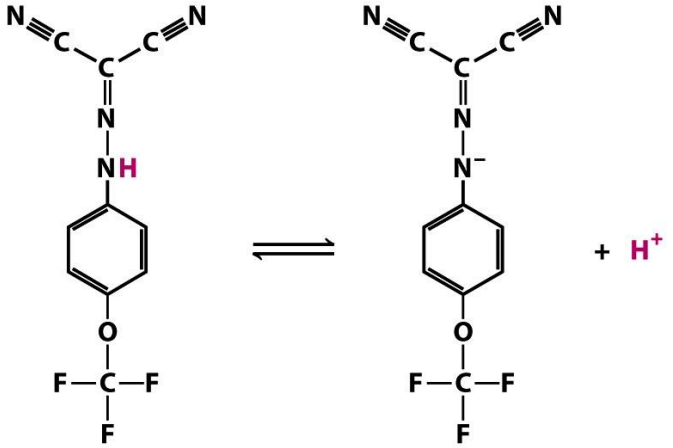
Inhibition of ATP synthase

Uncoupling agent

Two chemical uncouplers of oxidative phosphorylation



2,4-Dinitrophenol
(DNP)



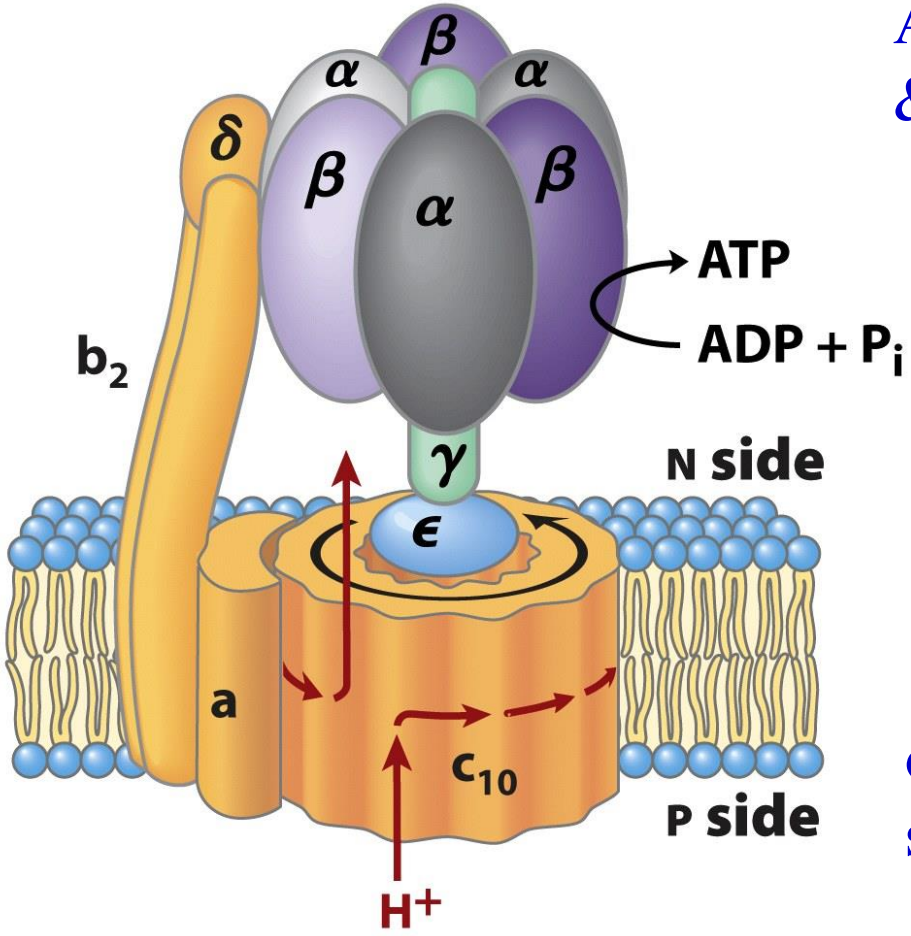
Carbonylcyanide-*p*-
trifluoromethoxyphenylhydrazone
(FCCP)

Ionophores

- Weak acids with hydrophobic properties
- Diffuse freely across the membrane

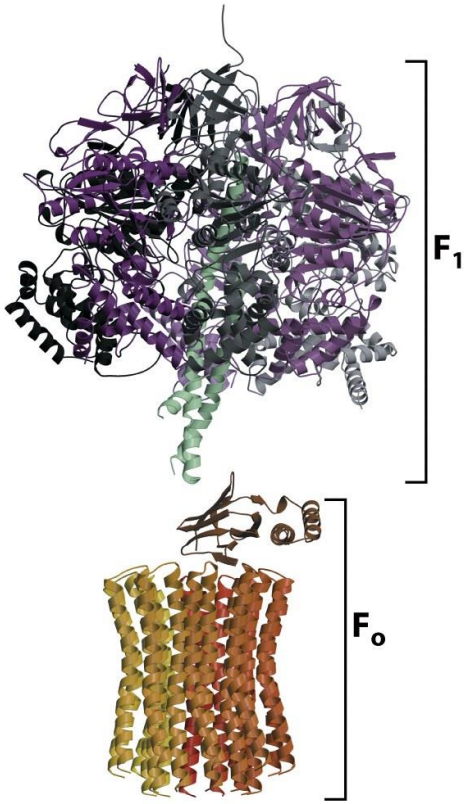
Mechanism of ATP synthesis

Structure of ATP synthase



ATP synthesis
& hydrolysis

F_1



Fo

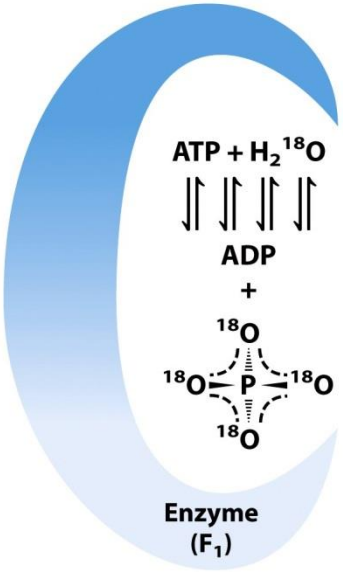
oligomycin
sensitive

Q. How is it possible to attach Pi to ADP to produce ATP?

Q. Is the energy needed for ATP synthesis or ATP release ?

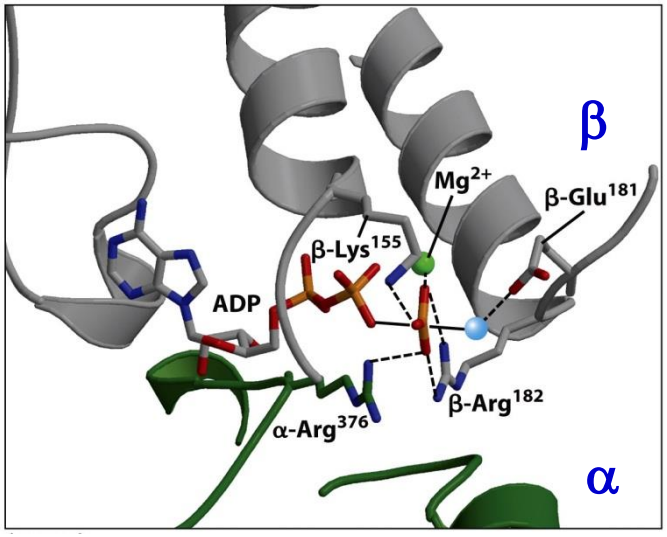
Catalytic mechanism of F_1

Isotope exchange exp → 4 O^{18} Pi



Enz-ATP ⇌ Enz-(ADP + Pi)
on F1:
K'eq = 2.4
$\Delta G'^{\circ} \approx 0$ → Reaction is readily reversible

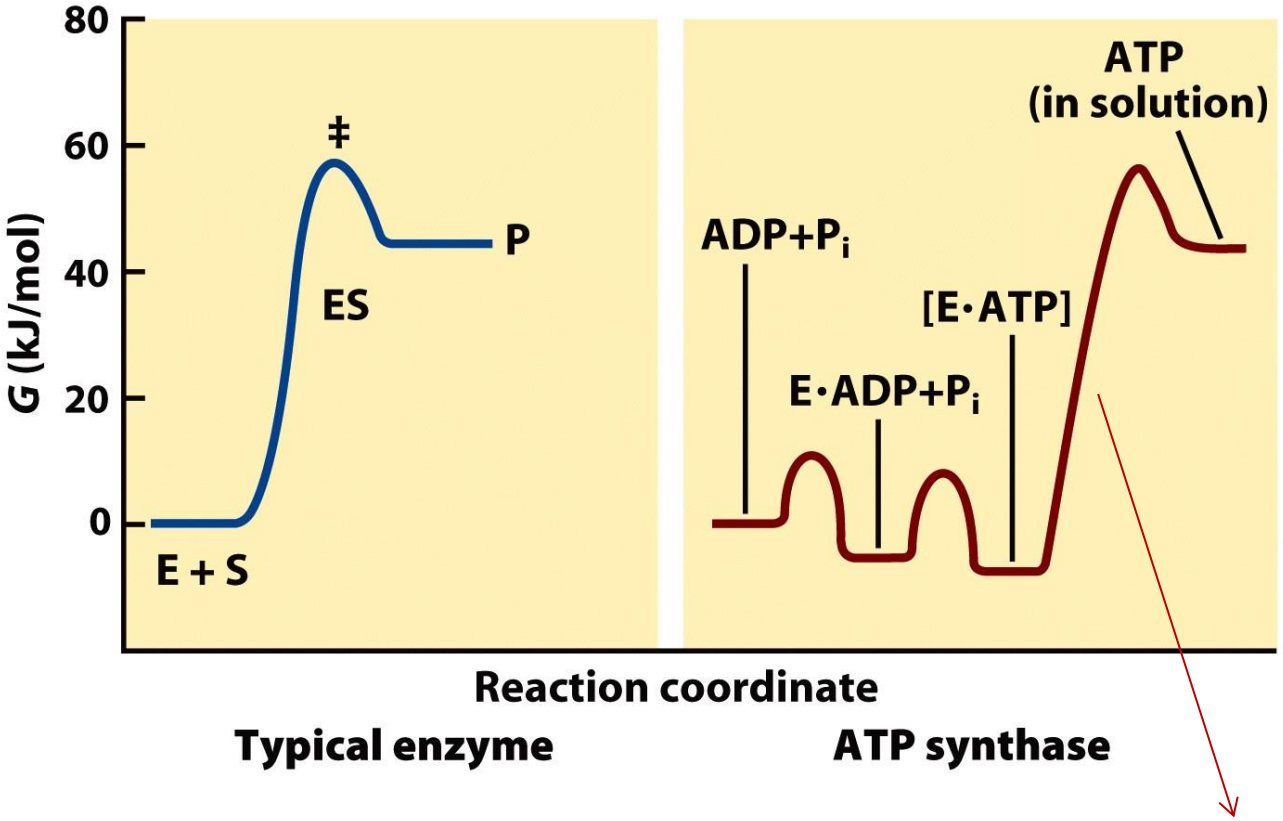
In solution:
$ATP + H_2O \rightleftharpoons ADP + Pi$
$\Delta G'^{\circ} = -30.5$ kJ/mol



F_oF_1 binds ATP with very high affinity ($K_d \leq 10^{-12}$ M), while ADP with lower affinity ($K_d \approx 10^{-5}$ M)

→ **Drive the *Eq* toward to ATP**

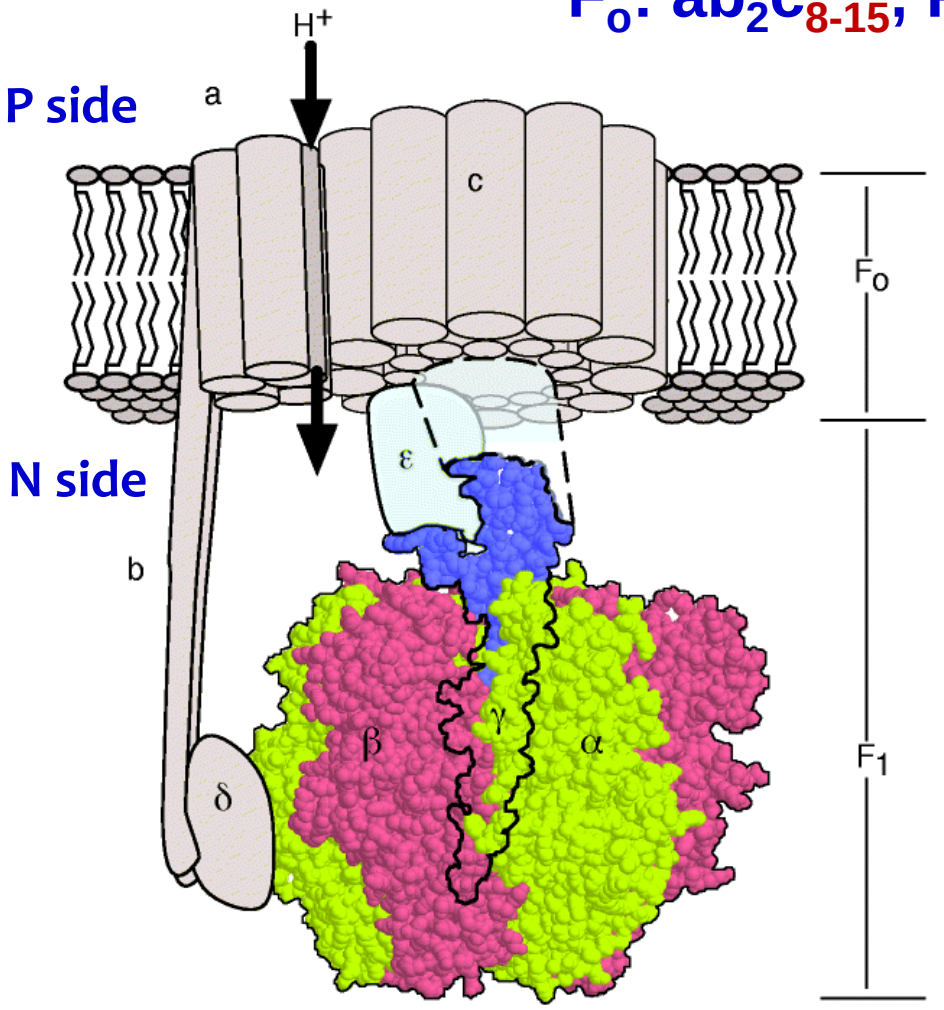
Reaction coordinate diagrams for ATP synthase and for a more typical enzyme



Need of energy for releasing ATP
Provided by the proton flow

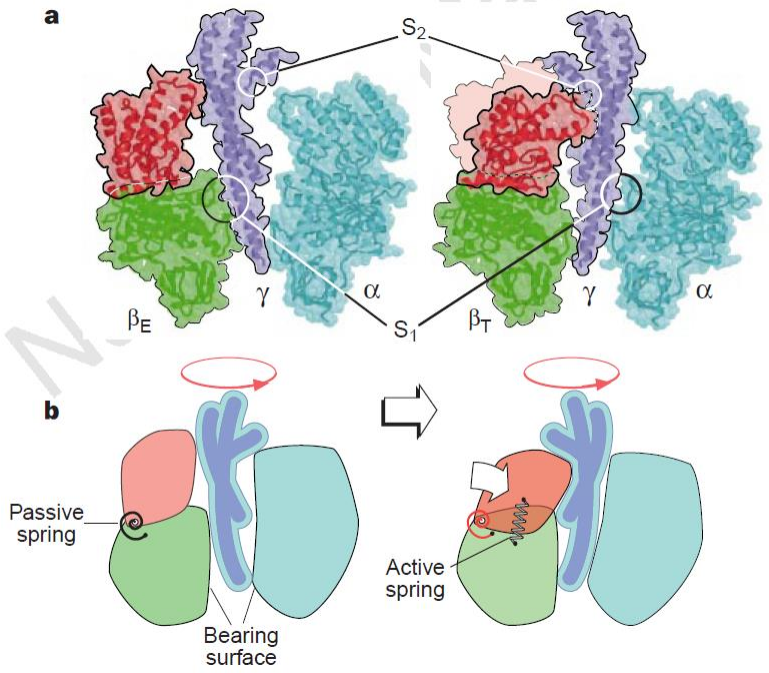
ATP synthase has two functional domains, F_o and F_1

F_o: ab_2c_{8-15}, F_1: $\alpha_3\beta_3\gamma\delta\varepsilon$



Nature. (1998) 396:279-82.

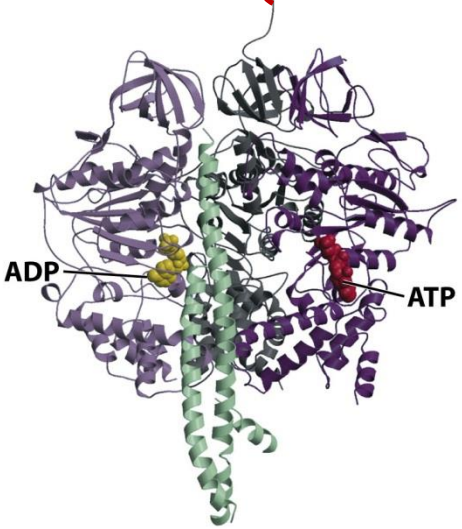
γ as central shaft



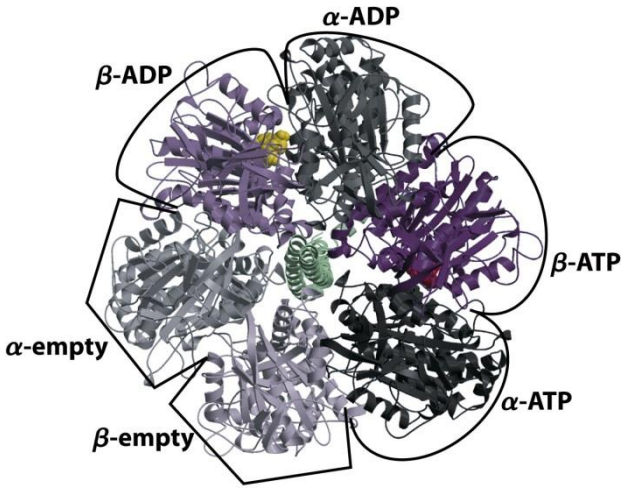
F_o : oligomycin sensitive
Proton pore

F_1: ATP synthesis & hydrolysis

ATP is released by the conformational change of β subunit (rotational catalysis)

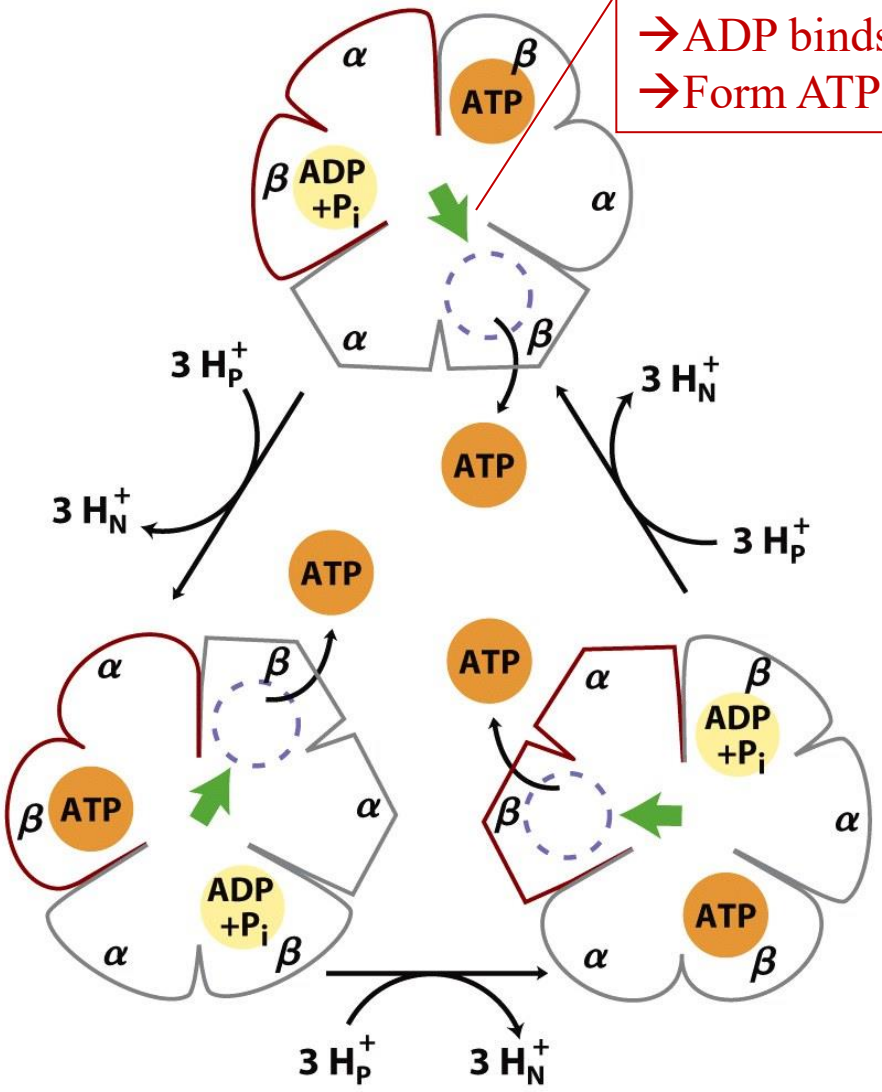


conformations of β differ

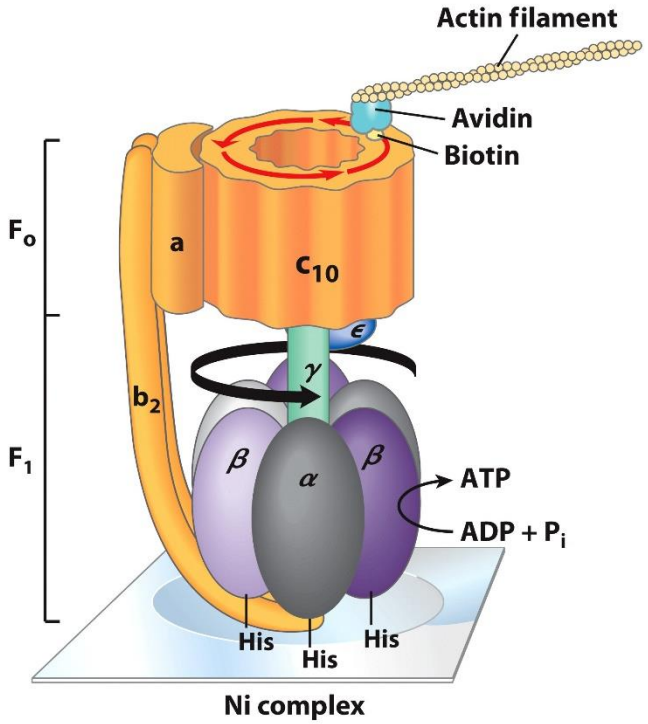


γ associated β-empty

Rotation of γ
→β-empty form
→Release ATP
→ADP binds
→Form ATP

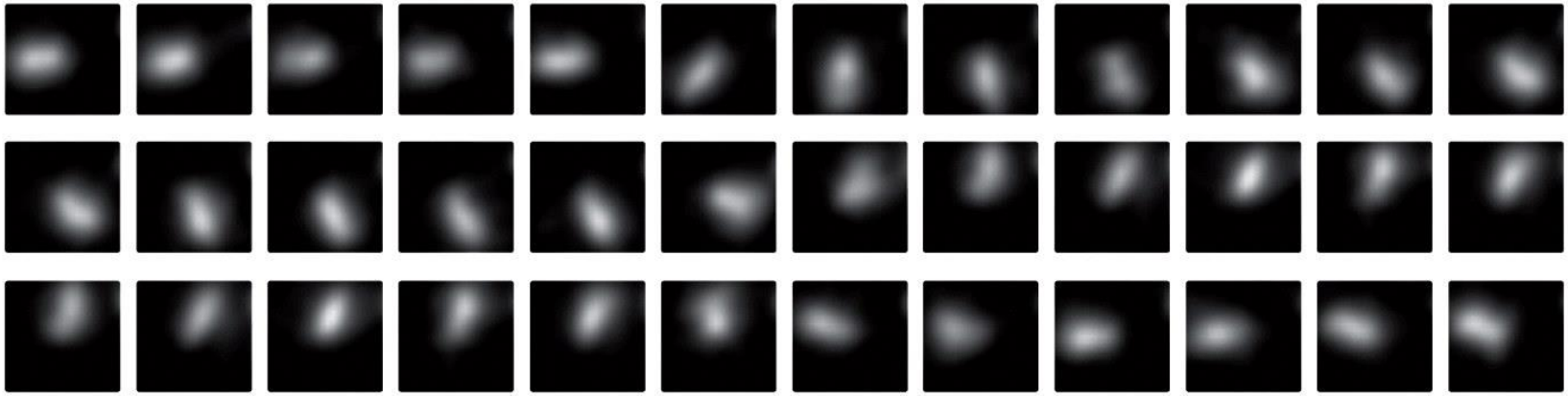


Experimental demonstration of rotation of F_o and γ



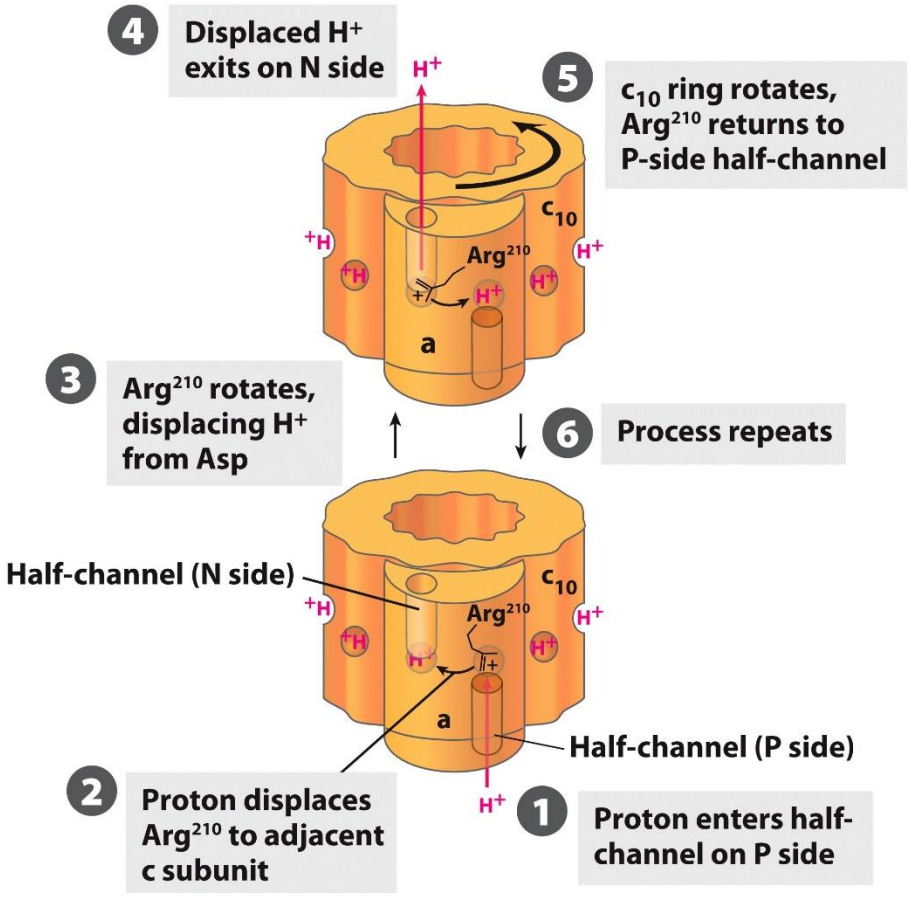
- F_1 engineered to contain His on a Ni complex;
- Biotin is attached to a c subunit of F_o.
- Avidin binds biotin & attached to actin (labeled with a fluorescent probe)

→ **γ rotates in one direction**

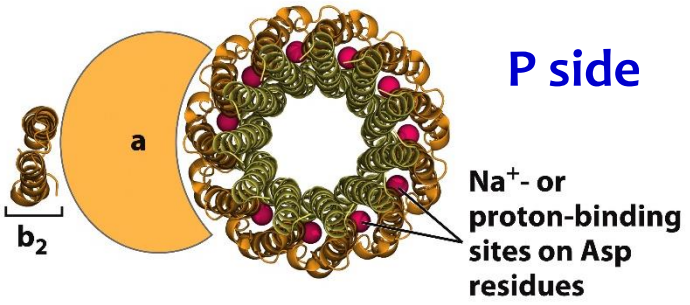


Experimental demonstration of rotation of F_o and γ

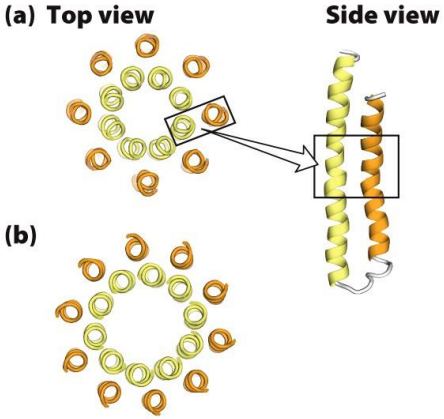
N side



P side

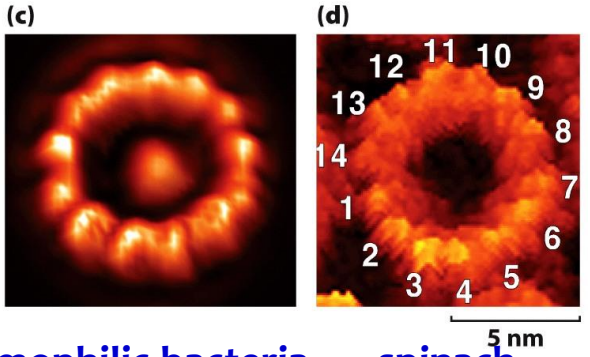


P side



Bovine

Yeast



Thermophilic bacteria

spinach

Nonintegral stoichiometries of O_2 consumption and ATP synthesis

How many ATPs are formed by an NADH ? (**P/O ratio *x***)

$$x\text{ADP} + x\text{Pi} + \frac{1}{2}O_2 + H^+ + \text{NADH} \rightarrow x\text{ATP} + H_2O + \text{NAD}^+$$

Chemiosmotic paradigm:

P/O is not necessarily integral !

Measurement of proton flux :

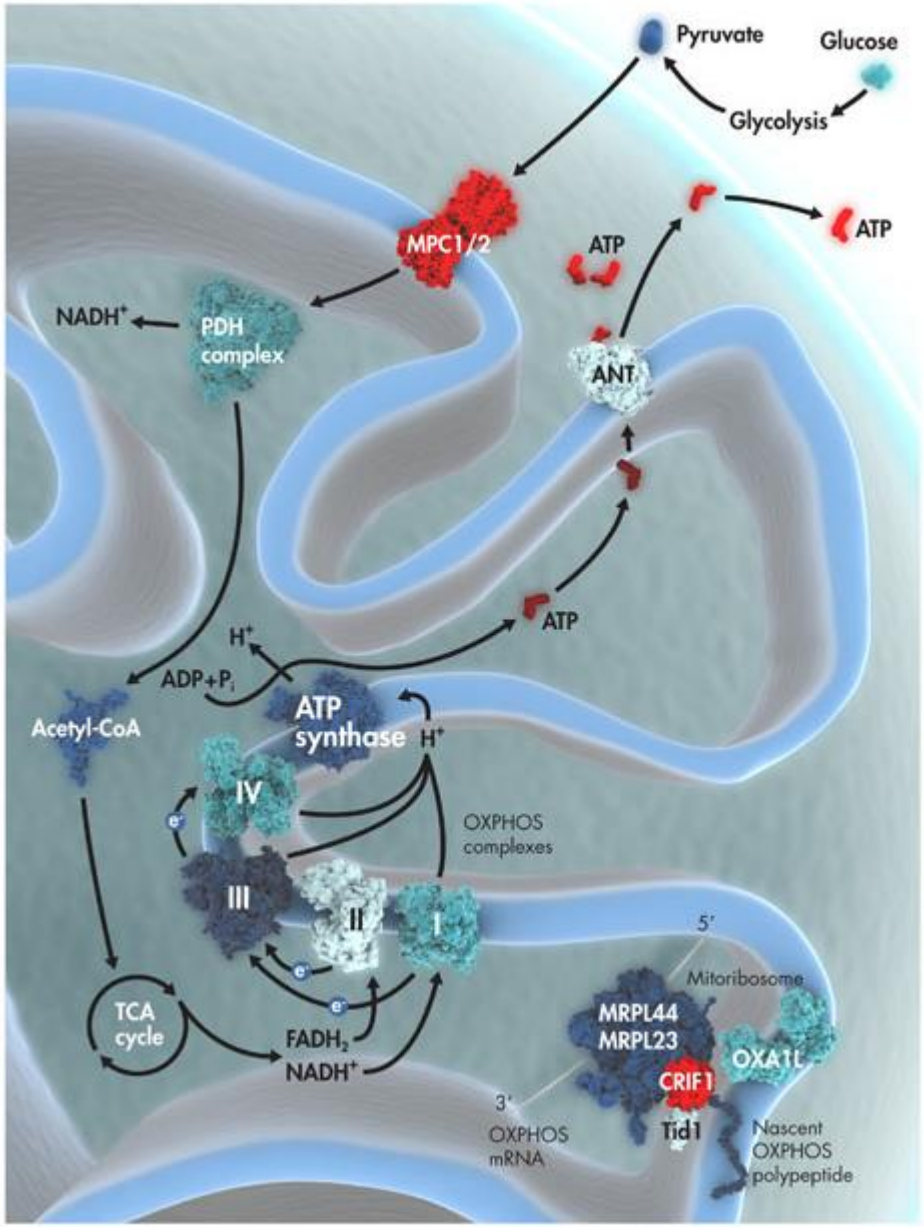
10 protons for NADH and 6 for succinate / a pair of electrons

4 protons for one ATP

→2.5 ATPs for NADH and 1.5 for succinate ($FADH_2$)

But, 3 for NADH and 2 for $FADH_2$ are still common in literatures

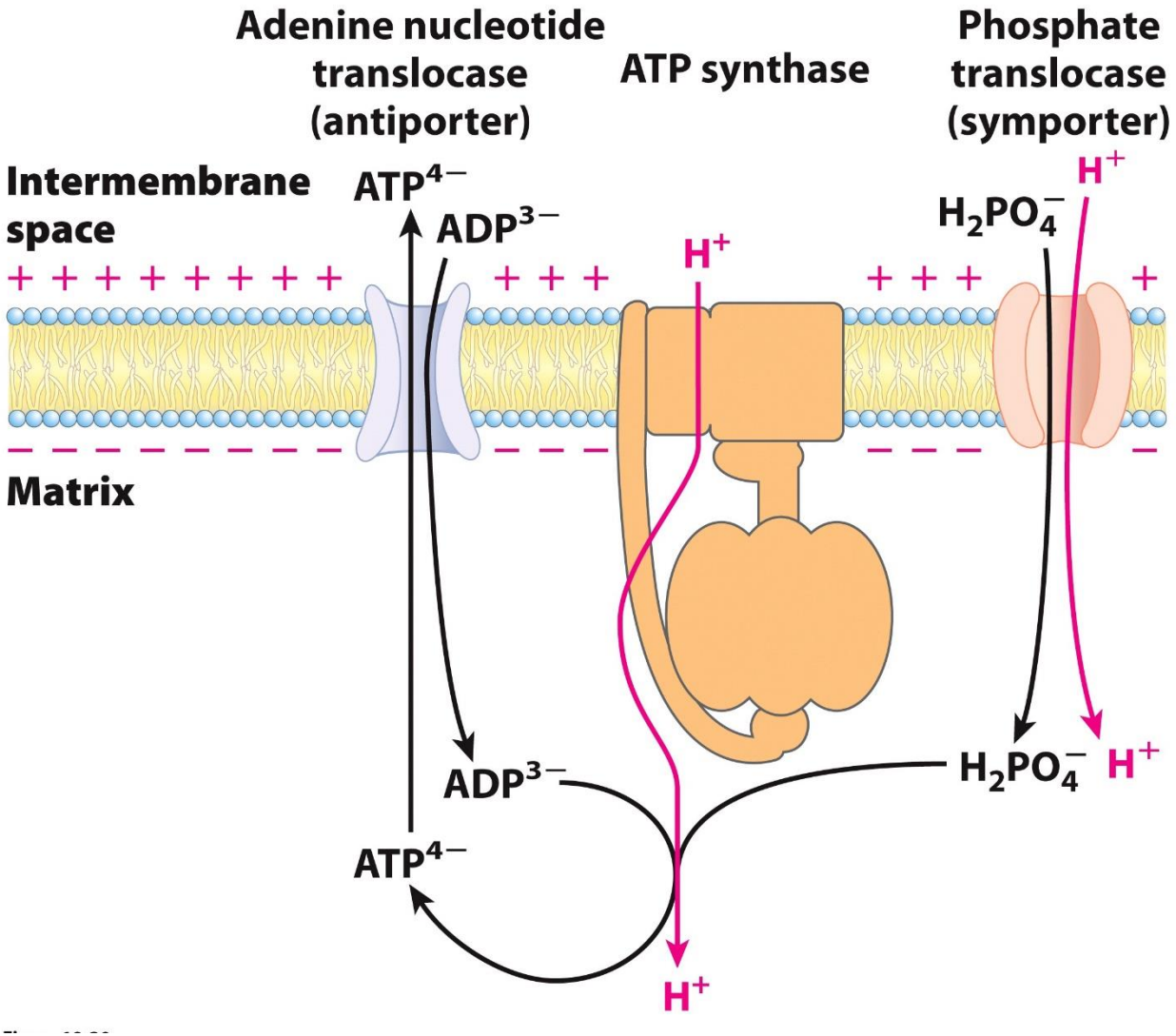
Mitochondiral pyruvate carrier (MPC)



Adenine nucleotide translocator (ANT)

Mitochondrial ribosome complex

Transport the synthesized ATPs to cytosol



Summary

1. Proton gradient generated by the complex I, III, and IV provides the energy for ATP synthesis
2. ATP synthase carries out “rotational catalysis” with two functional domains Fo and F1
3. ATP is stabilized relative to ADP on the surface of F1
4. Chemiosmotic coupling allows nonintegral stoichiometries of O2 consumption and ATP synthesis
5. Shuttle systems indirectly convery cytosolic NADH into mitochondria for oxidation

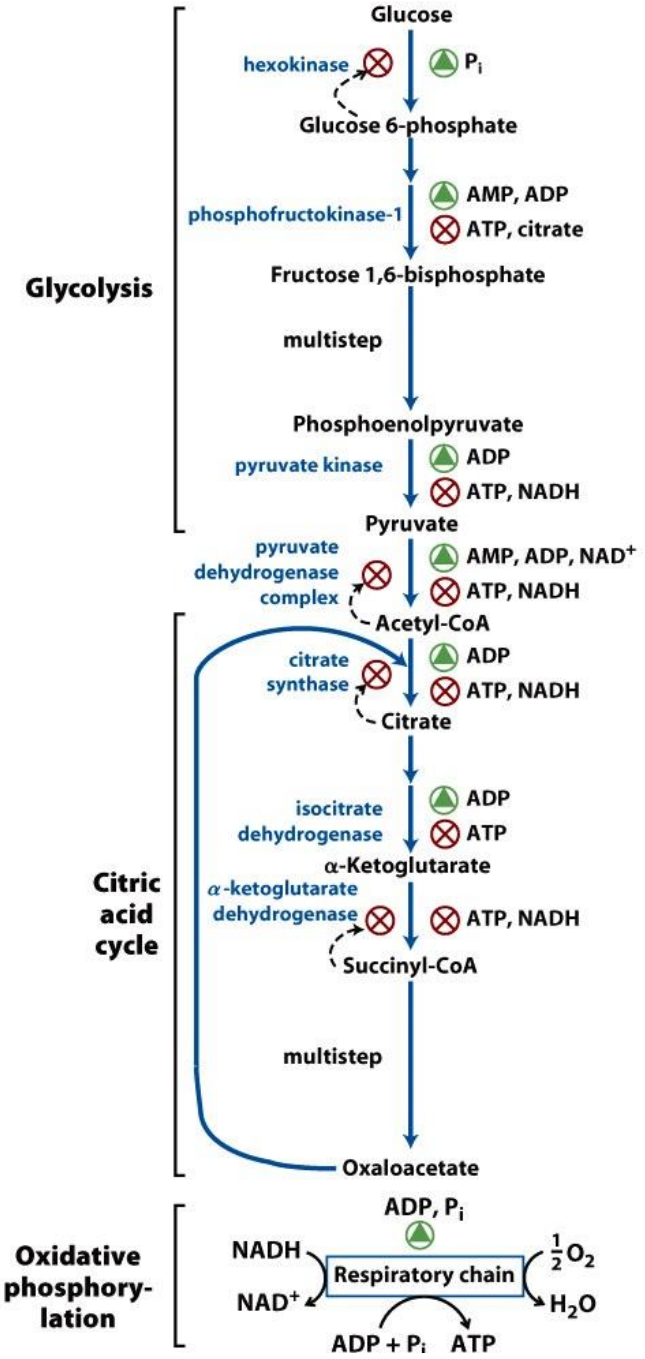
19.3 Regulation of oxidative phosphorylation

Oxidative phosphorylation is regulated by cellular energy needs

1. **Acceptor control** of respiration : by the availability of **ADP**
2. Acceptor control ratio : maximal rate of ADP-induced O_2 consumption to basal rate ≥ 10
3. Mass-action ratio : [ATP]/([ADP][Pi])
4. Energy requiring process → mass-action ratio decrease → the rate of respiration increases

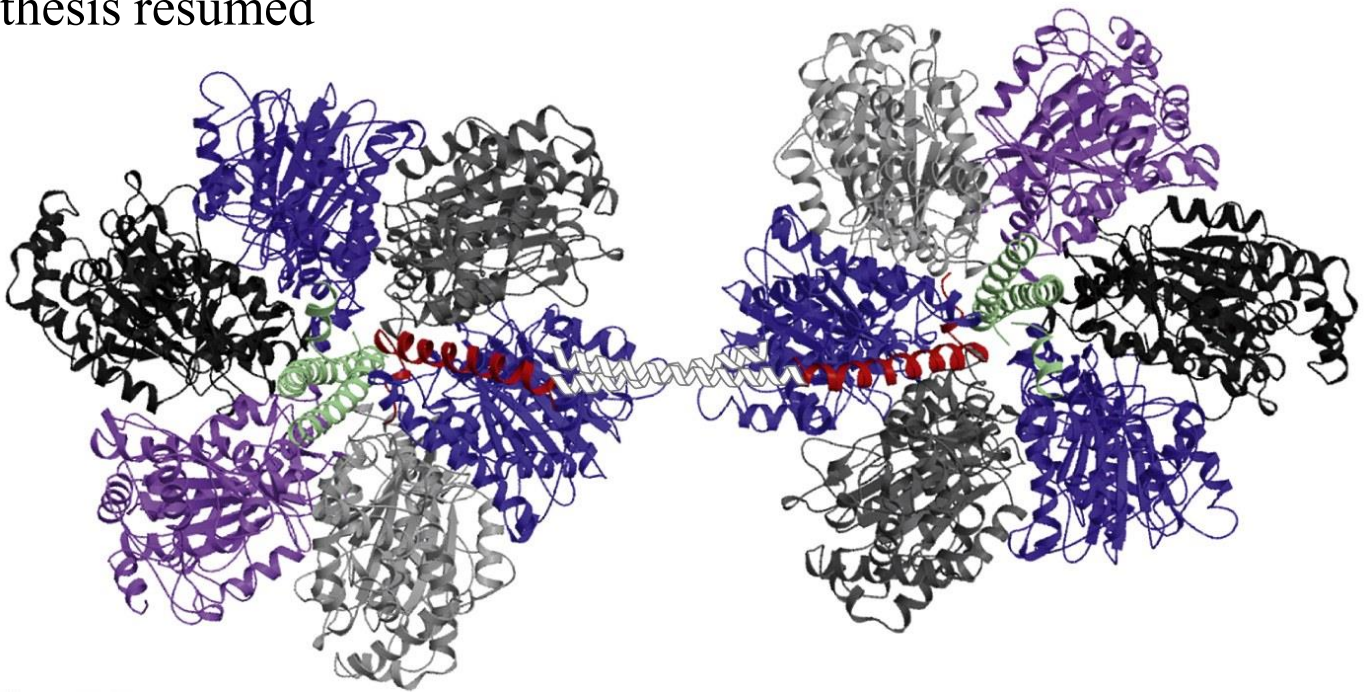
ATP producing pathways are coordinately regulated

[ATP]/([ADP][Pi])
[NADH]/[NAD+]
[citrate]



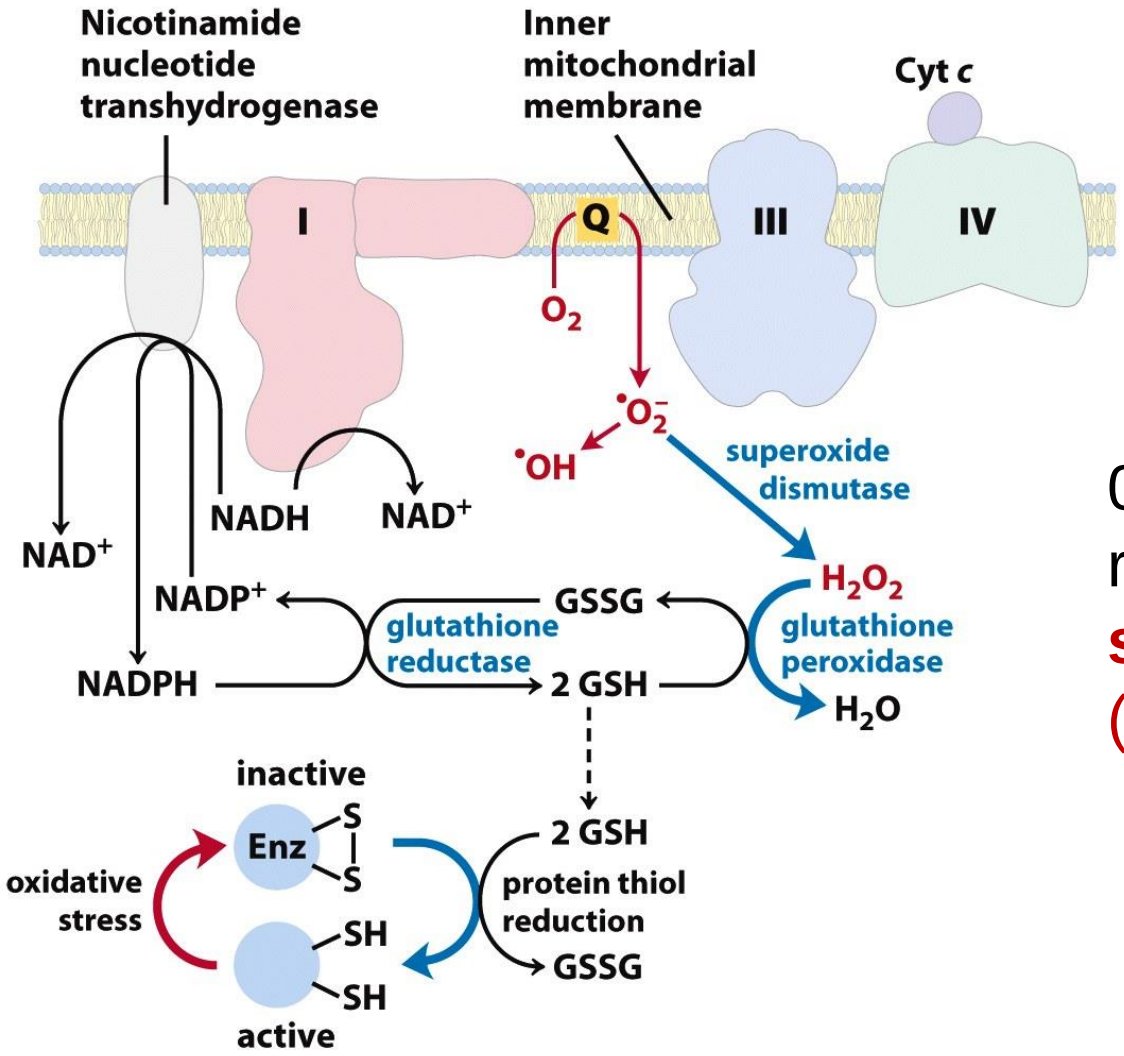
An inhibitory protein (IF_1) prevents ATP hydrolysis during hypoxia

- When a cell is hypoxic → electron transfer slows → ATP synthesis slows → ATP hydrolysis is prevented by IF1 (to save ATP)
- In oxygen defieciency → pyruvate and lactate increases → pH < 6.5 → IF1 dimerization → ATPase inhibition
- In aerobic condition → pH increases → IF1 dimer destabilized → ATP synthesis resumed

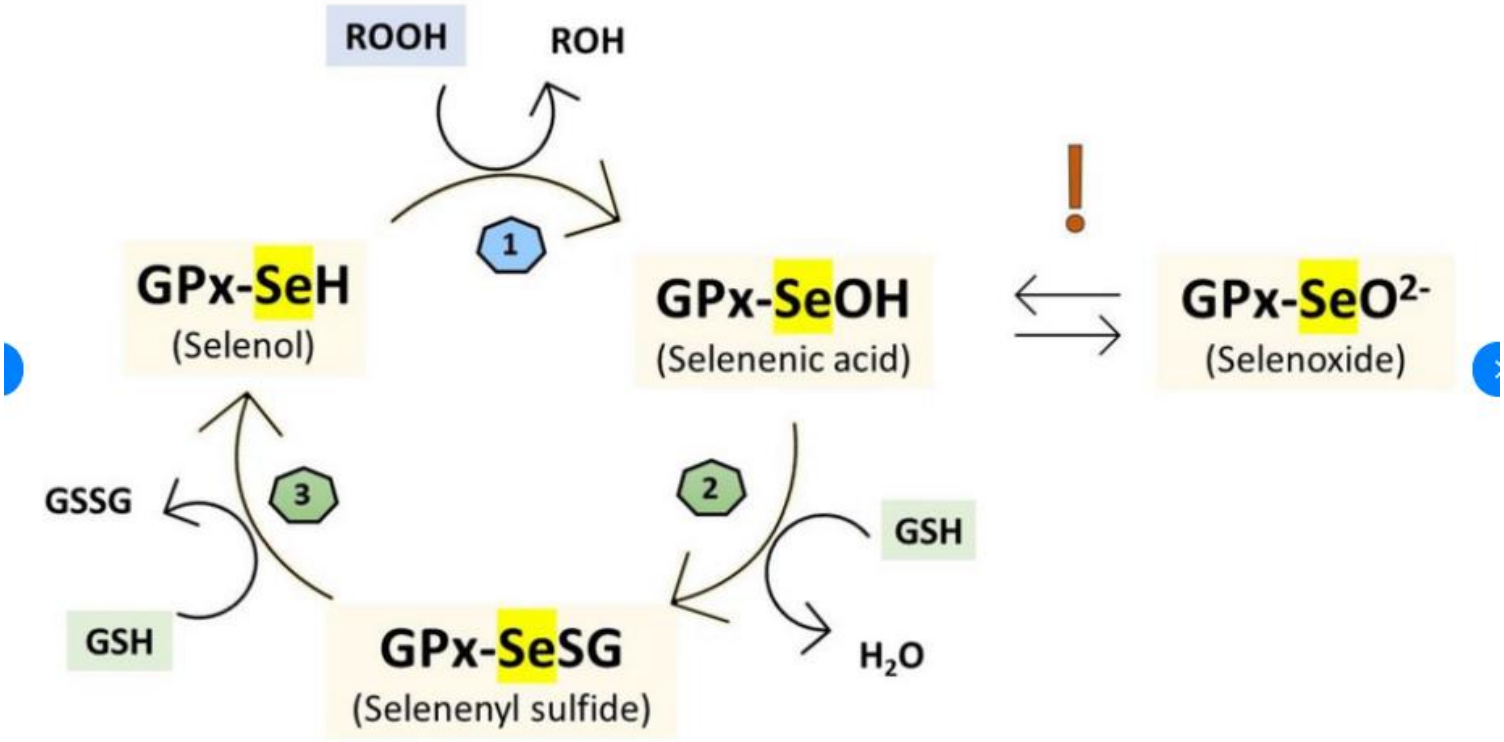


Structure of bovine F_1-ATPase in a complex with its regulatory protein IF_1

Reactive oxygen species are generated during oxidative phosphorylation



0.1~4% of the O_2 in respiration forms **superoxide radical** ($\bullet O^-_2$)



Glutathione peroxidase (GPx) redox cycle. The main reaction that glutathione peroxidase (GPx) catalyzes is the reduction of hydrogen peroxide and organic hydroperoxides (ROOH) to water and corresponding alcohol (ROH) [122]. The reaction is catalyzed by an active site on GPx which is a selenocysteine residue—selenol. During the reaction, GPx undergoes a redox cycle, where (1) selenol is oxidized by hydroxide peroxide to selenenic acid (GPx-SeOH) [130]. Selenenic acid is then reduced back to the selenol by a two-step process. In the first step, (2) selenenic acid reacts with GSH to form selenenyl sulfide (GPx-SeOH). In the second step, (3) the second GSH reduces selenenyl sulfide back to selenol, producing GSSH as an oxidized form of GSH [131]. Selenenic acid can also be oxidized to Se oxides, which will rapidly be reduced back to continue the active catalytic pathway. This is due to very weak π bonds in Se oxides [129]

O_2

superoxide radical ($\bullet O^-_2$)

hydrogen peroxide (H_2O_2)

hydroxyl radical ($\bullet OH$)

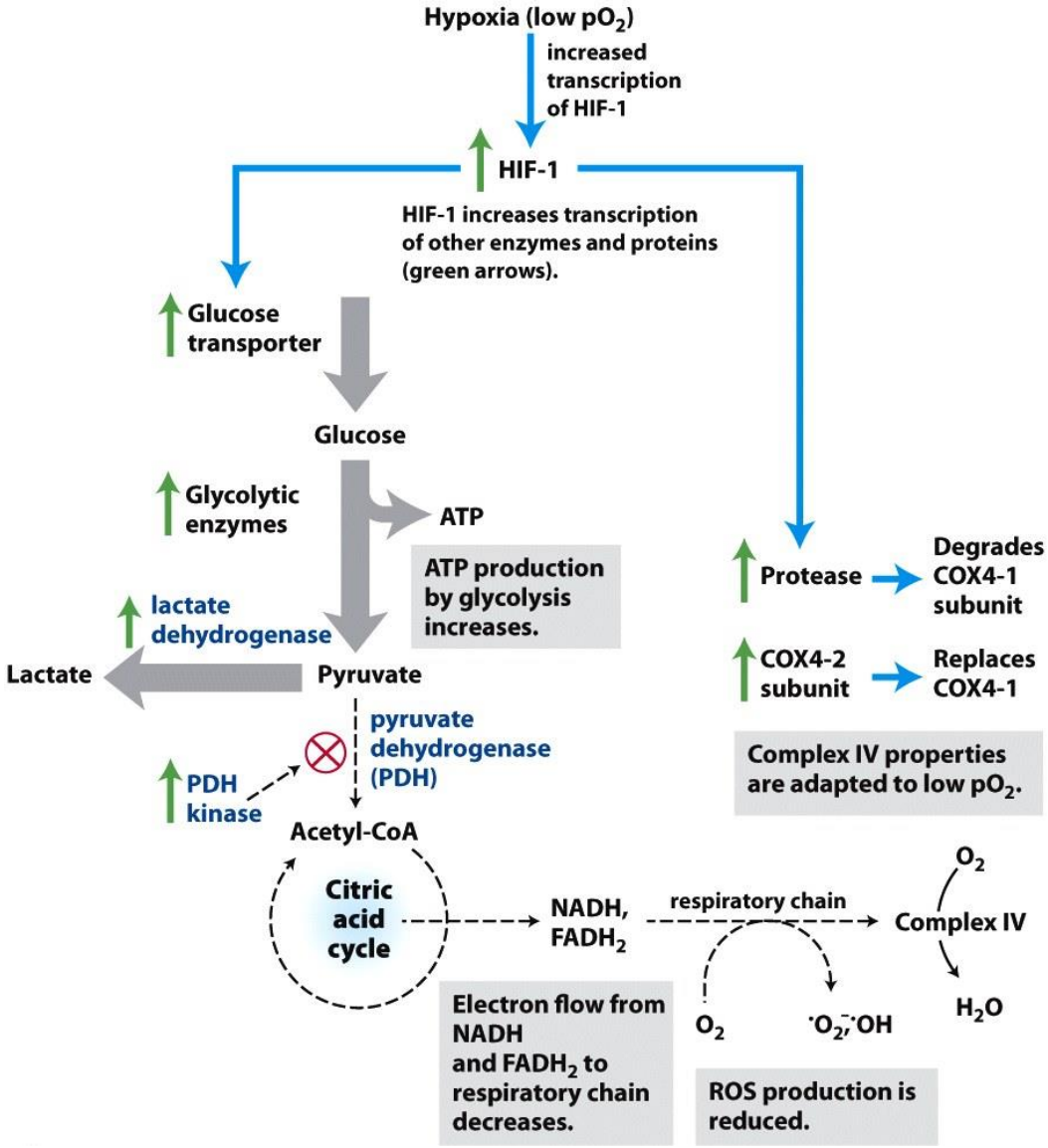
H_2O

What is reactive oxygen species (ROS) ?

Oxidative phosphorylation

Radiation
Inflammation
Metals
Drugs

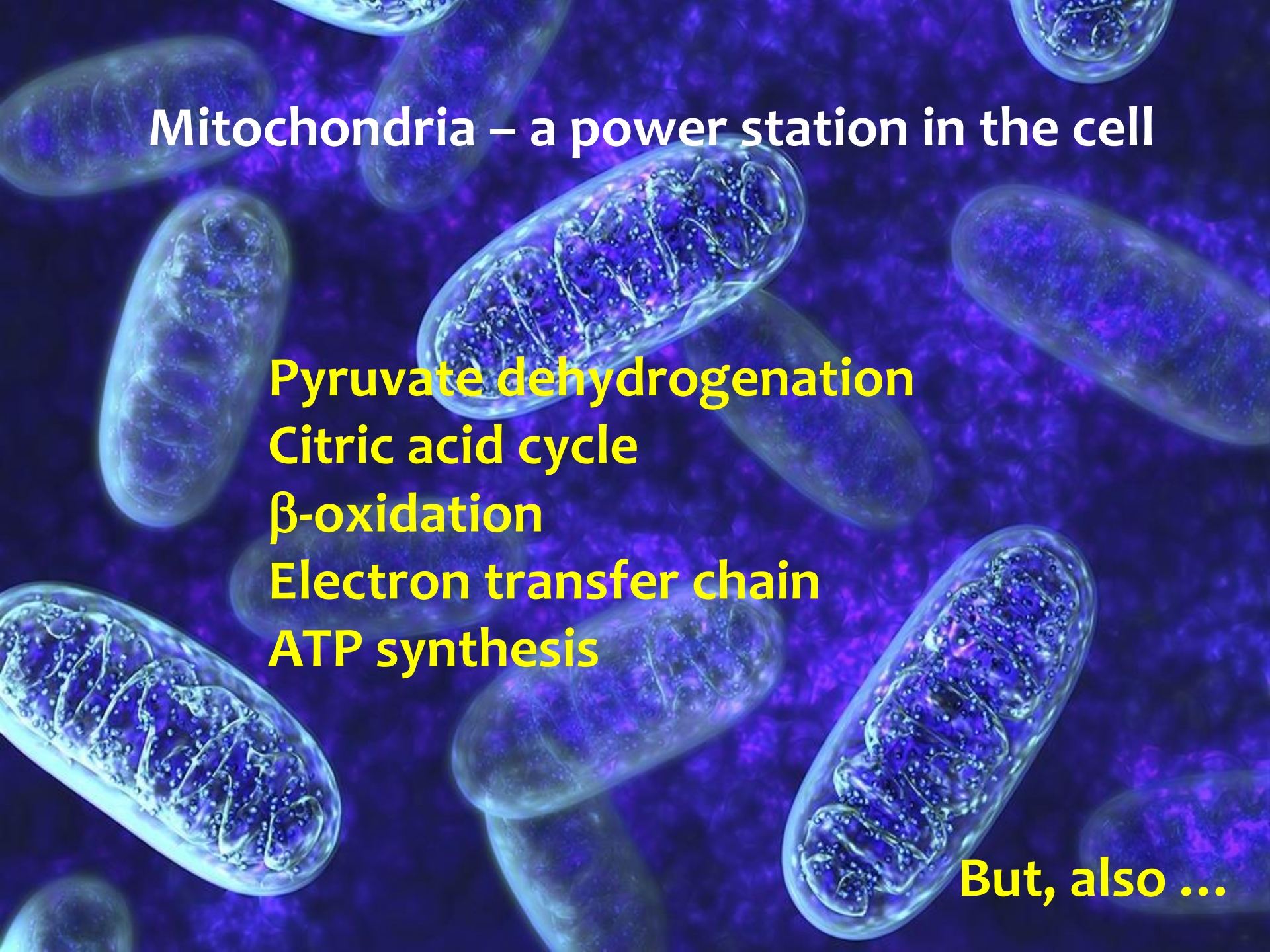
Hypoxia-inducible factor (HIF-1) regulates gene expression to reduce ROS formation



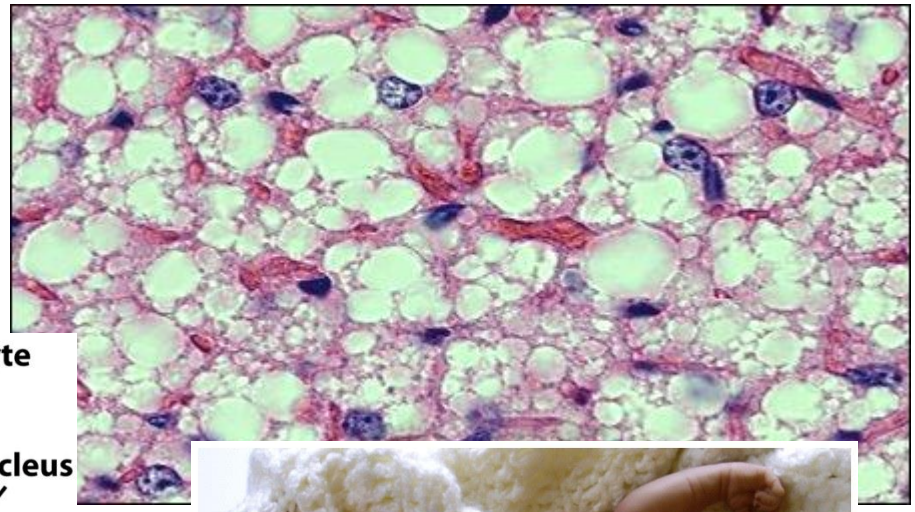
In hypoxic cells
→Imbalance between input e- & e- transfer
→ROS increase
→ mitochondrial damage

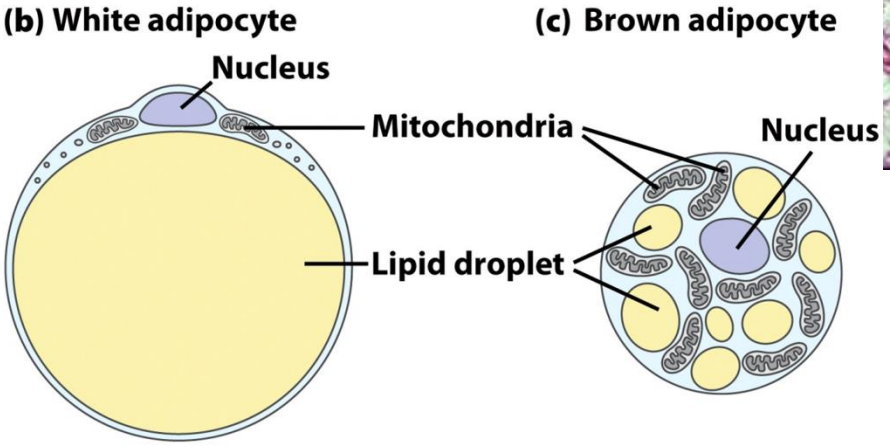
Defense mechanism :
HIF-1 expression →
1.Glycolysis upregulated
2.PDH down
3.ROS reduced
4.Complex IV effective

19.4 Mitochondira in thermogenesis, steroid synthesis, and apoptosis

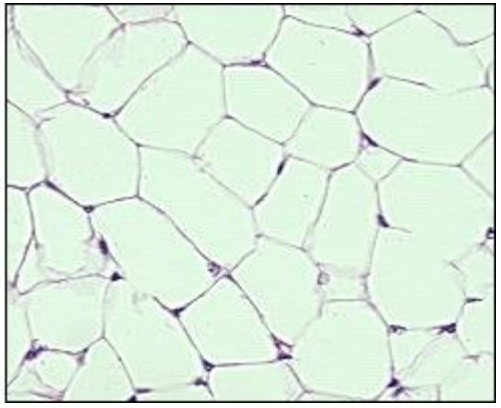

Mitochondria – a power station in the cell
Pyruvate dehydrogenation
Citric acid cycle
β-oxidation
Electron transfer chain
ATP synthesis
But, also …

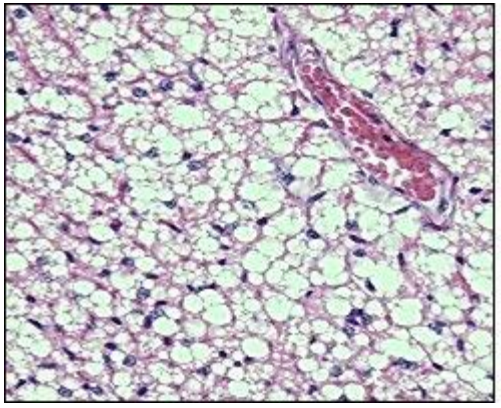
(1) Thermogenesis













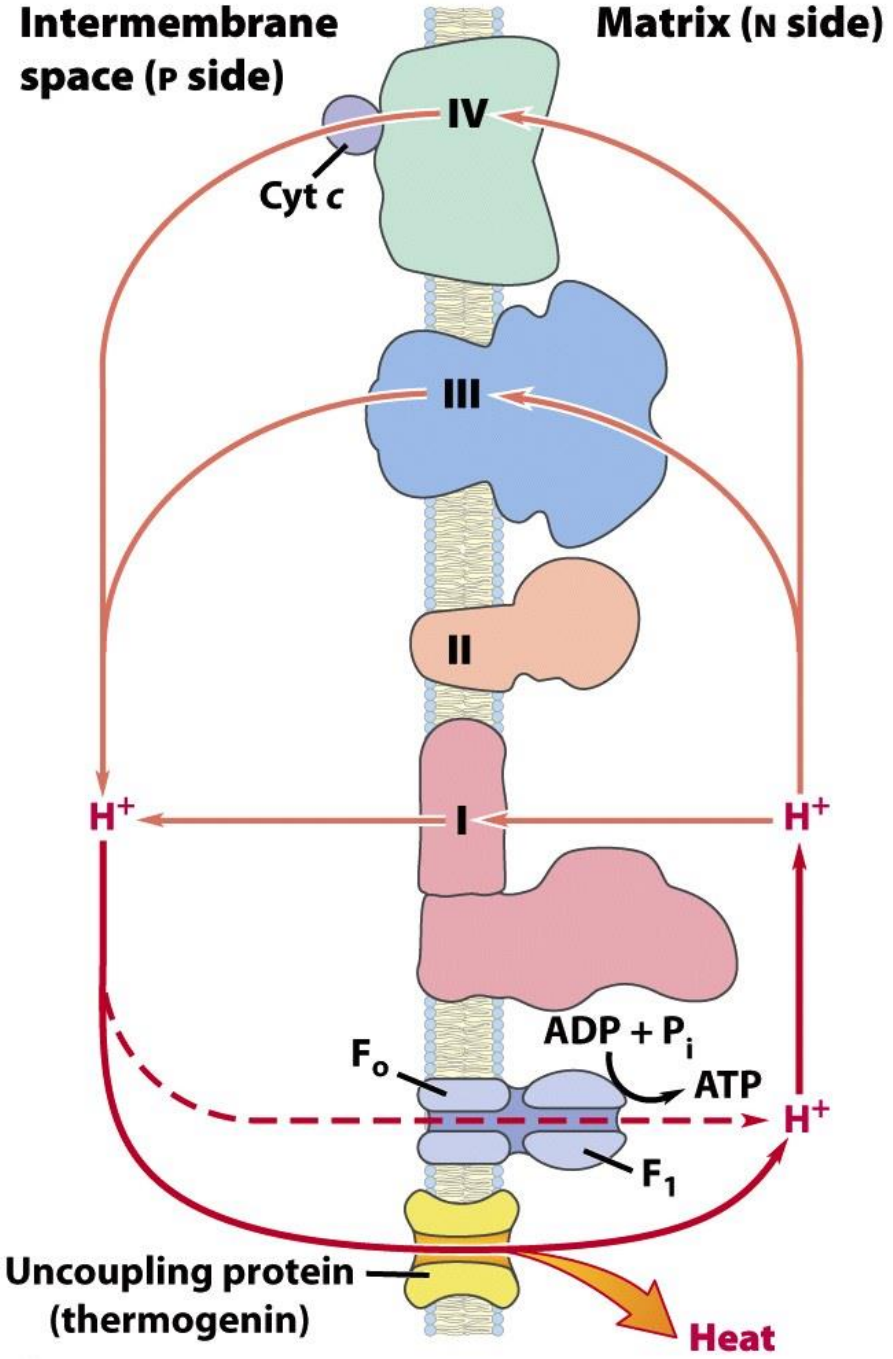
Brown Adipose Tissue

Heat generation by uncoupled mitochondria

Thermogenin provides a path for proton

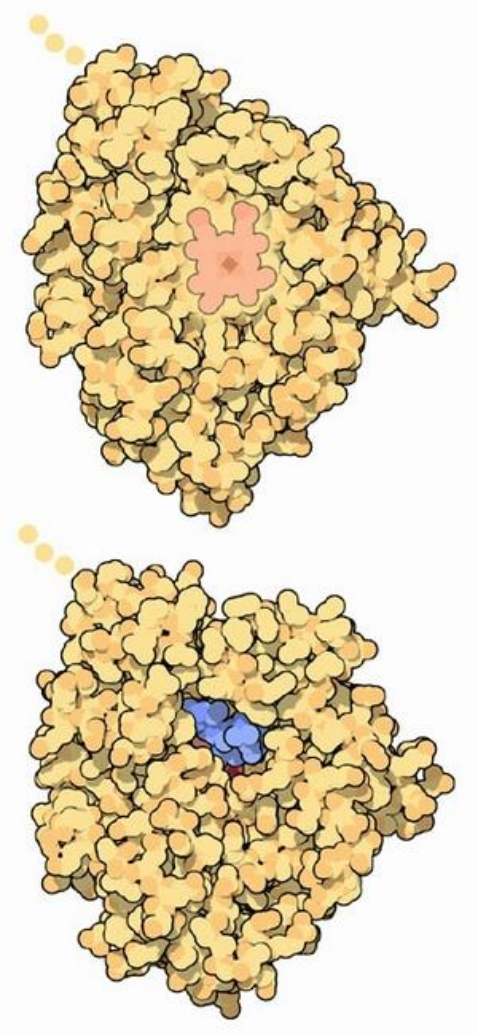
→Uncoupling of H+ pumping and ATP synthesis

→Energy dissipated as heat



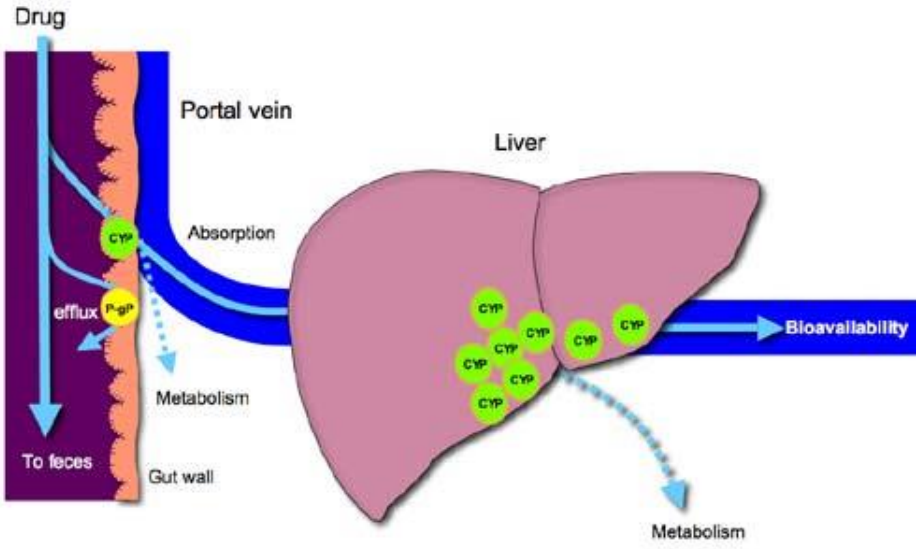
(2) Detoxification

by cytochrome p450



Detoxification : the first line of defense
Adding oxygen (oxygenation) : solubilize → excretion

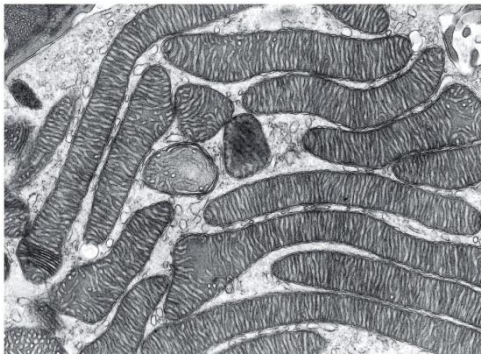
Sometimes it generate more toxic molecule (tyrenol → NAPQI (N-acetyl-p-benzoquinoneimine), by CYP2E1)



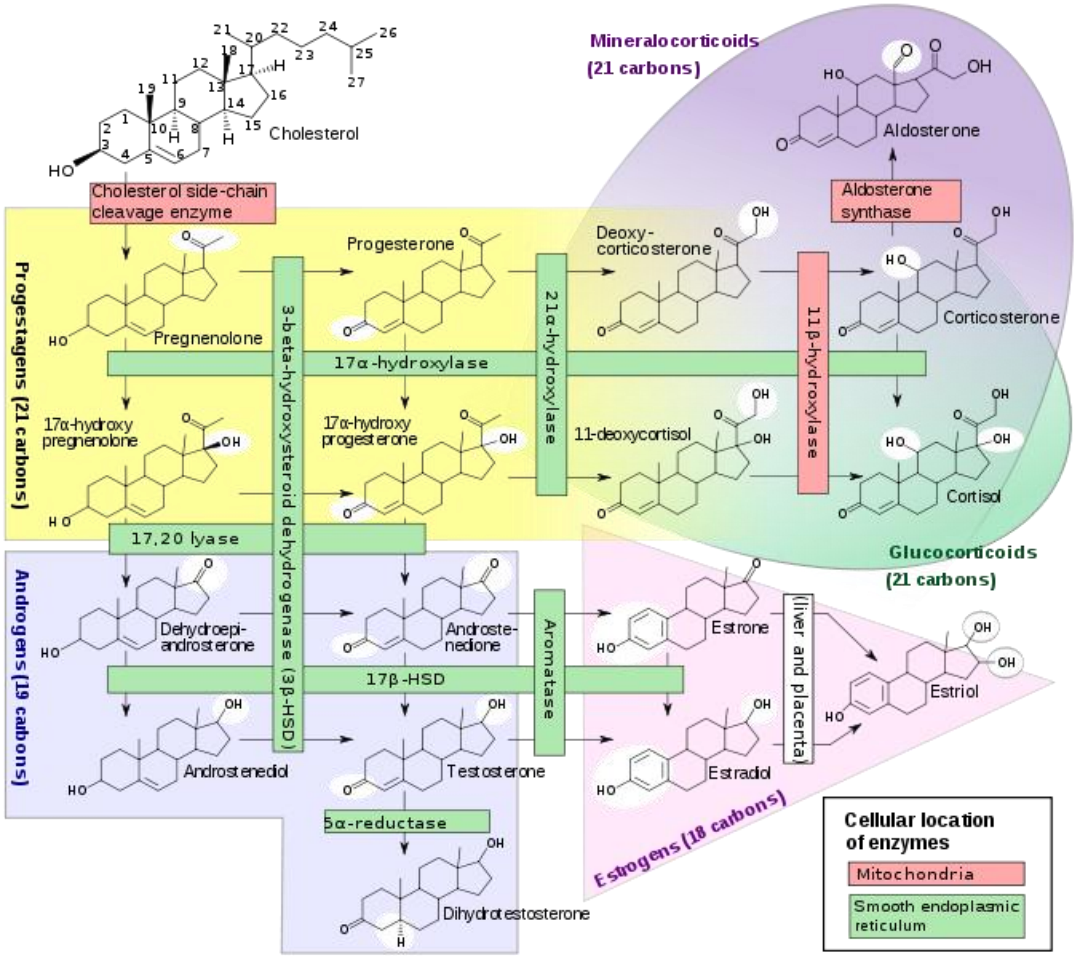
(3) Steroid synthesis

Mitochondrial P-450 oxygenases catalyze steroid hydroxylations

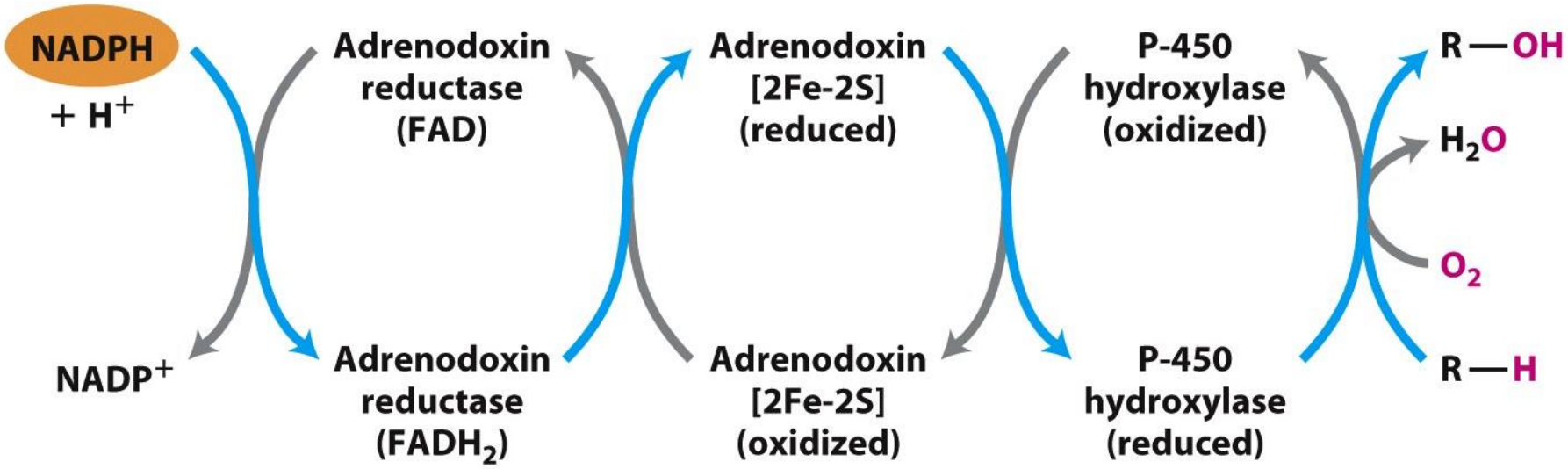




Mitochondria of adrenal gland, specialized for steroid synthesis



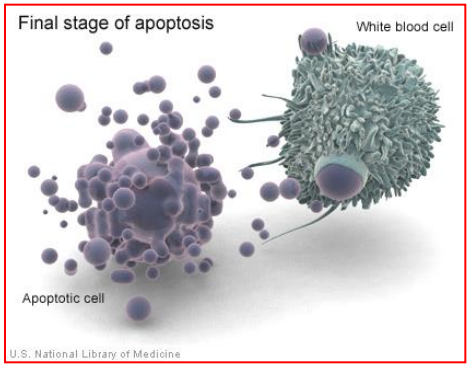
Path of electron flow in mitochondrial cytochrome P-450 reactions in adrenal gland



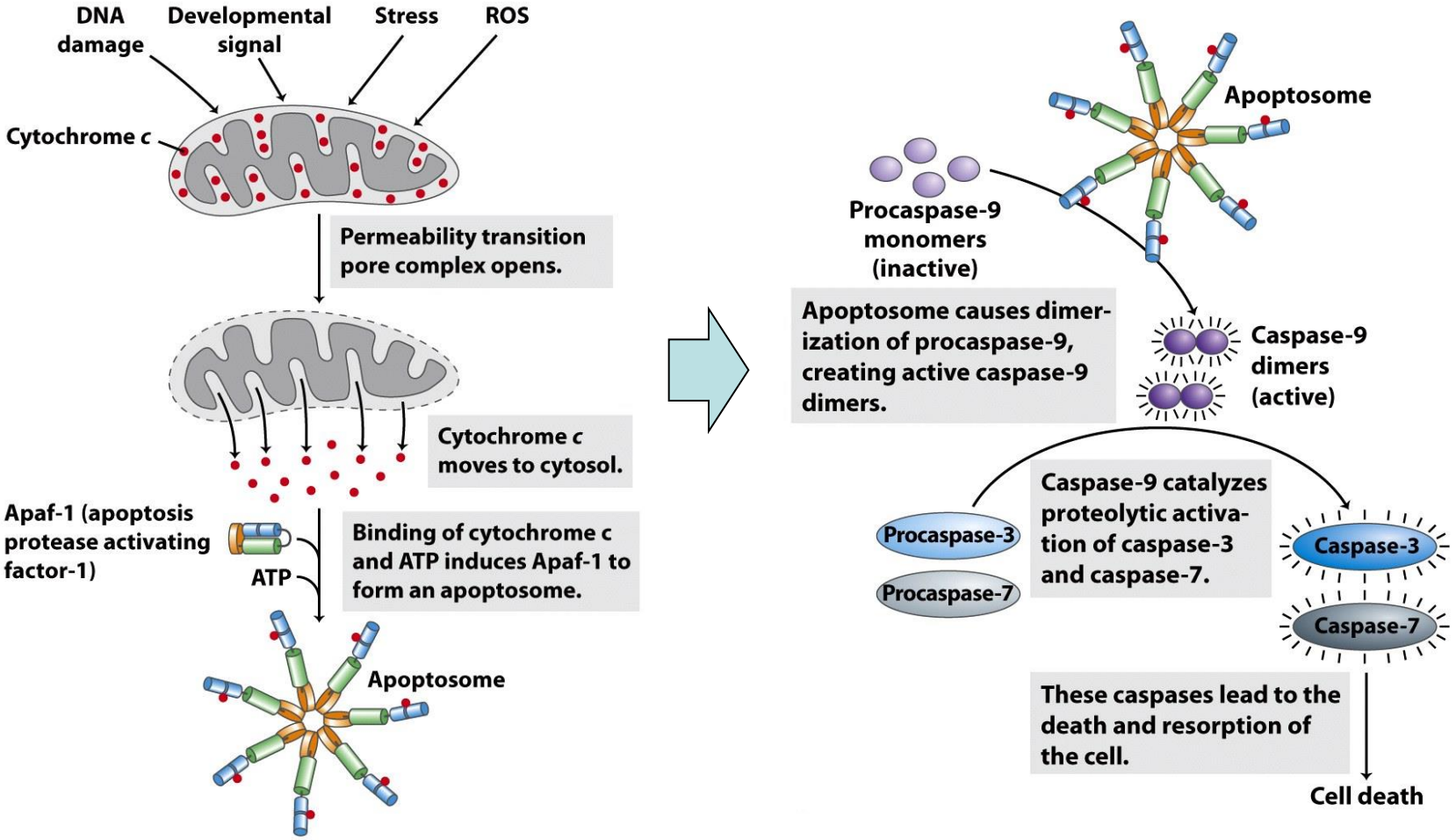
Cytochrome P-450 hydroxylase interacts directly with O_2 and the substrate (R—H) to form the products, H_2O and R—OH.

(4) Apoptosis :

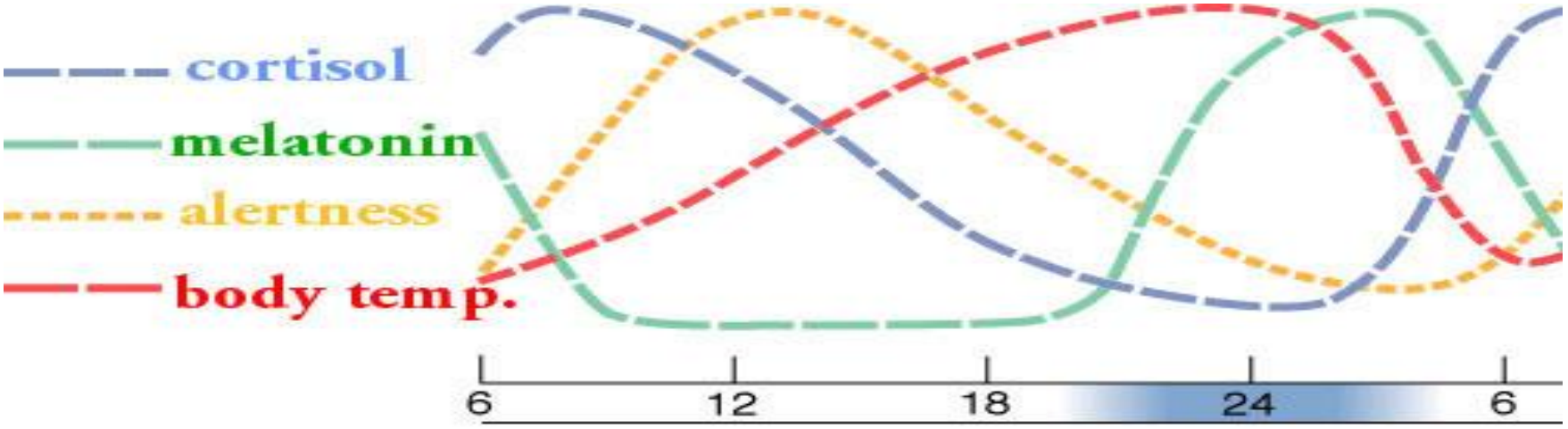
programmed cell death

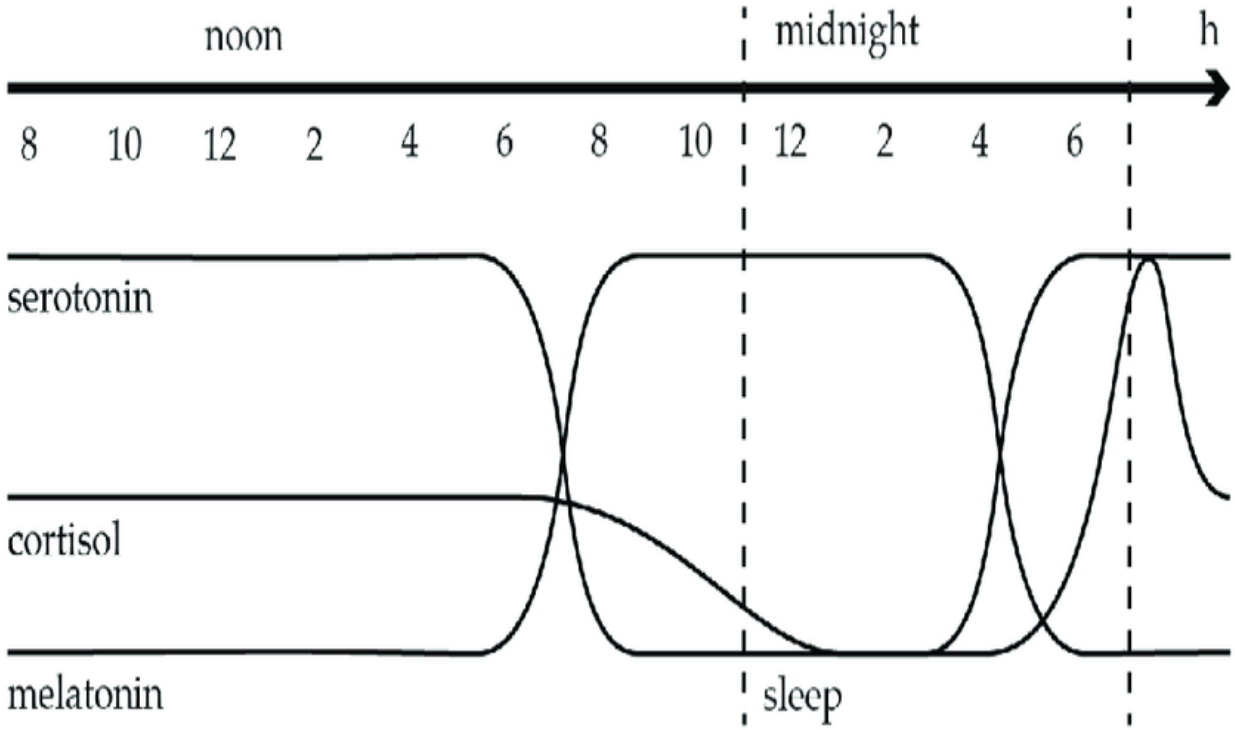


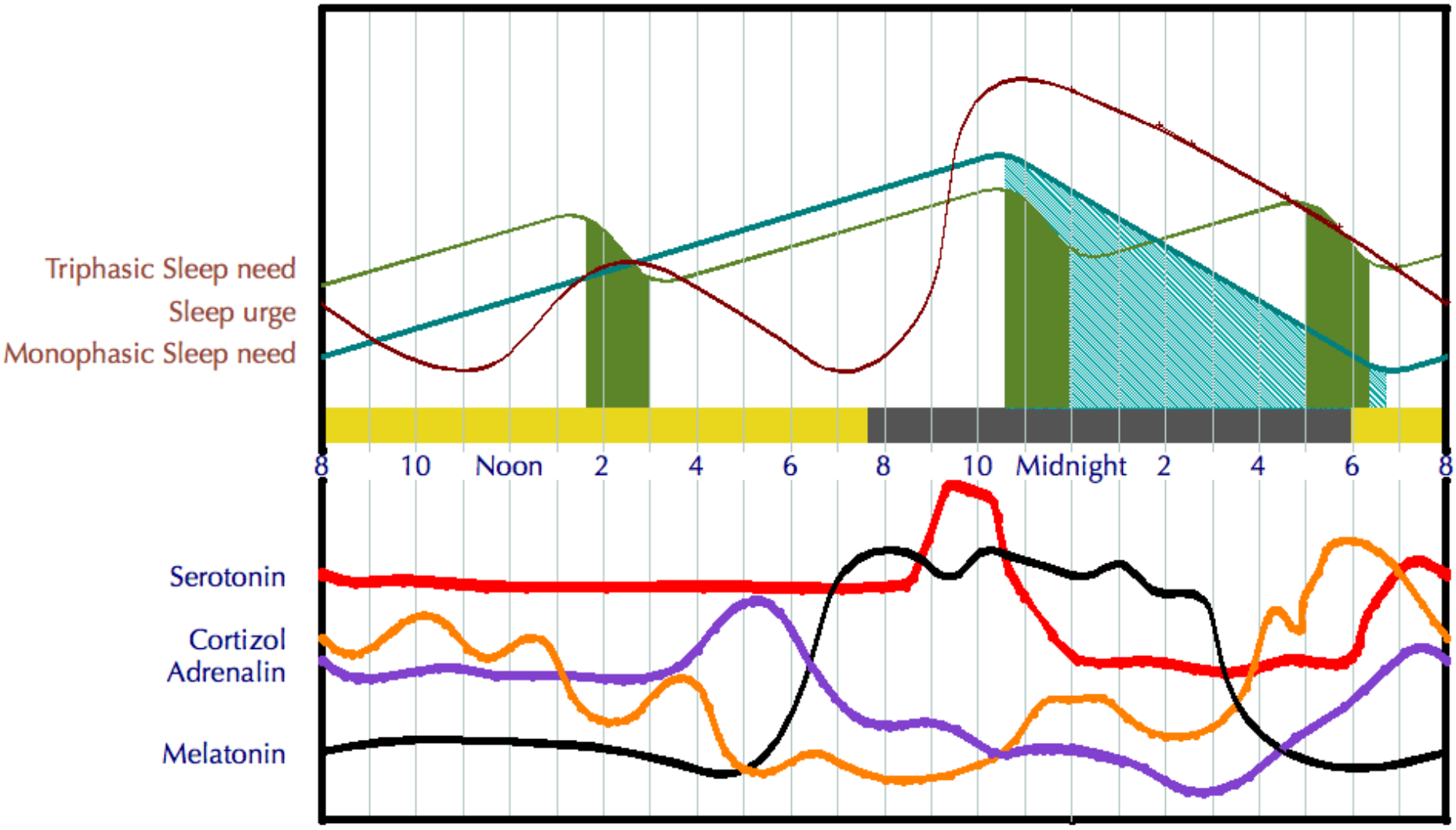
Role of cytochrome *c* in apoptosis



밤에 쓴 편지


cortisol
melatonin
alertness
body temp.
6
12
18
24
6

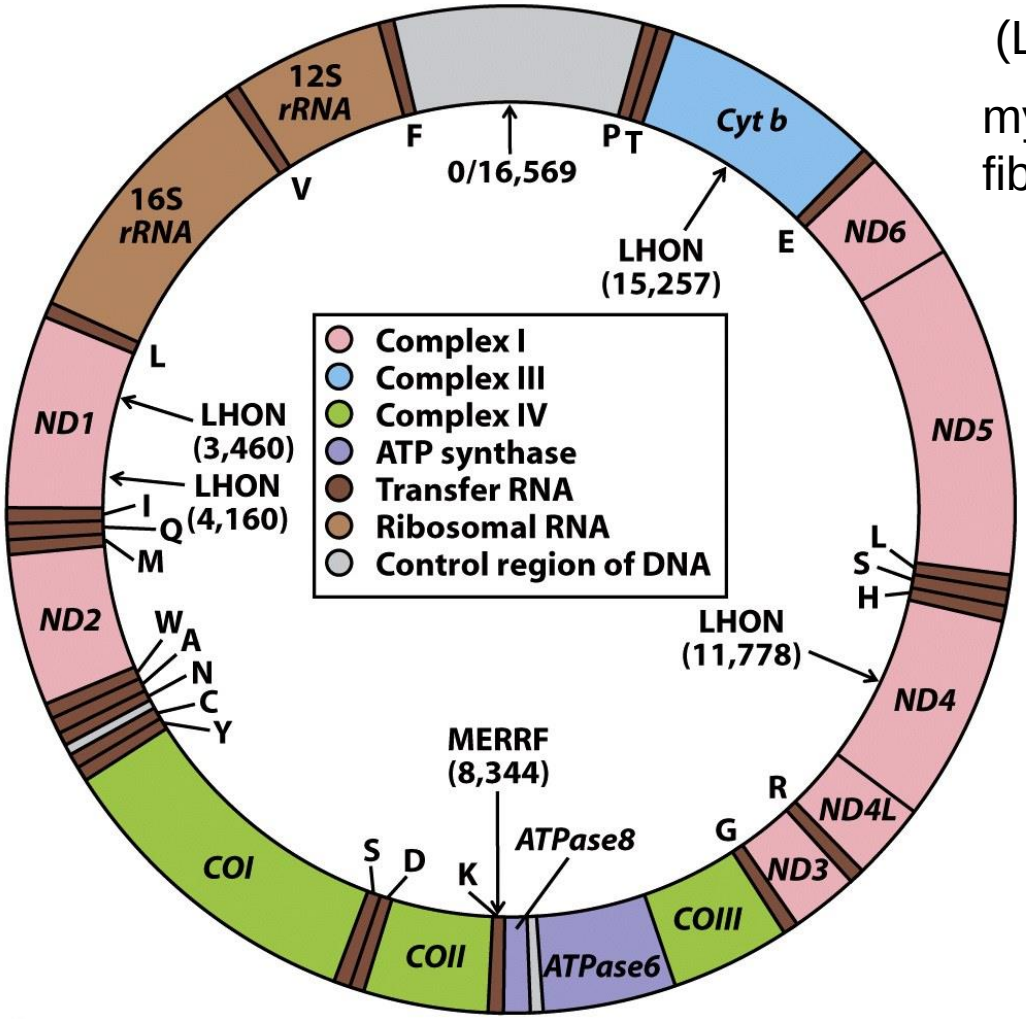

noon
midnight
h
8
10
12
2
4
6
8
10
12
2
4
6
serotonin
cortisol
melatonin
sleep


Triphasic Sleep need
Sleep urge
Monophasic Sleep need
8
10
Noon
2
4
6
8
10
Midnight
2
4
6
8
Serotonin
Cortizol
Adrenalin
Melatonin

19.5 Mitochondiral genes: their origin and the effect of mutations

Mutations in mitochondria accumulate throughout the LIFE

Mitochondrial genes and mutations



Leber's hereditary optic neuropathy (LHON)

myoclonic epilepsy and ragged-red fiber disease (MERRF)

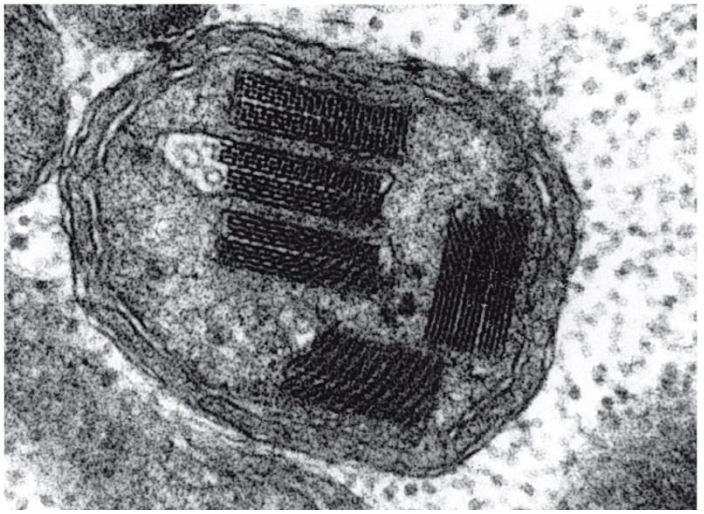
mitochondrial encephalomyopathies：affect primarily the brain and skeletal muscle. These diseases are invariably inherited from the mother.

Leber's hereditary optic neuropathy (LHON)

: single base change in ND4 (complex I) or cytochrom b
→affects the central nervous system
→ATP supply is not enough
→Leading to blindness

myoclonic epilepsy and ragged-red fiber disease (MERRF)

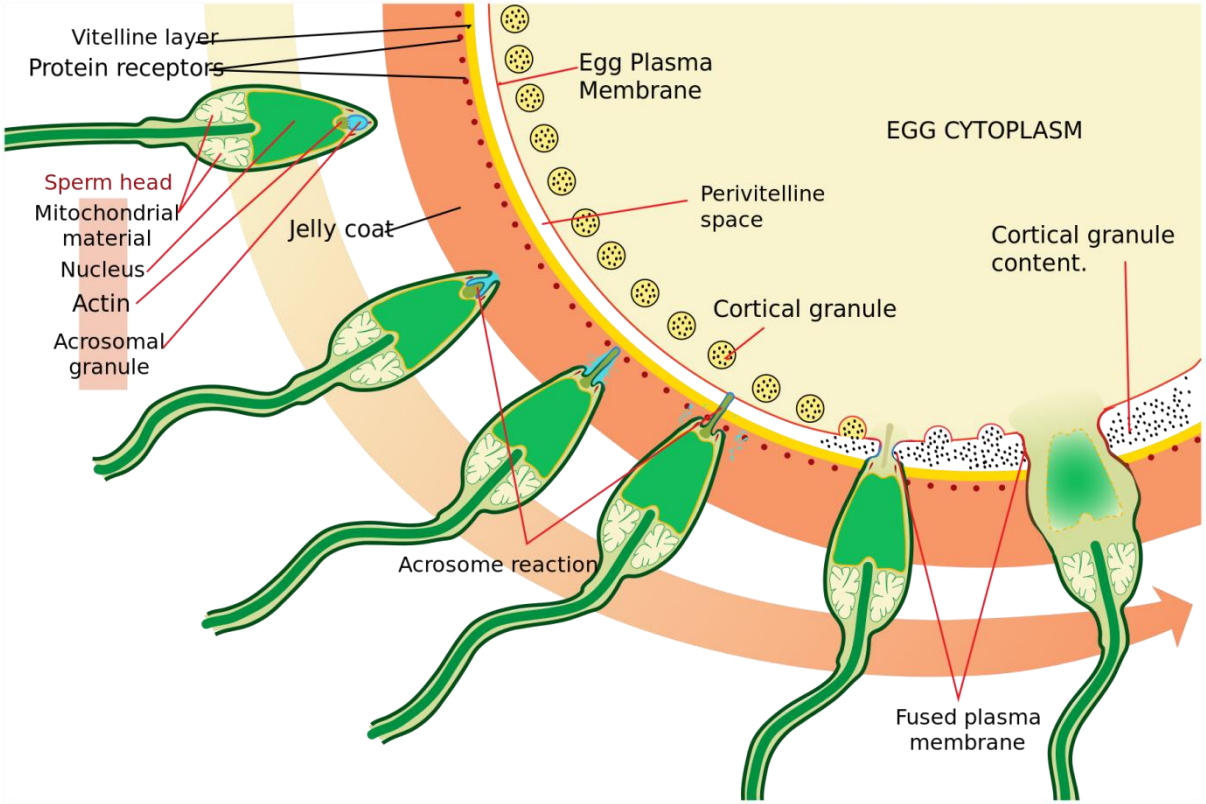
: mutation in tRNA ^{Lys}
→ Abnormal shape of mitochondria
→ Muscular weakness & uncontrollable muscular jerking



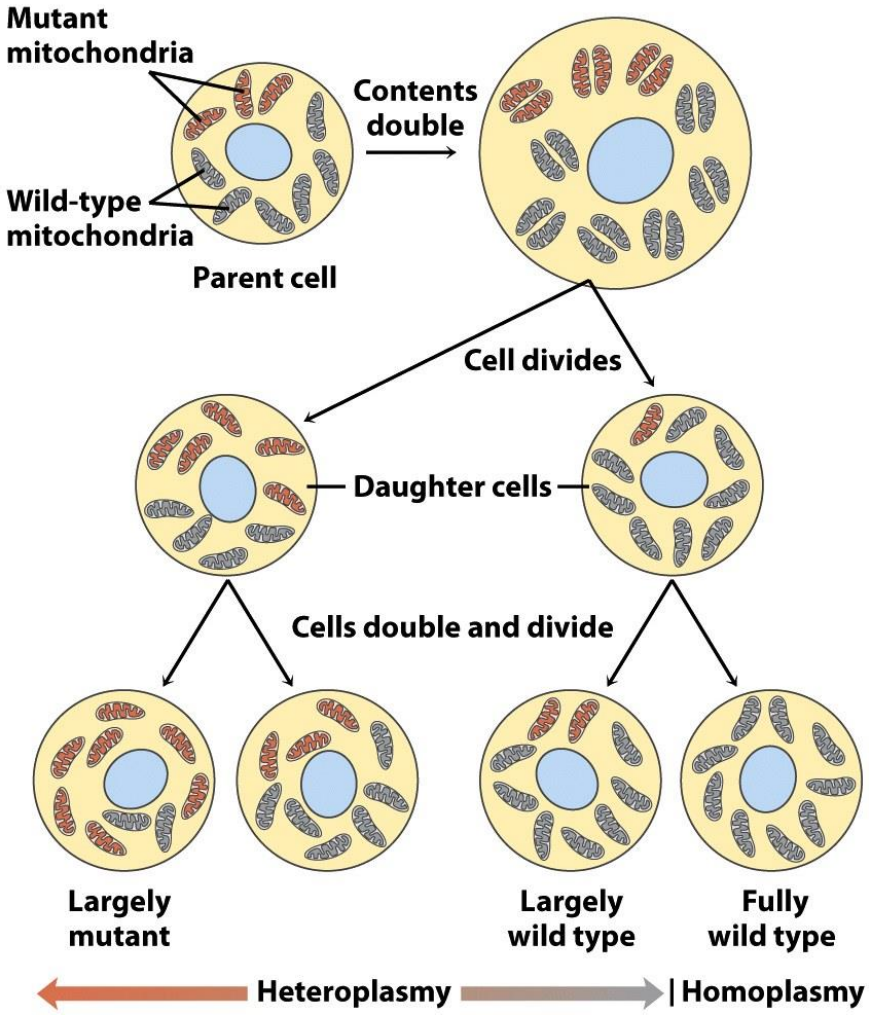
an abnormal mitochondrion from the muscle of an individual with MERRF

Mitochondria are normally inherited exclusively from the mother;

The mitochondria in mammalian sperm are usually destroyed by the egg cell after fertilization.



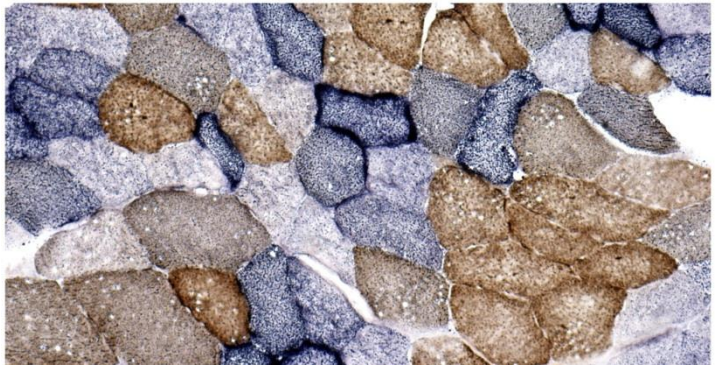
Heteroplasmy in mitochondrial genomes



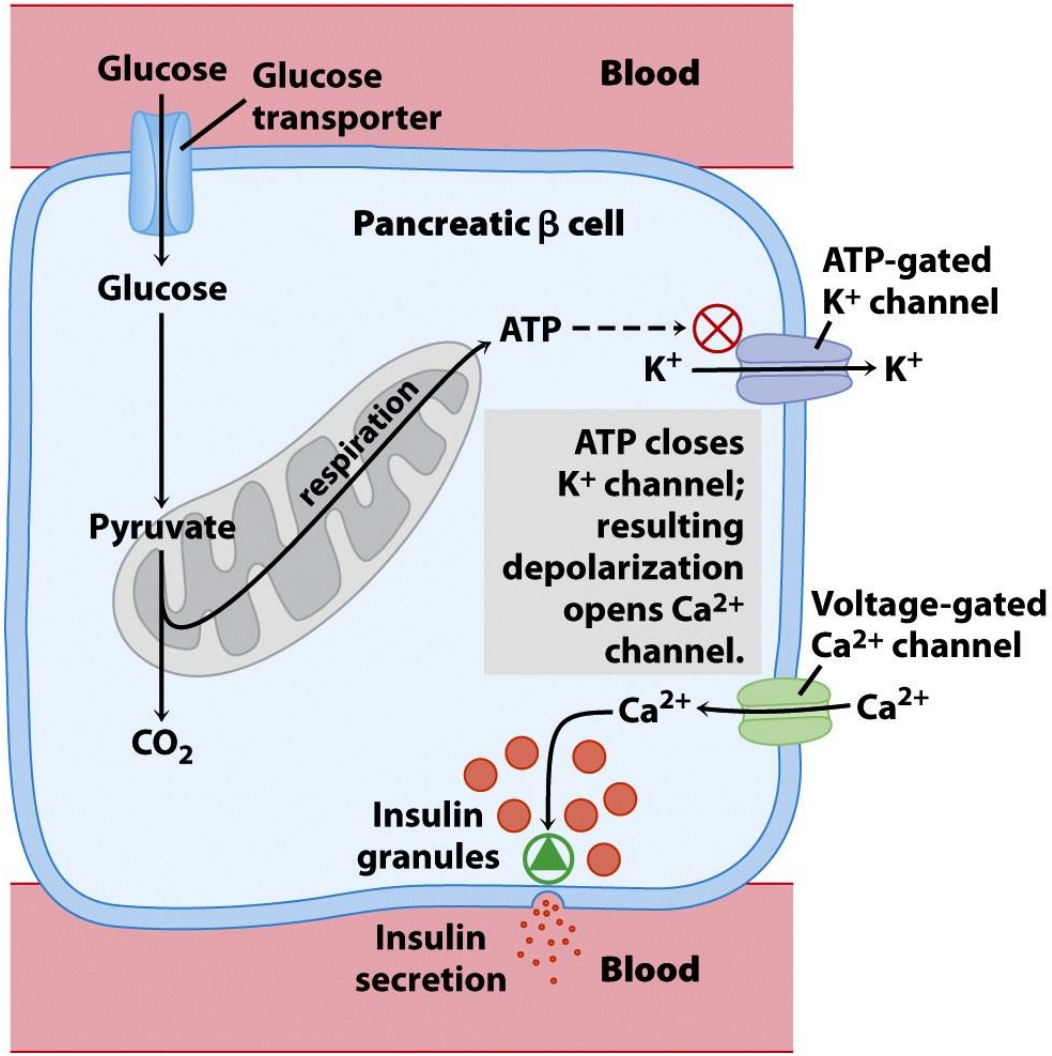
Mitochondira: high ROS production, less effective at repairing DNA damage

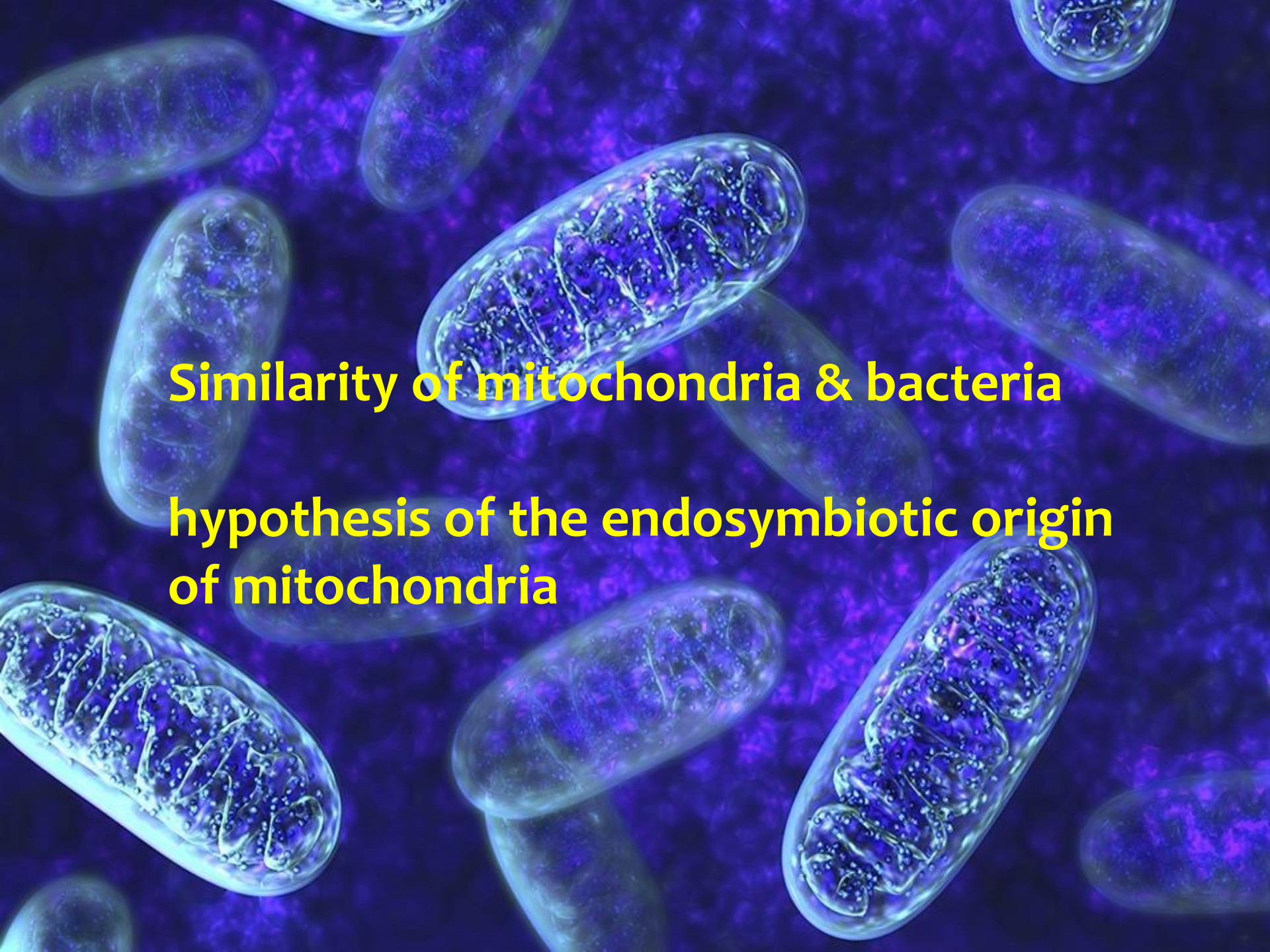
Aging theory:
accumulation of damage of mito → weakening of skeletal and heart muscle

Heteroplasmy :
Mitochondria of daughter cell are maternal → mutant phenotypes of differing severity;



Defective oxidative phosphorylation in pancreatic β cells blocks insulin secretion




Similarity of mitochondria & bacteria
hypothesis of the endosymbiotic origin of mitochondria

Summary

1. Uncoupled mitochondria in Brown adipose tissue produce heat
2. Mitochondrial P-450 oxygenases catalyze steroid hydroxylations
3. Mitochondria are central to the initiation of apoptosis
4. Mutations in mitochondrial DNA accumulate throughout the life of the organism
5. Some mutations in mitochondrial genomes cause disease
6. Diabetes can result from defects in the mitochondria of pancreatic β cells

Today in this class ...

Photosynthesis



Carbohydrate synthesis

Photophosphorylation

PHOTOSYNTHESIS: HARVESTING LIGHT ENERGY

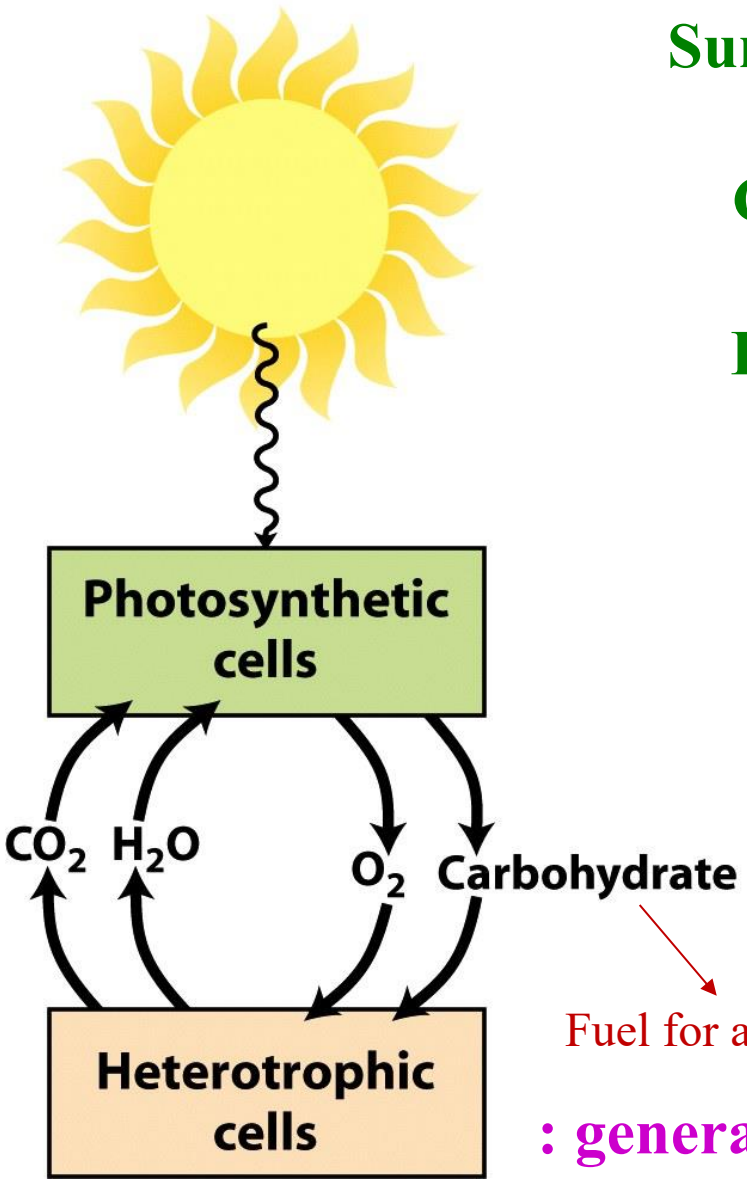
Life phenomena

How glucose is synthesized from sun light ?

How oxygen is synthesized from water ?

The answer is … “photosynthesis”

Photosynthesis



Sun light
+
CO_2
+
H_2O

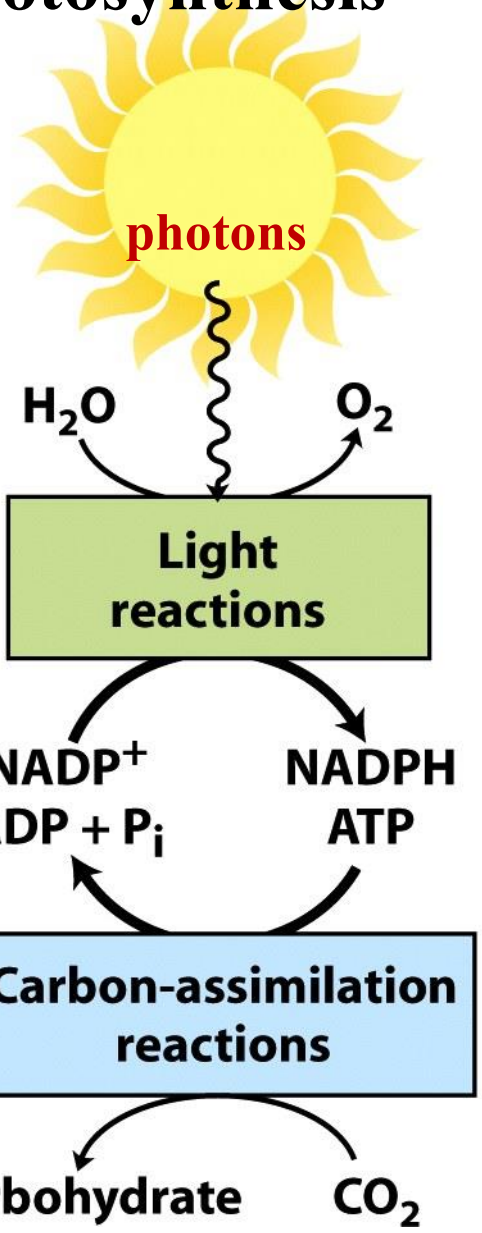
O_2
+
Carbohydrate

Key for life

Fuel for animal

: generate ATP and return CO_2

Photosynthesis



$nCO_2 + nH_2O + Photon \rightarrow (CH_2O)_n + nO_2$

Light dependent reactions

: Energy sources

Light independent reactions
(Dark reactions, carbon-fixation reactions)

: Calvin cycle → glyceraldehyde-3-P (G-3-P, PGAL)

Light energy

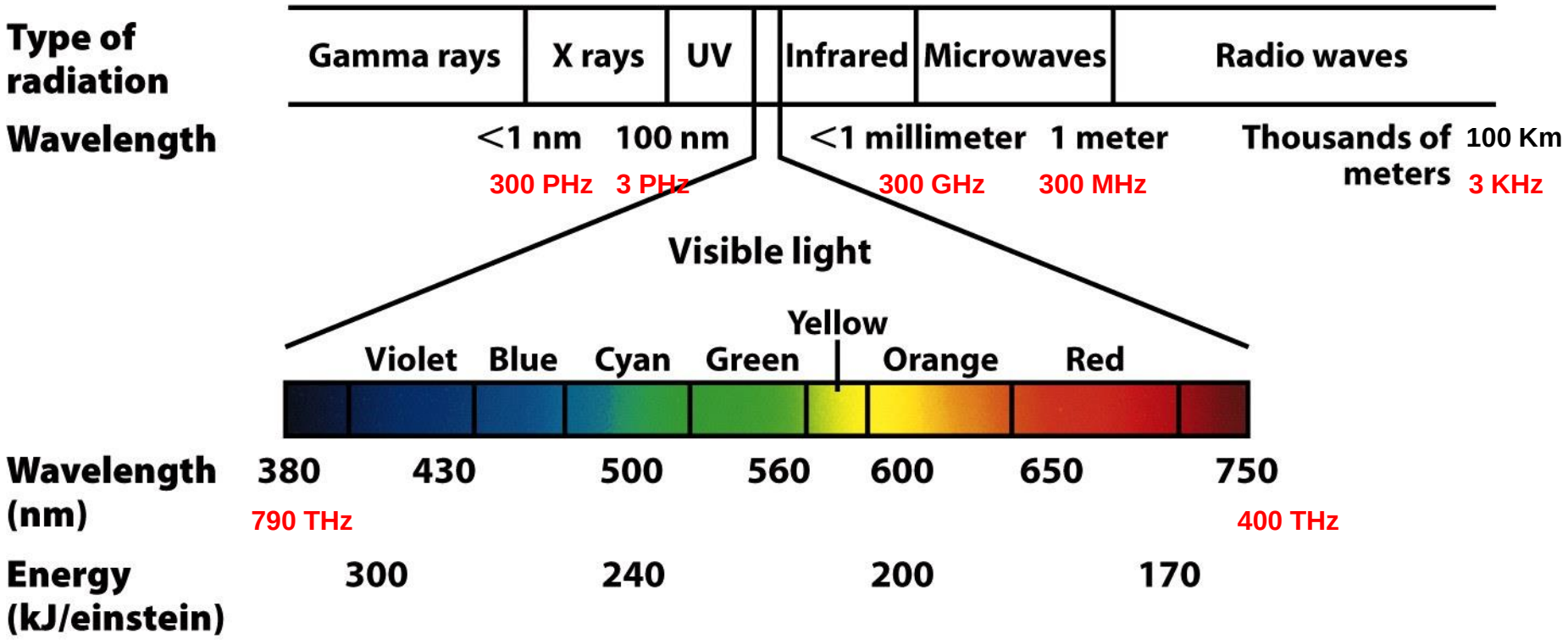
$E = h\nu = hc/\lambda$

h = 6.626 × 10 $^{-34}$ J·s , Planck's constant

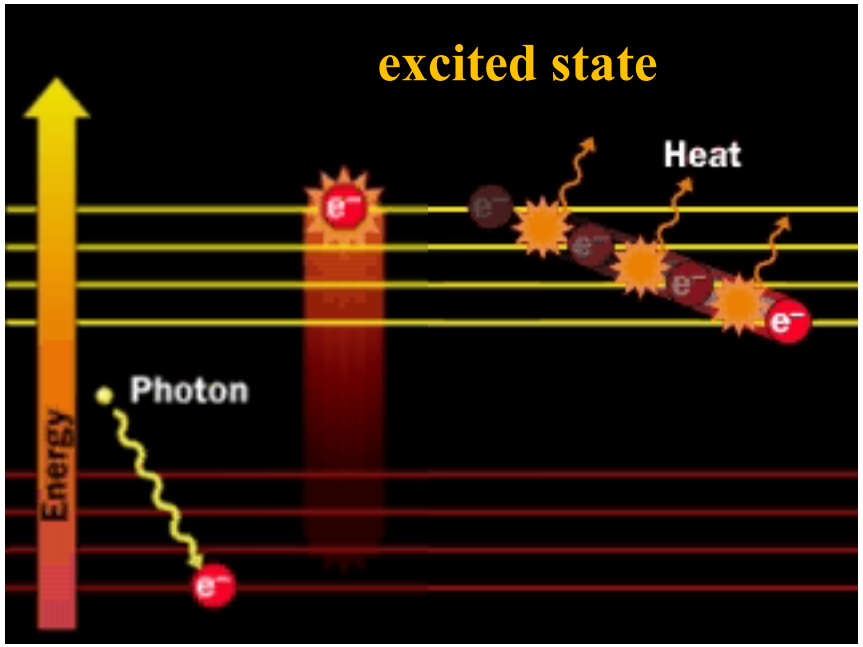
ν= cycles/s, frequency

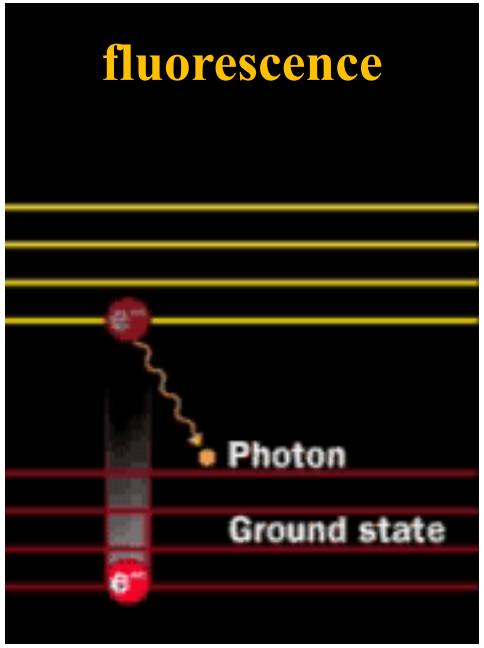
c = 3.00 × 10 ^{8} m/s , speed of light

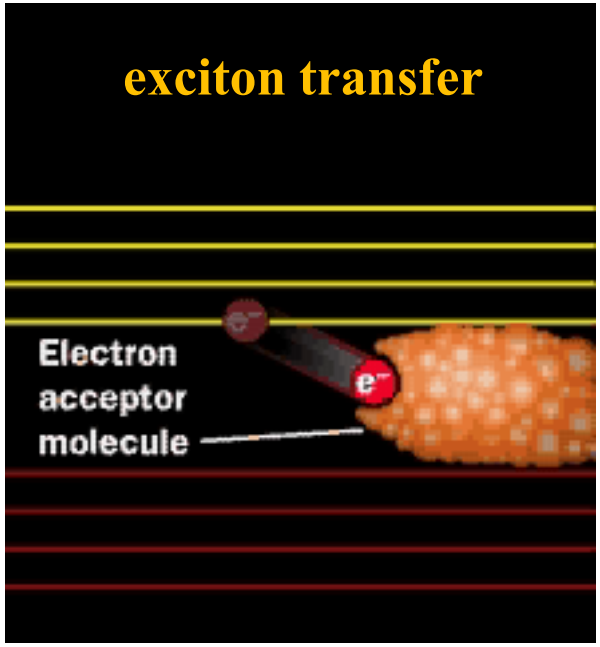
E of photon ~ 150 kJ/einstein for red light (1 einstein = 6.022 ×10 ^{23} photons)



Excitation and Energy transfer

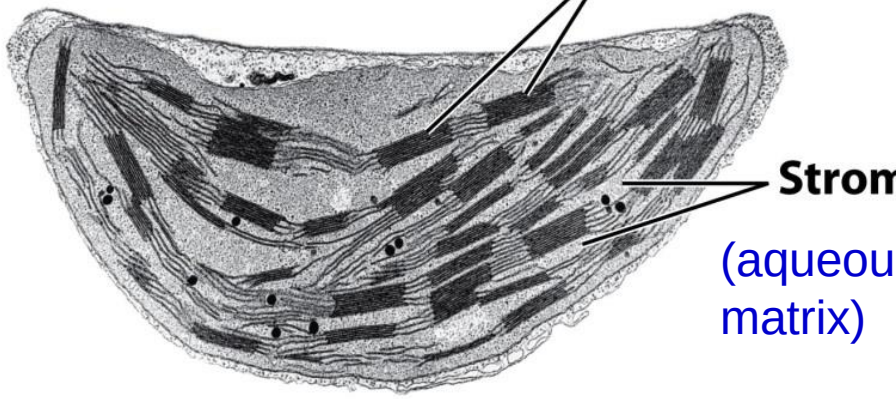




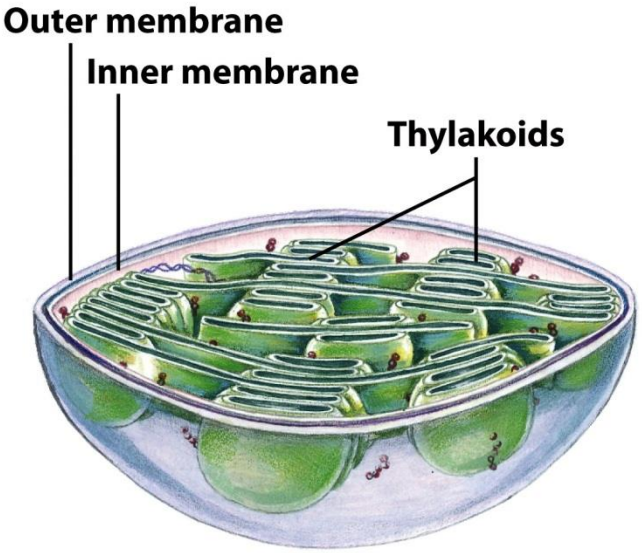


Quantum (quantity of energy) matches the energy transition

Photosynthesis in plants takes place in chloroplast

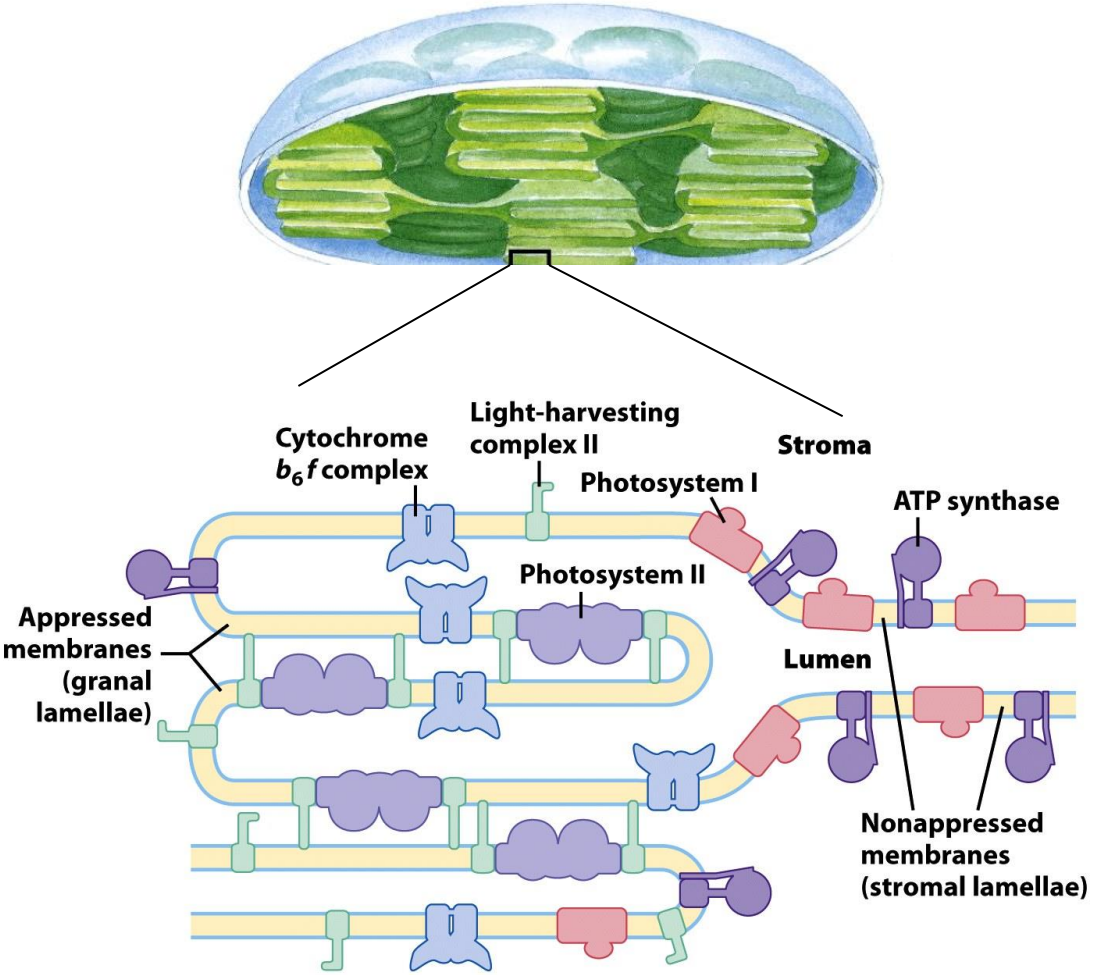


(aqueous fluid, matrix)



Chloroplast

Light-harvesting complex and photosystems in chloroplasts



Thylakoid

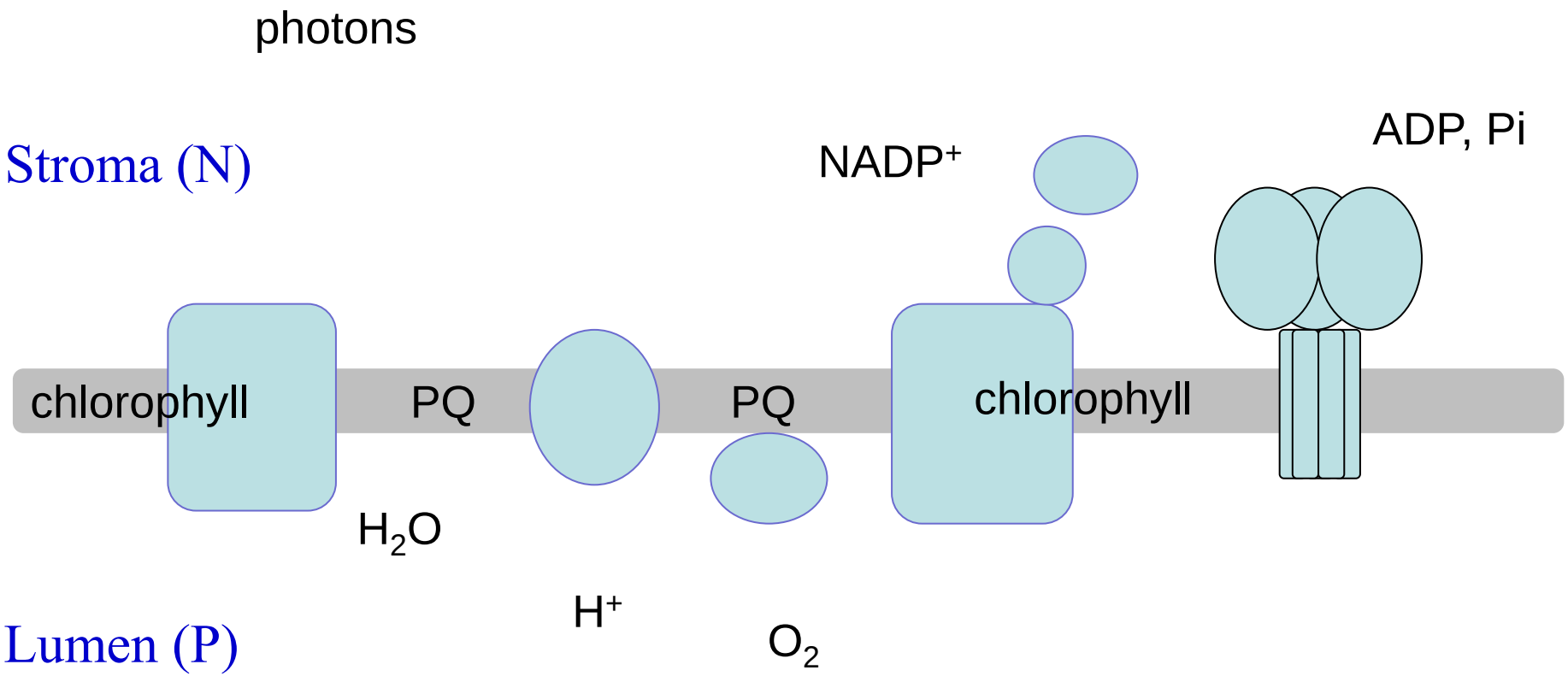
Grana

Stroma

(aqueous fluid)

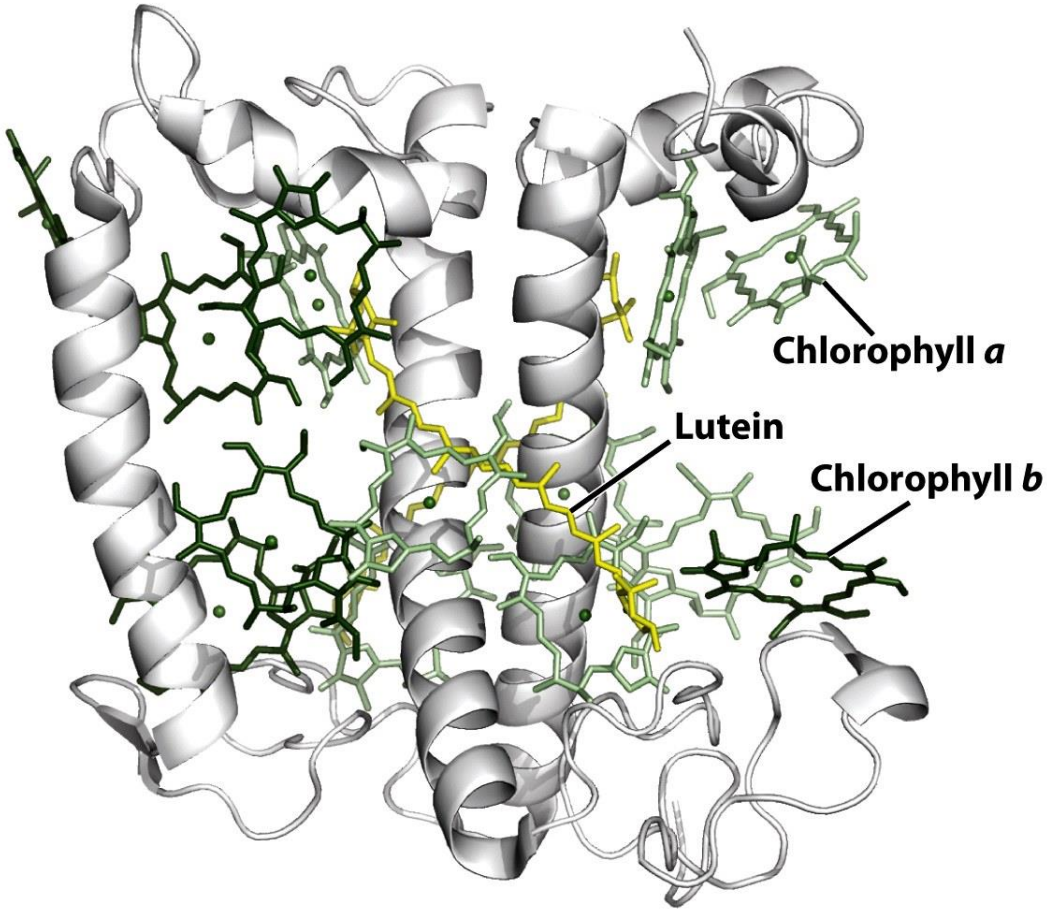
Lumen

(inner space)



https://youtu.be/hj_WKgnL6MI

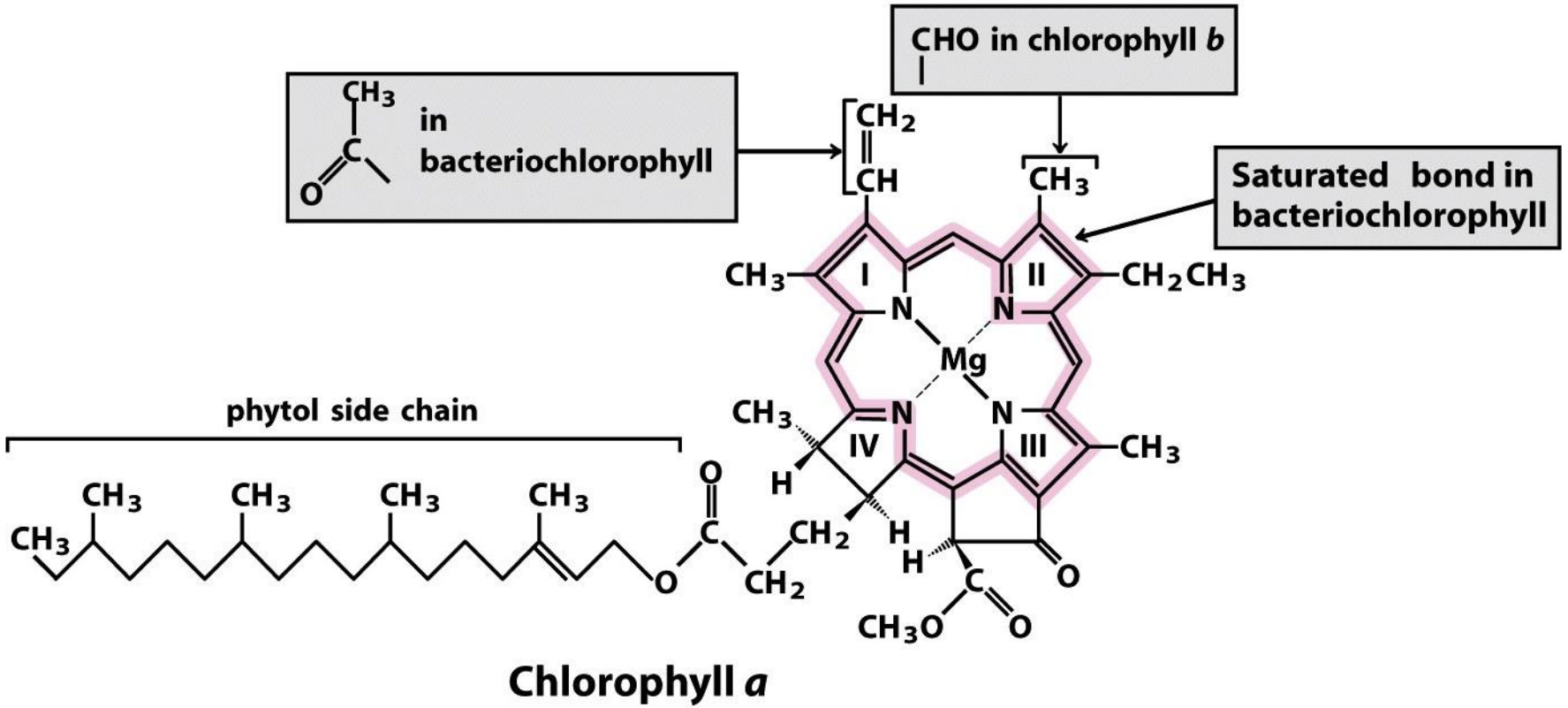
A light-harvesting complex, LHCII (monomer)



7 chlorophyll a, 5 chlorophyll b, and 2 luteins in 1 monomer

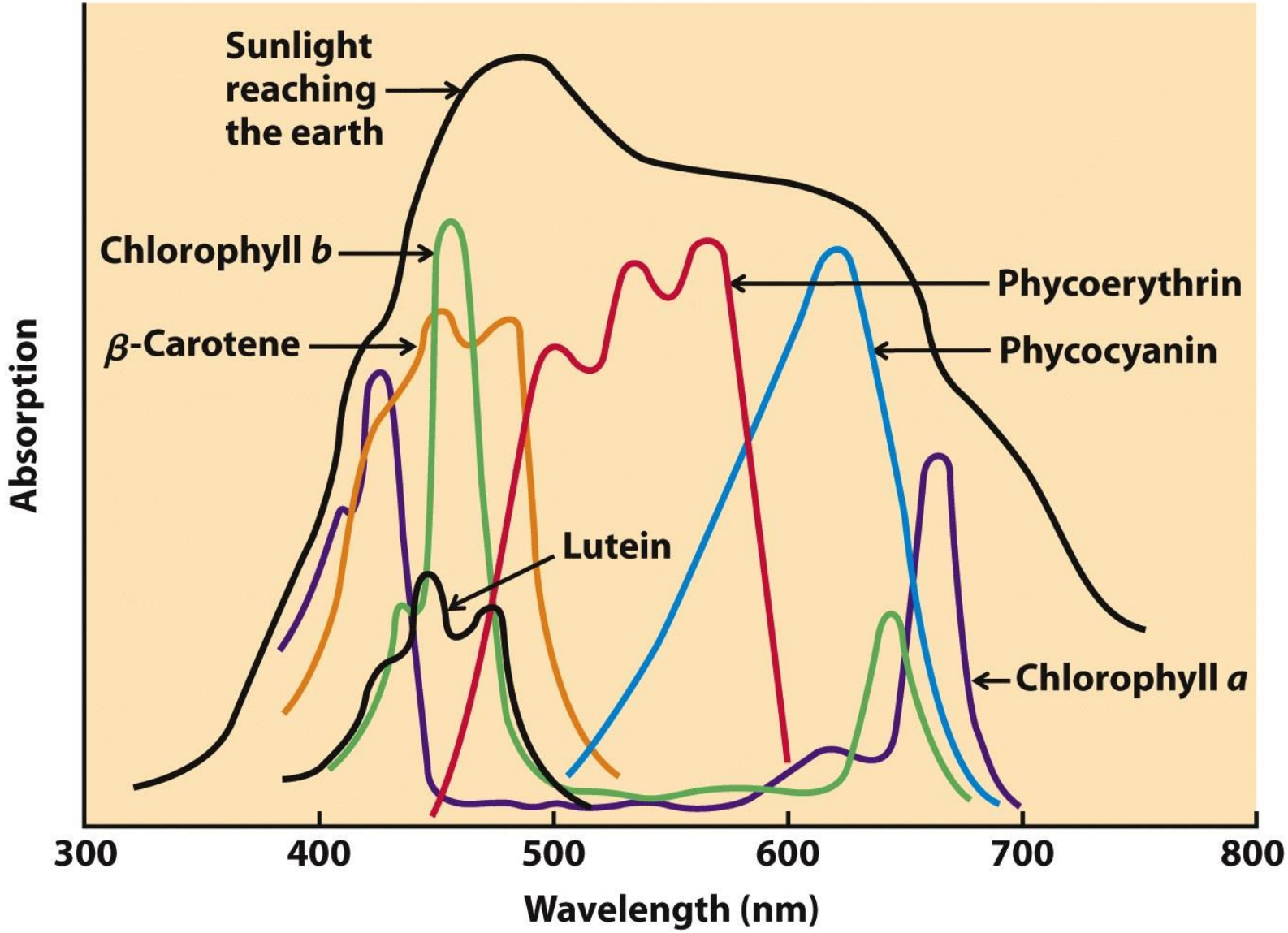
LHCII exist as trimer

Chlorophylls absorb light energy for photosynthesis

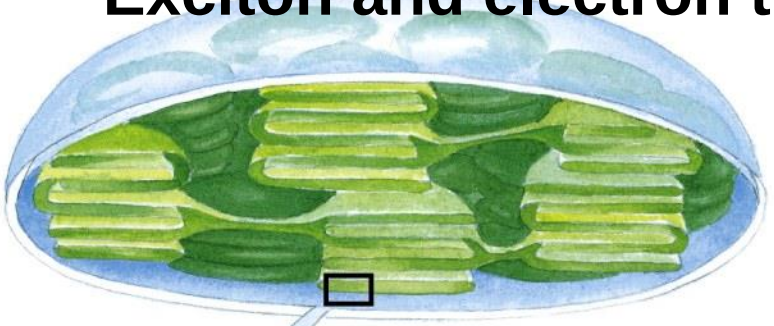


conjugated systems (alternating single and double bonds) account for the absorption of visible light.

Absorption of visible light by photopigments (chlorophylls and accessory pigments)



Exciton and electron transfer

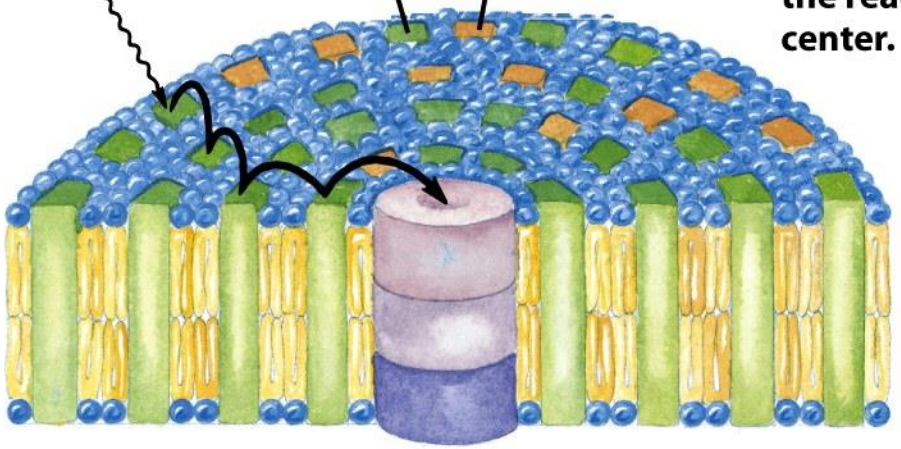


Antenna chlorophylls, bound to protein

Carotenoids, other accessory pigments

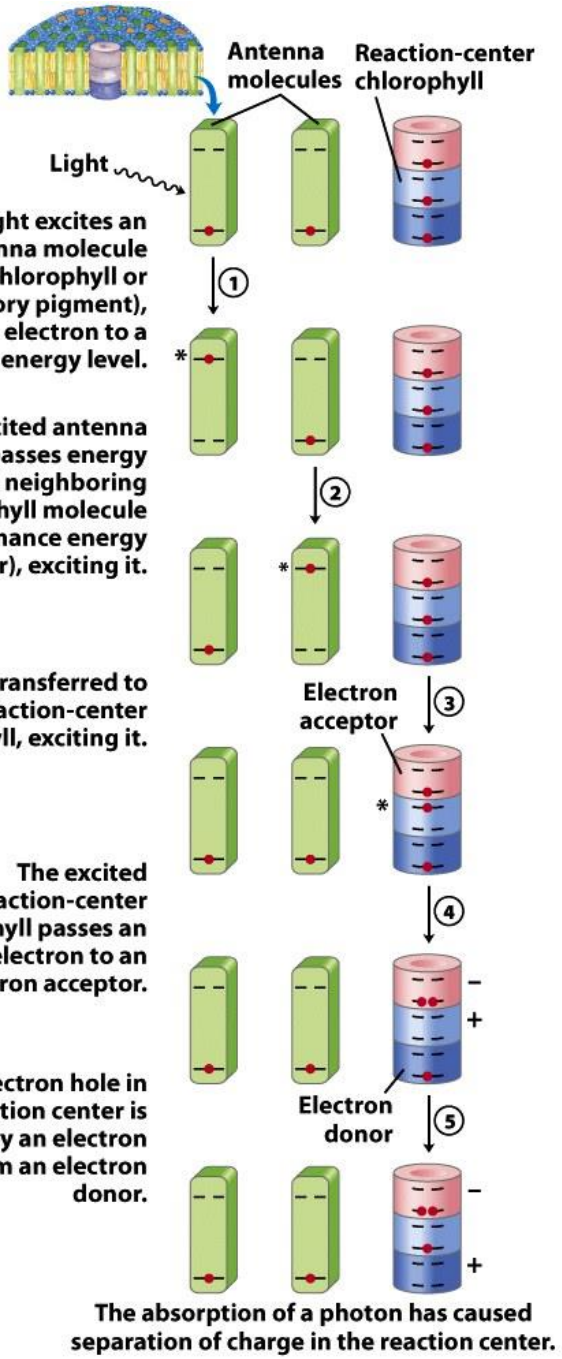
These molecules absorb light energy, transferring it between molecules until it reaches the reaction center.

Light

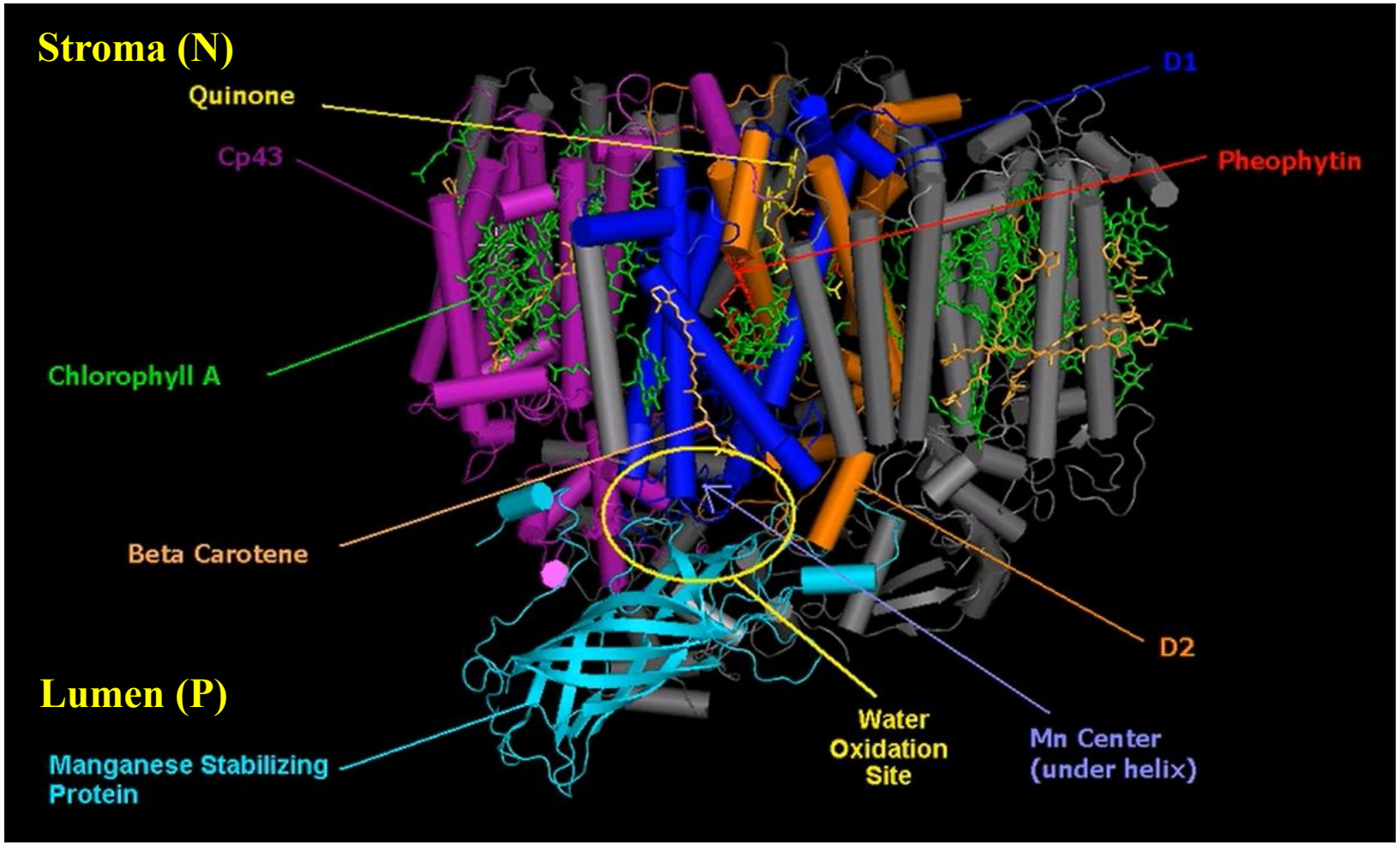


Reaction center

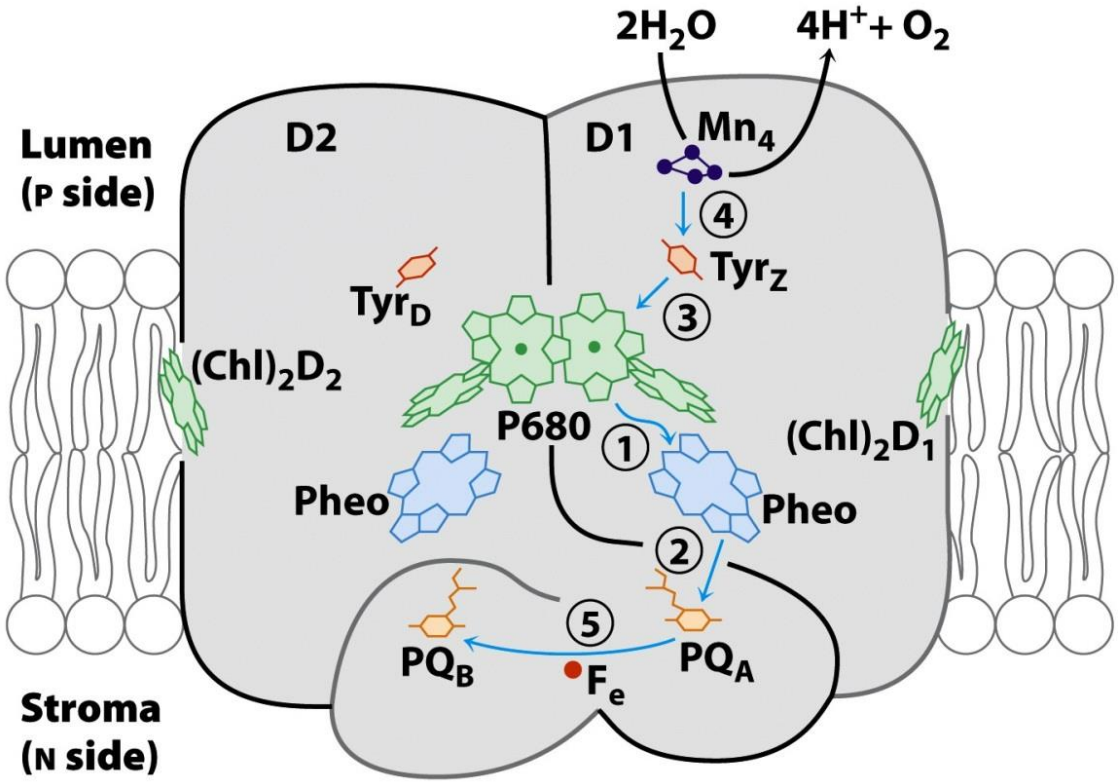
Photochemical reaction here converts the energy of a photon into a separation of charge, initiating electron flow.



Photosystem II of the cyanobacterium *Synechococcus elongates* (pheophytin-quinone system)



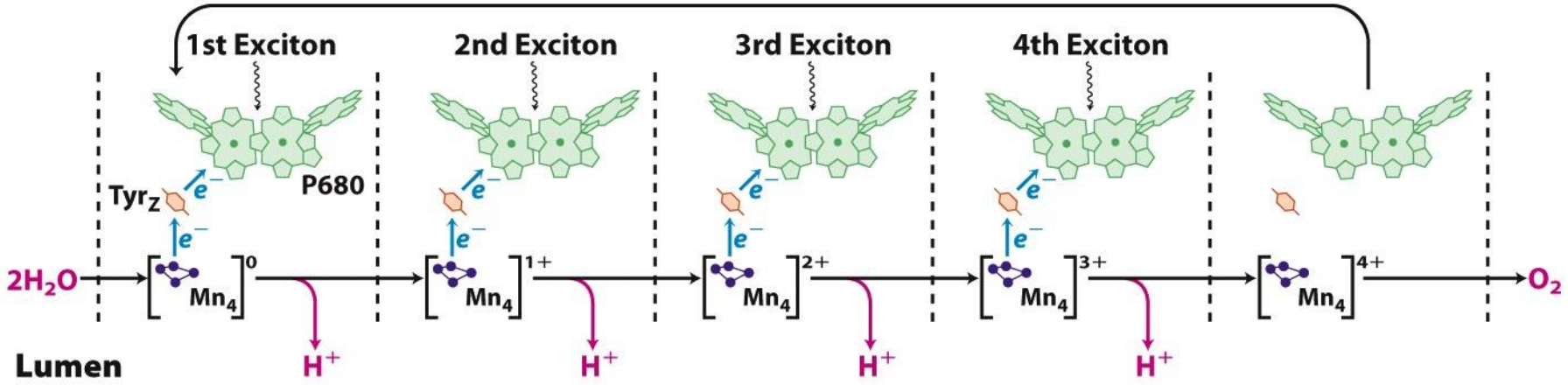
Photosystem II of the cyanobacterium *Synechococcus elongates* (pheophytin-quinone system)



$P680^+$
$^{\cdot}Pheo^-$

$4\ P680 + 4H^+ + 2PQ_B + 4\ photons \rightarrow 4\ P680^+ + 2PQ_BH_2$

Water-splitting activity of the oxygen-evolving complex



Oxidizing water : $2H_2O \rightarrow 4H^+ + 4e^- + O_2$

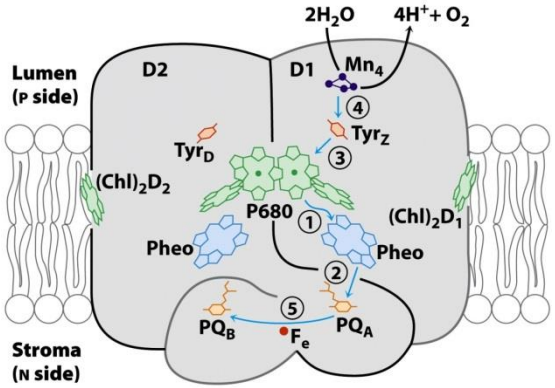
$4P680^+ + 4Tyr \rightarrow 4P680 + 4\ ^{\cdot}Tyr$

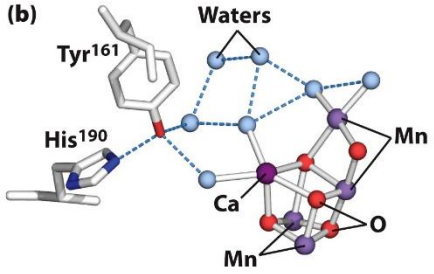
$4\ ^{\cdot}Tyr + [Mn\ complex]^0 \rightarrow 4Tyr + [Mn\ complex]^{4+}$

$[Mn\ complex]^{4+} + 2H_2O \rightarrow [Mn\ complex]^0 + 4H^+ + O_2$

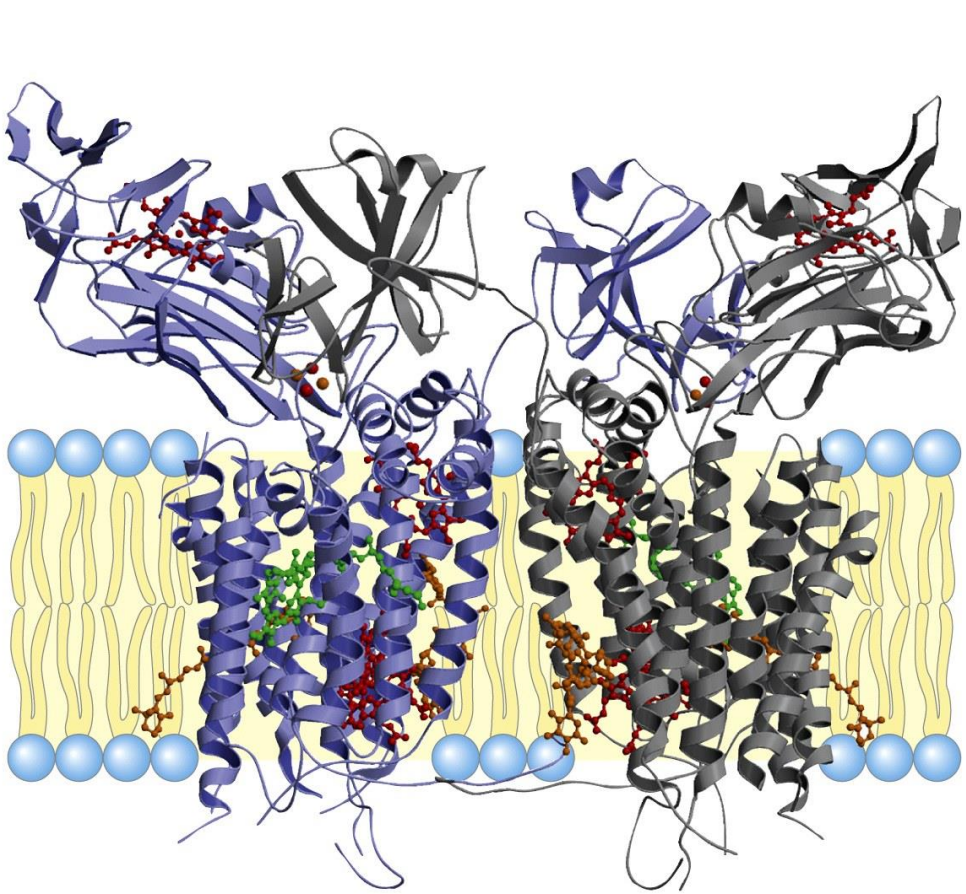
$4P680^+ + 4H^+ + 2PQ_B + 4\ photons \rightarrow 4P680 + 2PQ_BH_2$

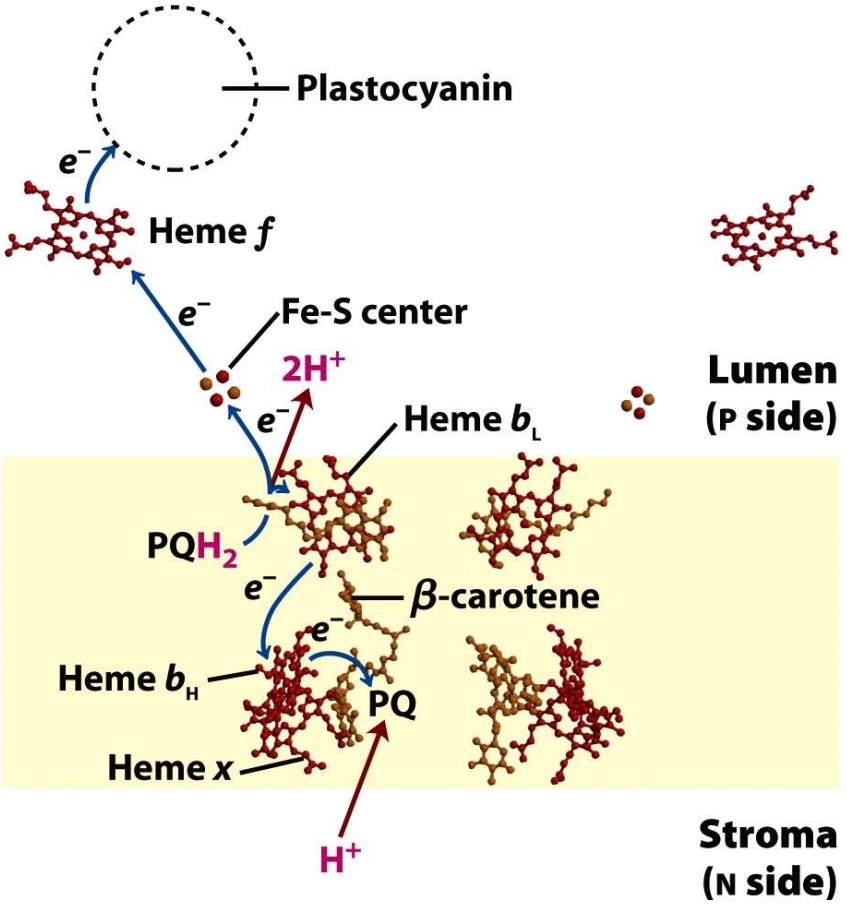
$2H_2O + 2PQ_B + 4\ photons \rightarrow O_2 + 2PQ_BH_2$





Electron and proton flow through the cytochrome b_6f complex





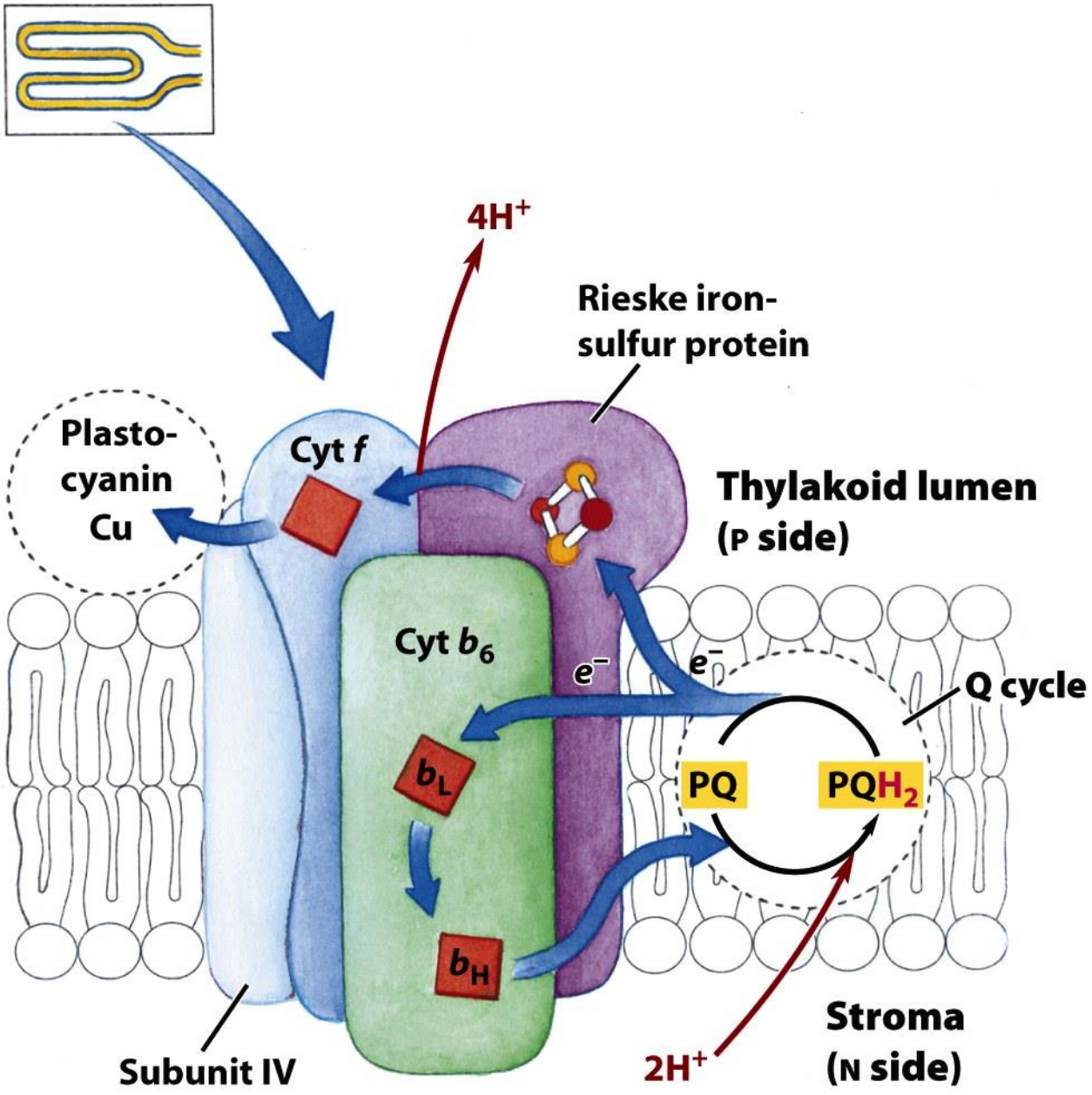
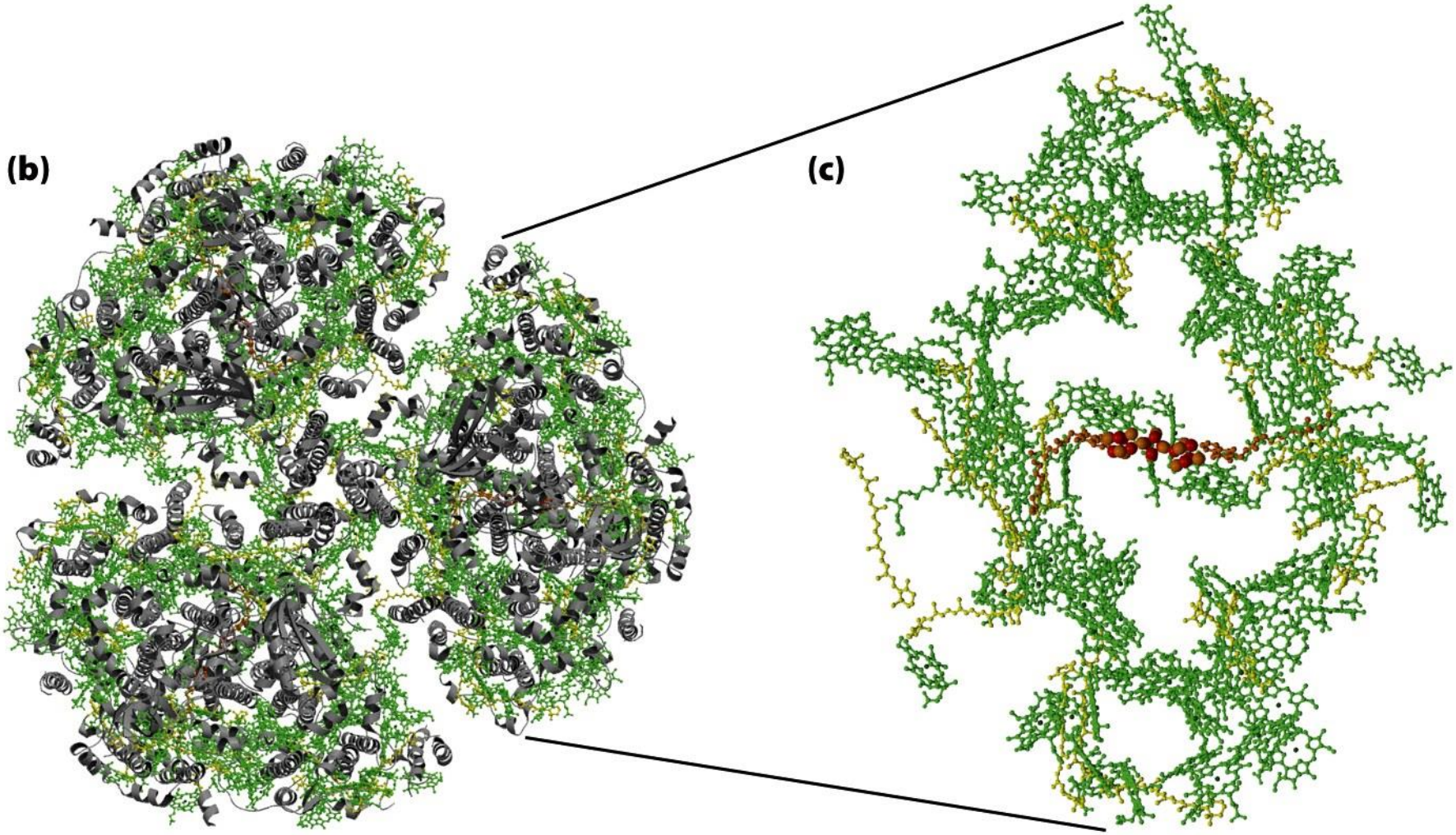


Figure 19-59c

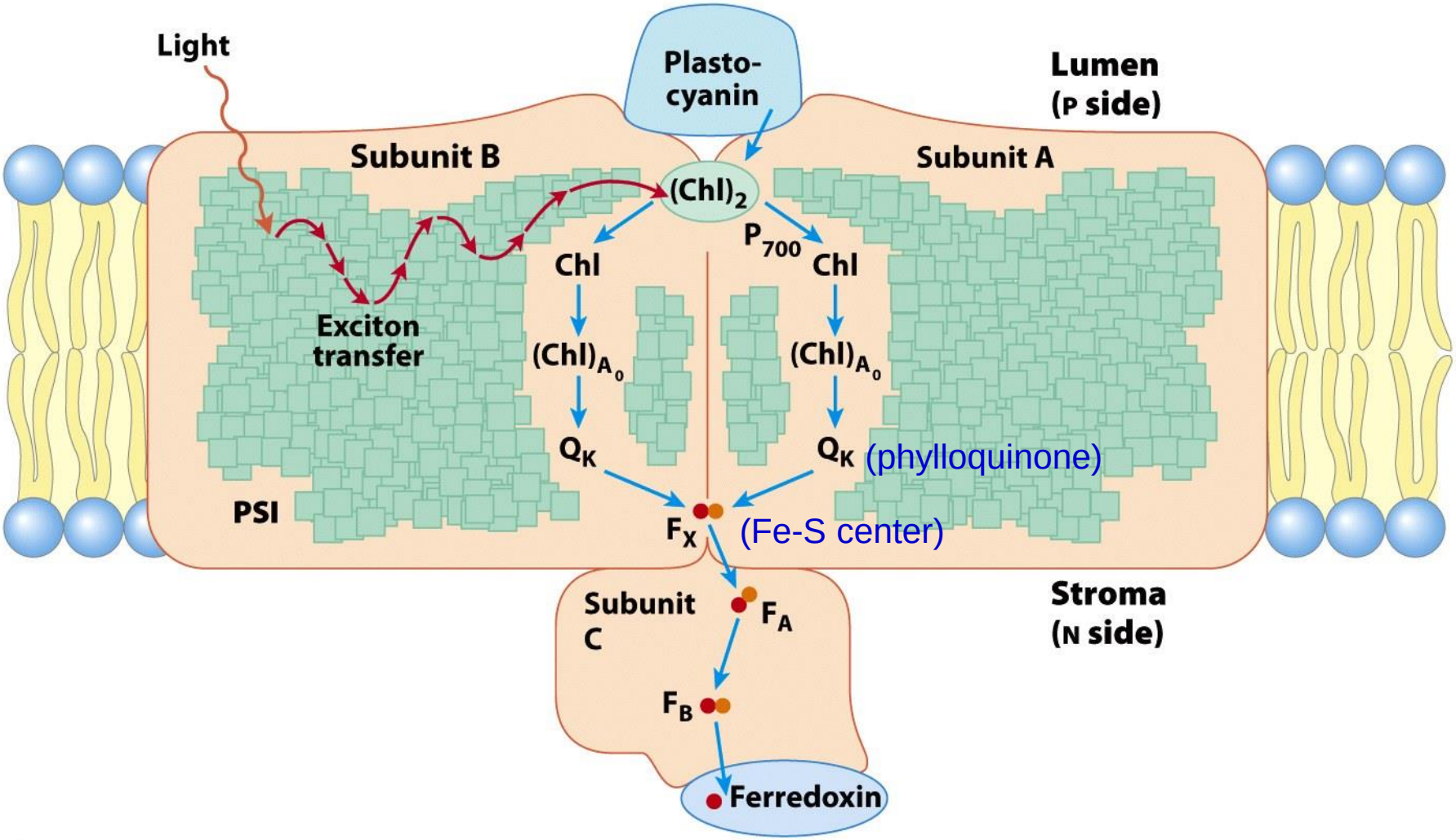
Lehninger Principles of Biochemistry, Fifth Edition

© 2008 W. H. Freeman and Company

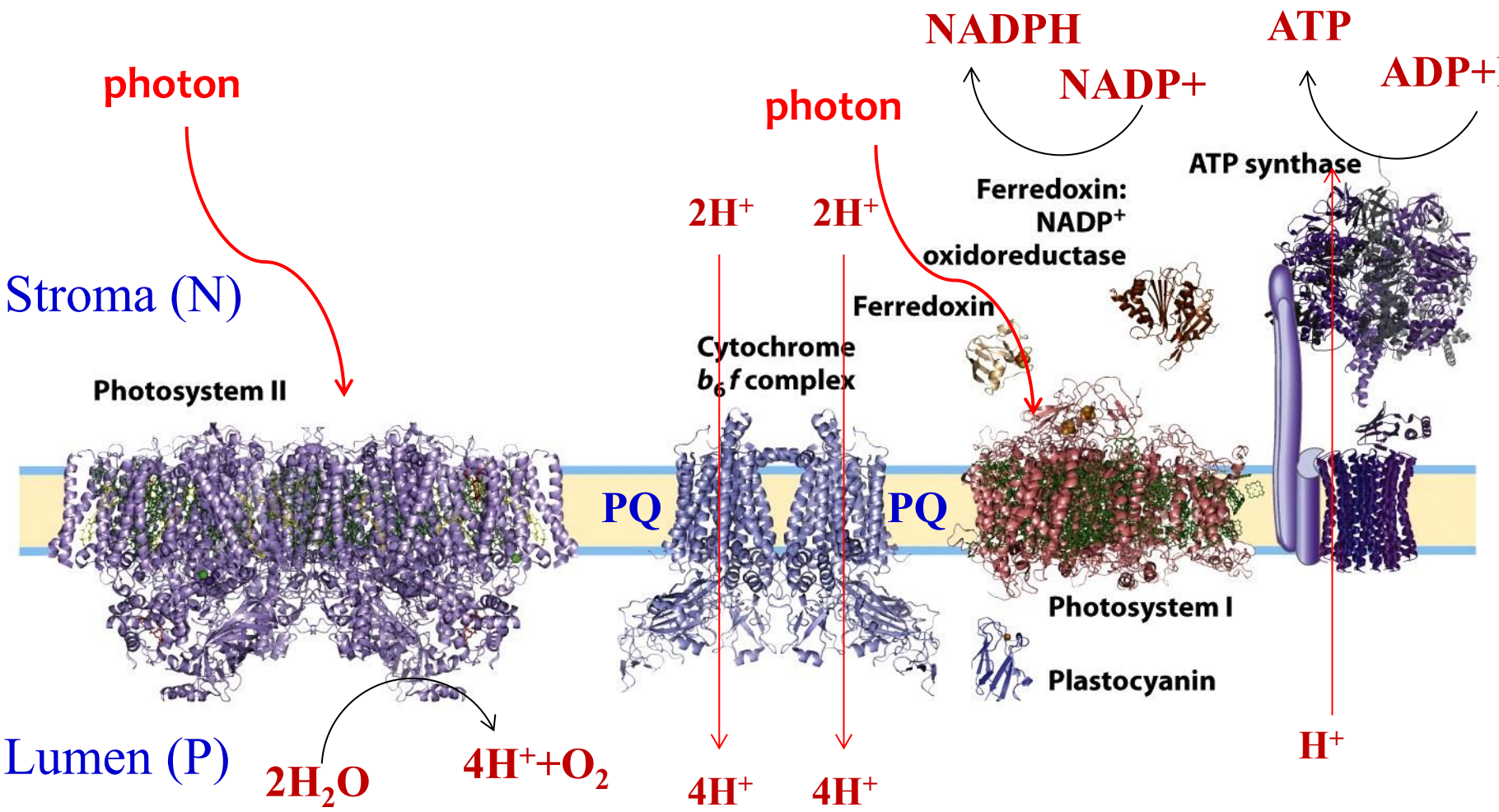
The supramolecular complex of PSI and its associated antenna chlorophylls



The supramolecular complex of PSI and its associated antenna chlorophylls



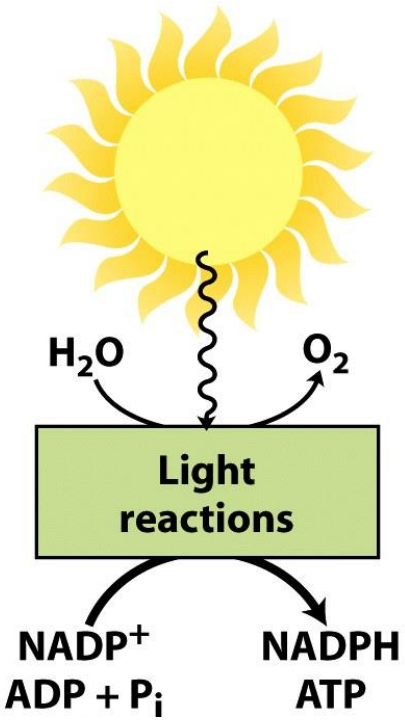
Photophosphorylation

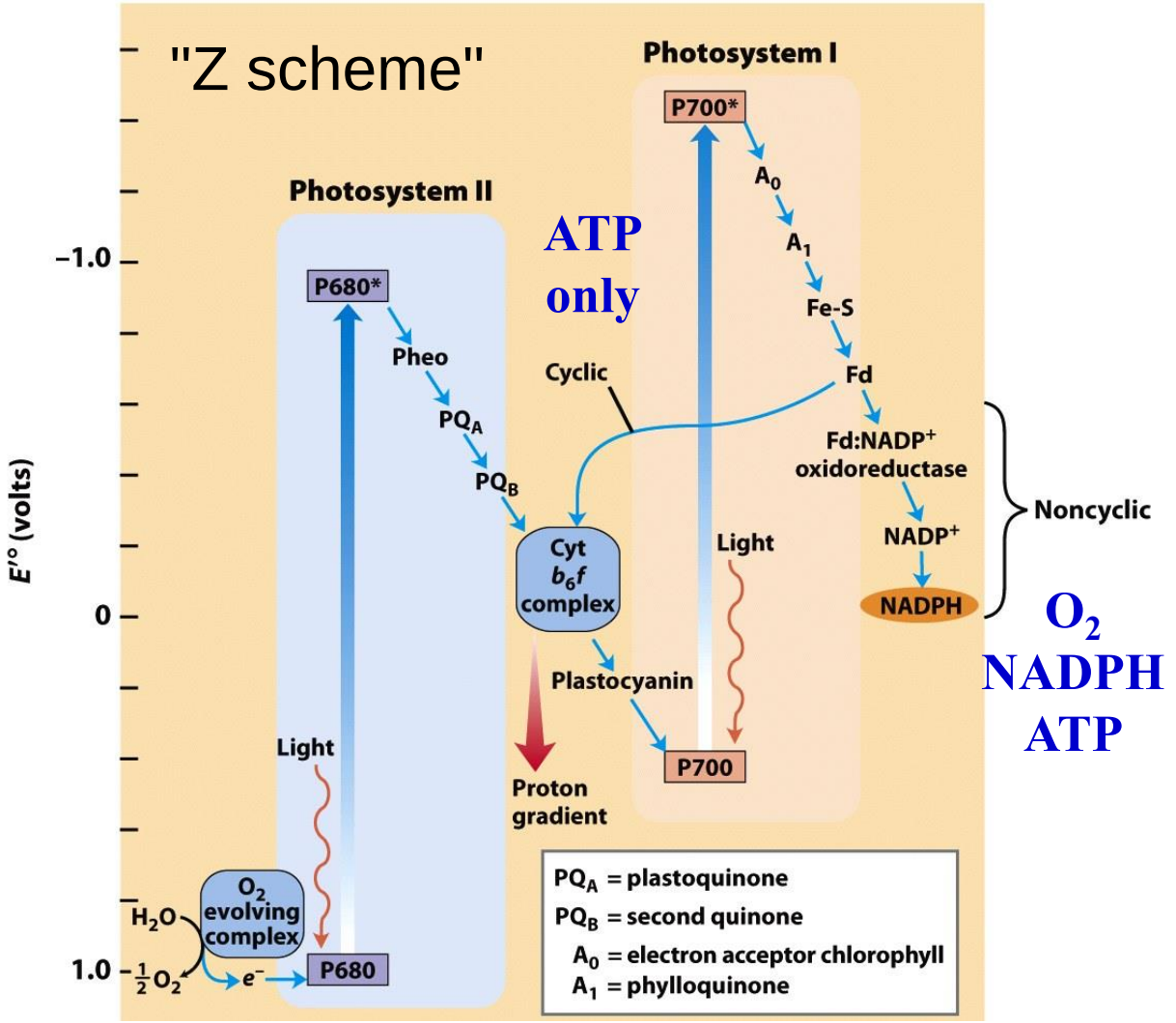


H_2O → P680 → Pheo → PQ → b_6f → PC → P700 → Fd → NADPH

PSII PSI

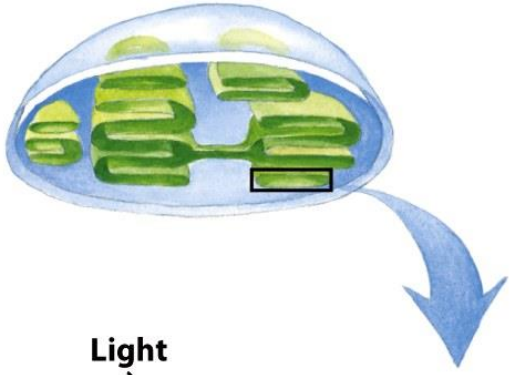
Cyclic and non-cyclic photosynthesis



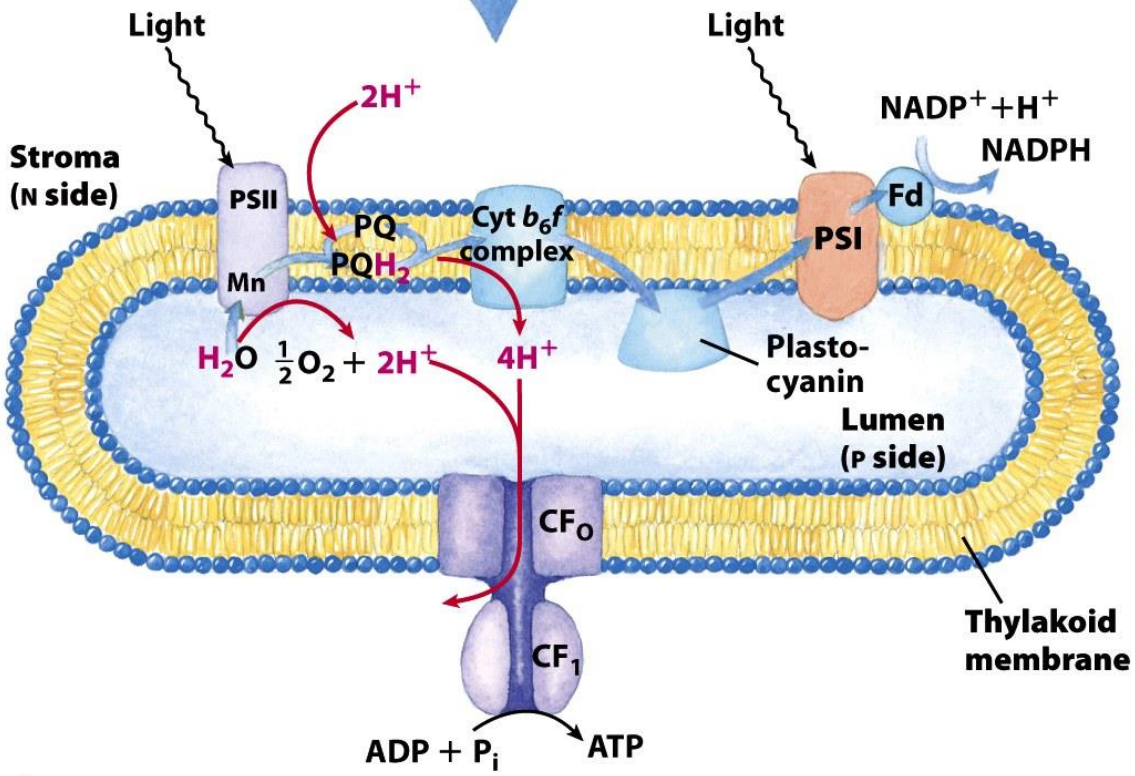


$2H_2O + 2NADP^+ + 8$ photons → $O_2 + 2NADPH + 2H^+$ ($4e^-$ transfer)

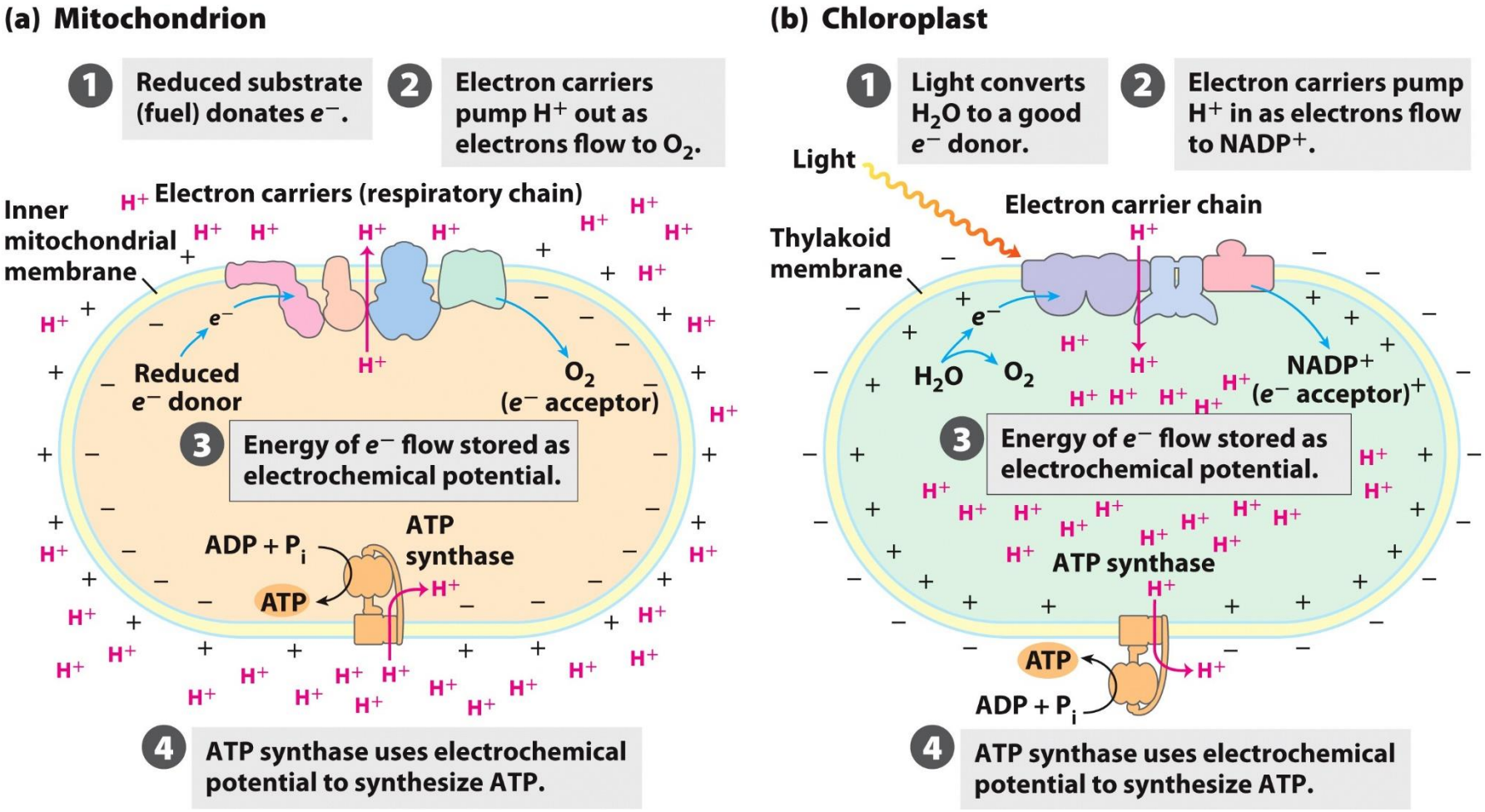
Proton and electron circuits in thylakoids (Photophosphorylation : ATP synthesis)



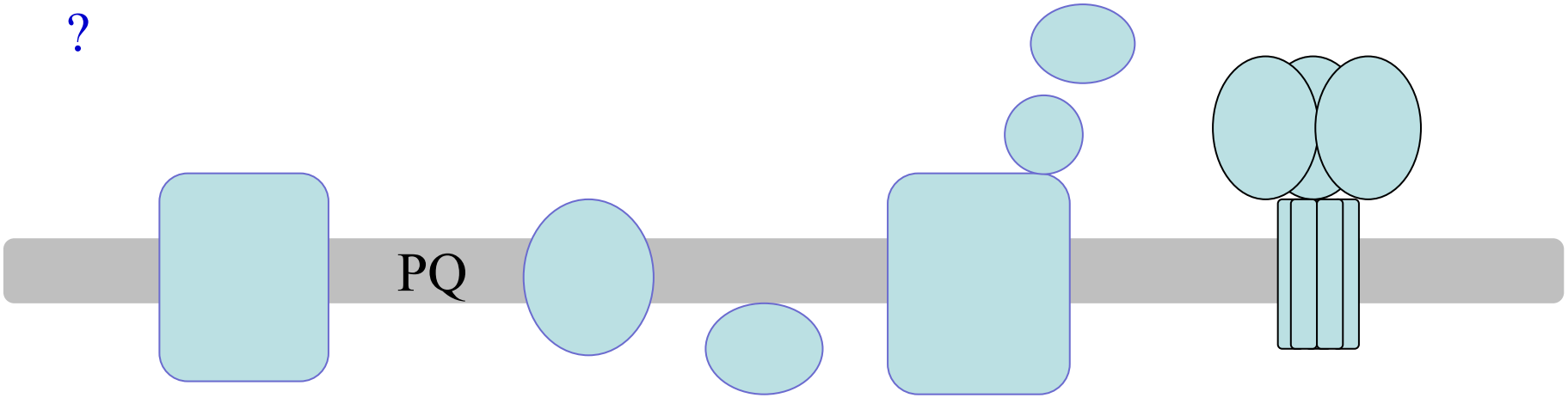
$2H_2O + 8$ photons $+ 2NADP^+ + \sim 3ADP + \sim 3Pi \rightarrow$
$O_2 + \sim 3ATP + 2NADPH$
(12 H^+ : $4H^+$ by OEC and $8H^+$ by b_6f, $\Delta pH = 3$)



Comparison of the chemiosmotic mechanisms for ATP synthesis



?



?

https://youtu.be/hj_WKgnL6MI

Summary

1. Photophosphorylation in the chloroplasts of green plants and in cyanobacteria involves electron flow through a series of membrane-bound carriers.
2. Chlorophyll funnels the absorbed energy to reaction centers by exiton transfer
3. In plansts, two reaction centers act in tandem
4. Antenna chlorophyll are tightly integrated with electron carriers
5. The cytochrome b6f complex links photosystem II and I
6. Water is split by the oxygen-evolving complex
7. A proton gradient couples electron flow and ATP synthesis

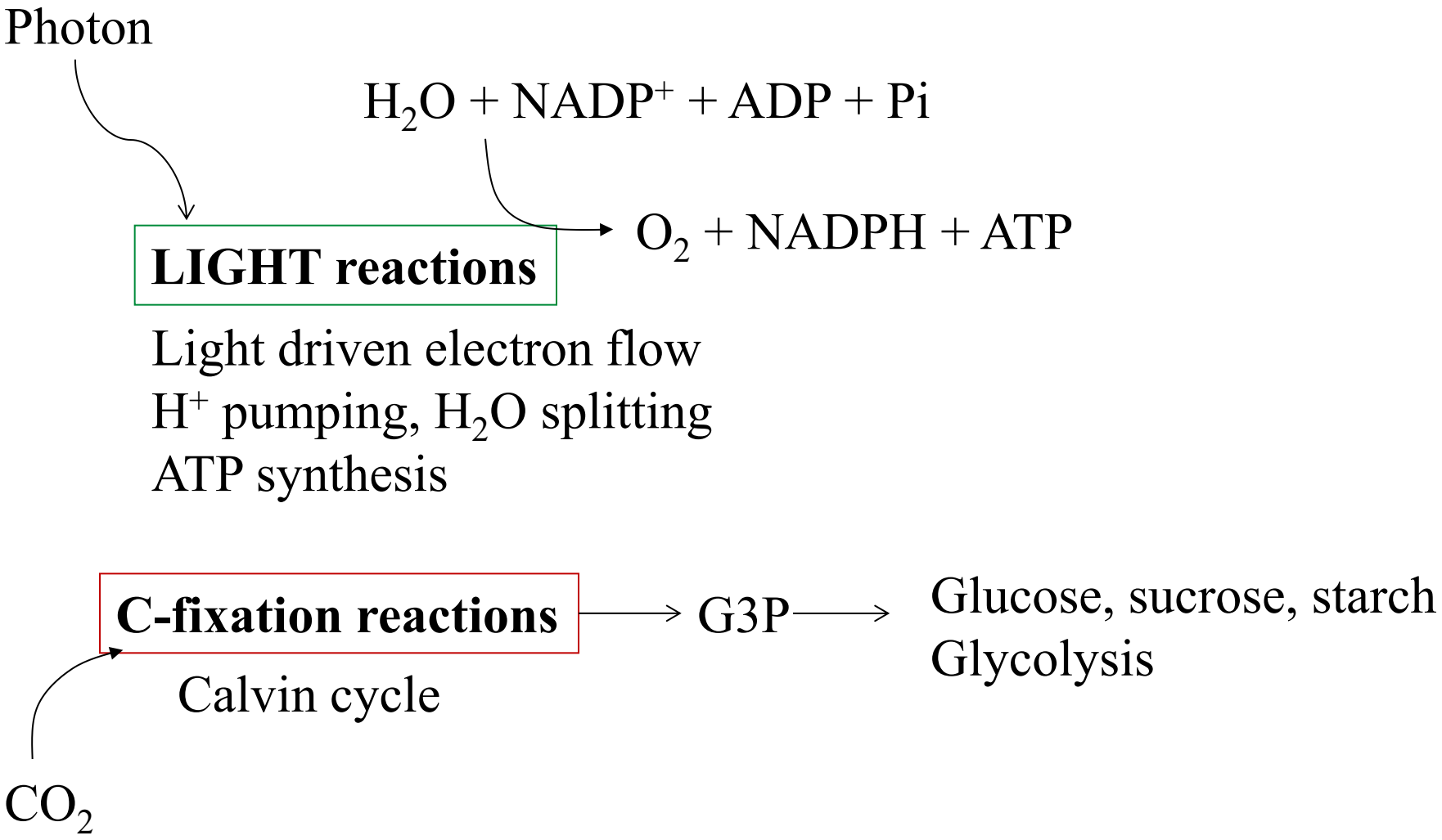
Carbohydrate biosynthesis

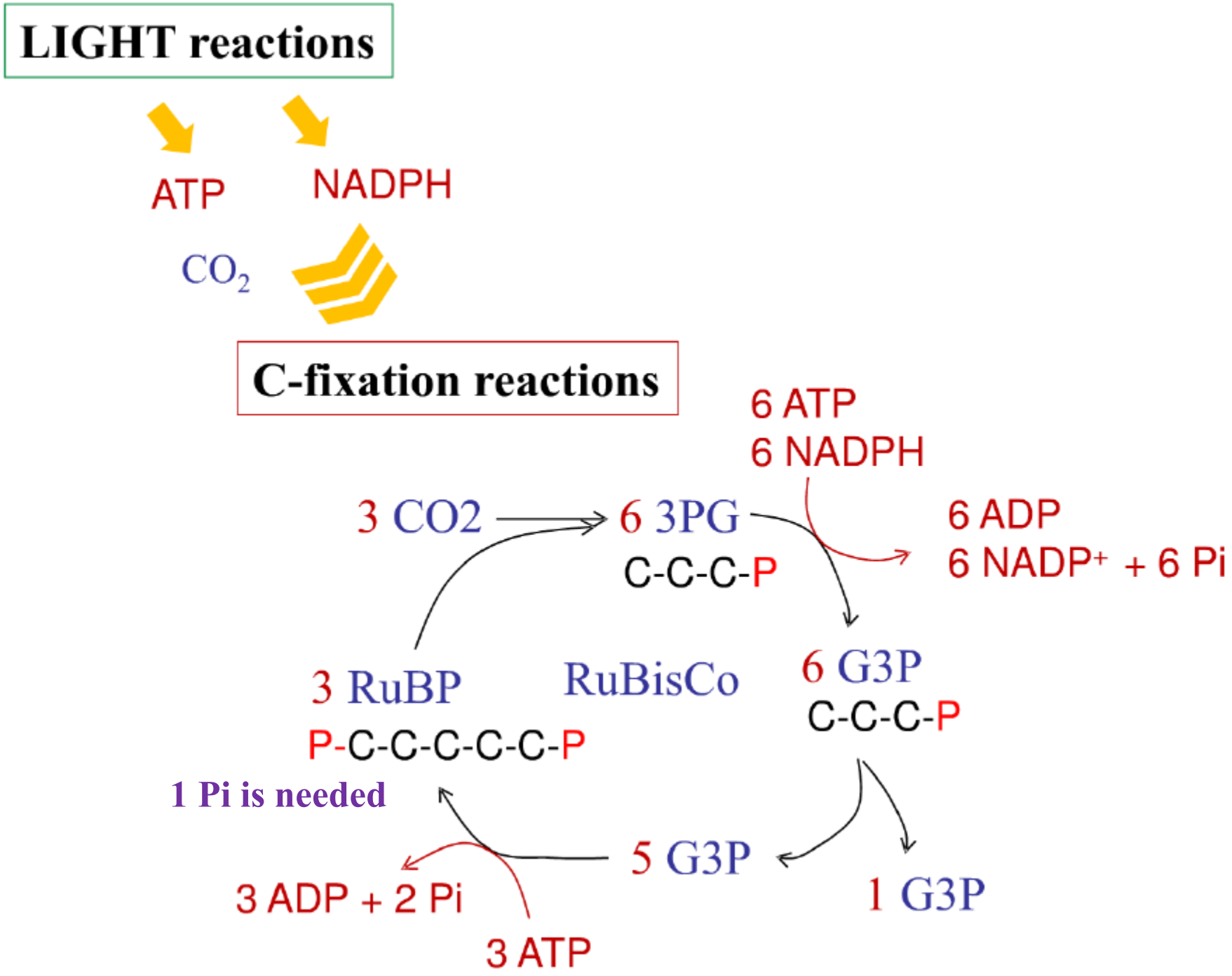
20.1 Photosynthetic Carbohydrate Synthesis

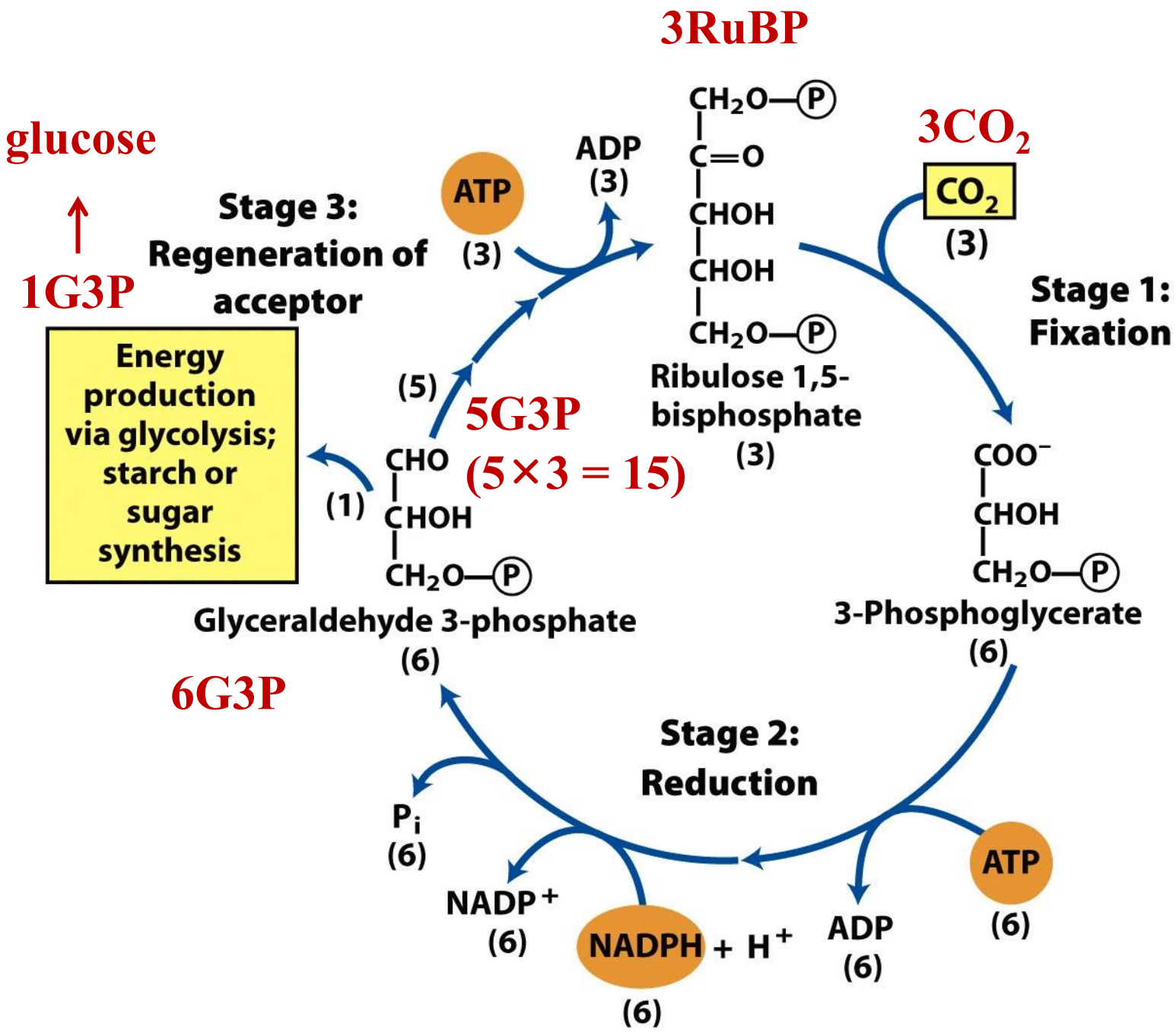




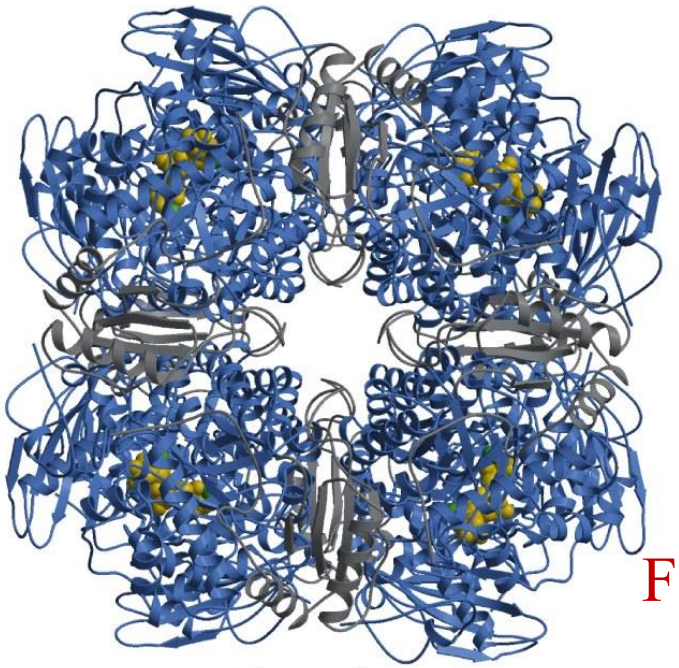
PHOTOSYNTHESIS




LIGHT reactions
ATP
NADPH
CO_2
C-fixation reactions
6 ATP
6 NADPH
3 CO2
6 3PG
C-C-C-P
6 ADP
6 NADP+ + 6 Pi
3 RuBP
P-C-C-C-C-C-P
RuBisCo
6 G3P
C-C-C-P
1 Pi is needed
5 G3P
1 G3P
3 ADP + 2 Pi
3 ATP

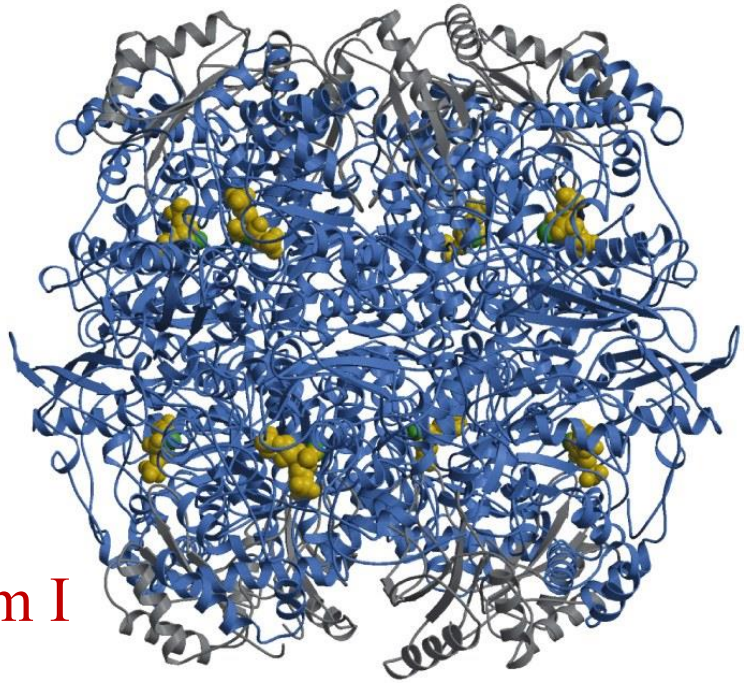

3RuBP
glucose
1G3P
3CO2
CO2
(3)
ATP
(3)
ADP
(3)
Stage 3:
Regeneration of
acceptor
Stage 1:
Fixation
Energy
production
via glycolysis;
starch or
sugar
synthesis
(5)
5G3P
(5×3 = 15)
Ribulose 1,5-
bisphosphate
(3)
(1)
Glyceraldehyde 3-phosphate
(6)
3-Phosphoglycerate
(6)
6G3P
Stage 2:
Reduction
Pi
(6)
NADP+
(6)
NADPH + H+
(6)
ADP
(6)
ATP
(6)

Ribulose 1,5-bisphosphate carboxylase (Rubisco)



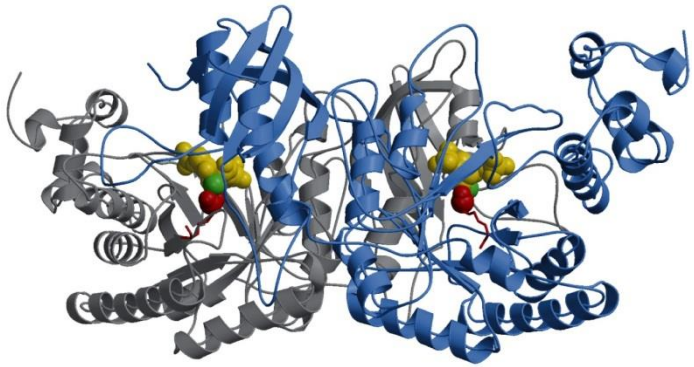
Top view

Form I

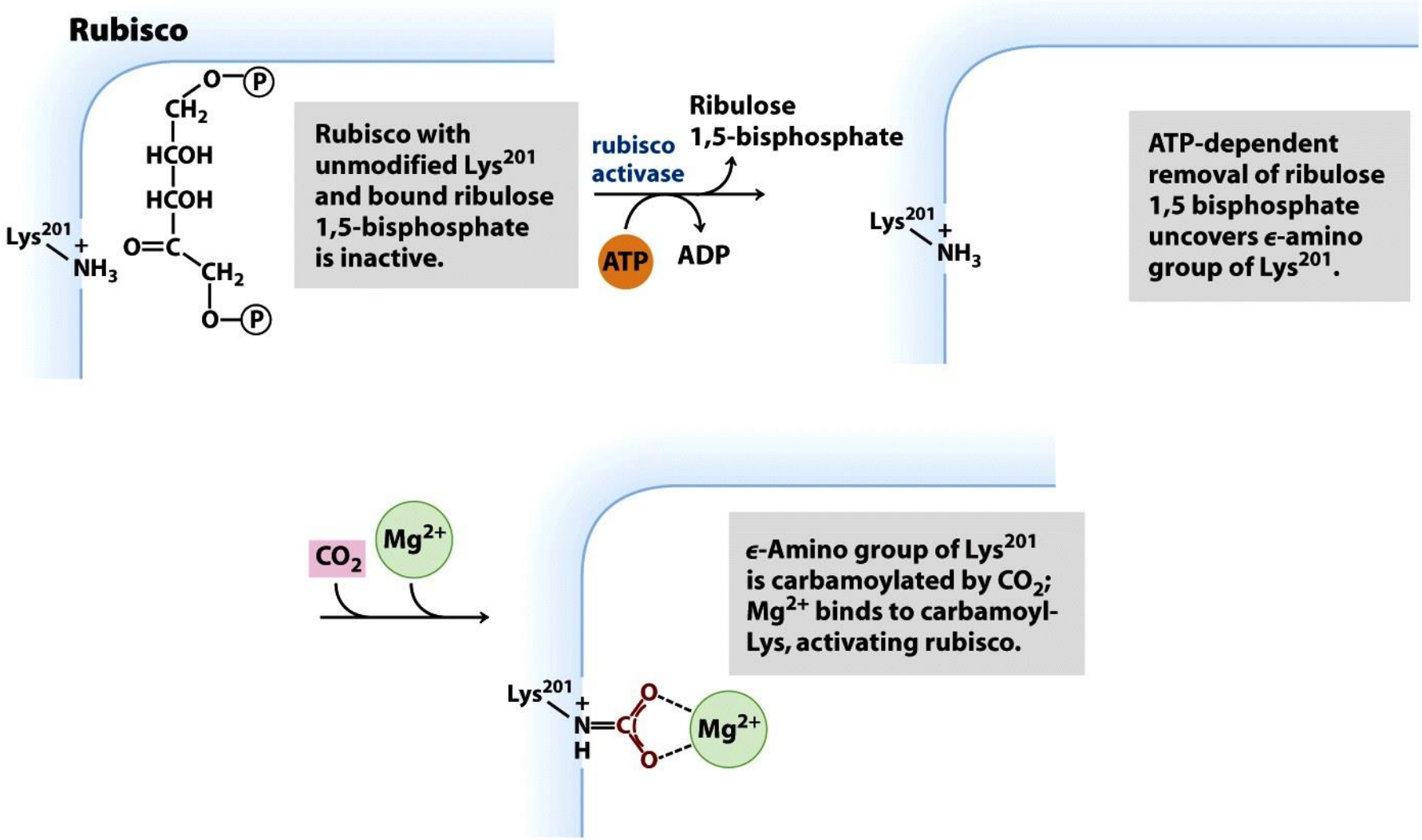


Side view

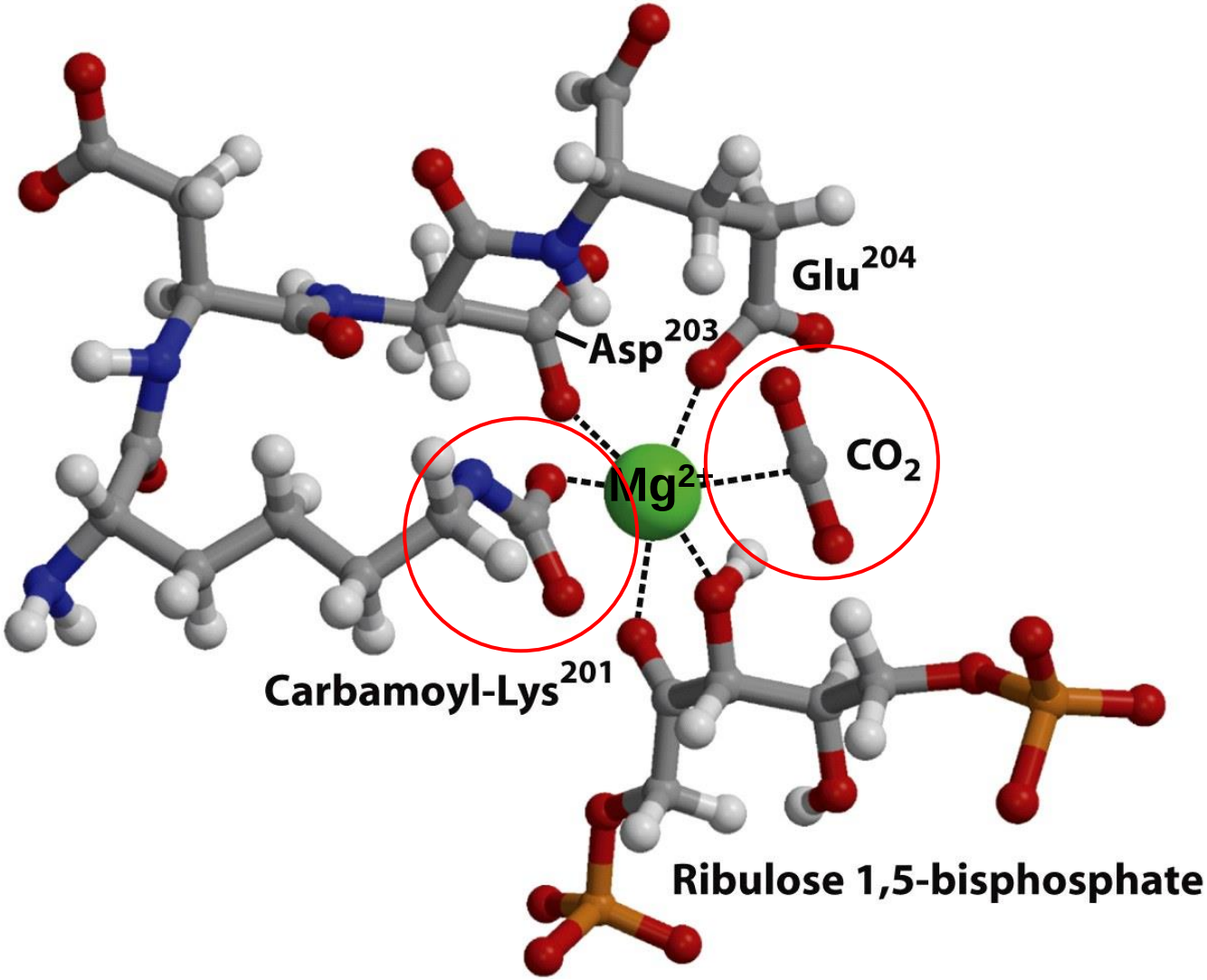
Form II



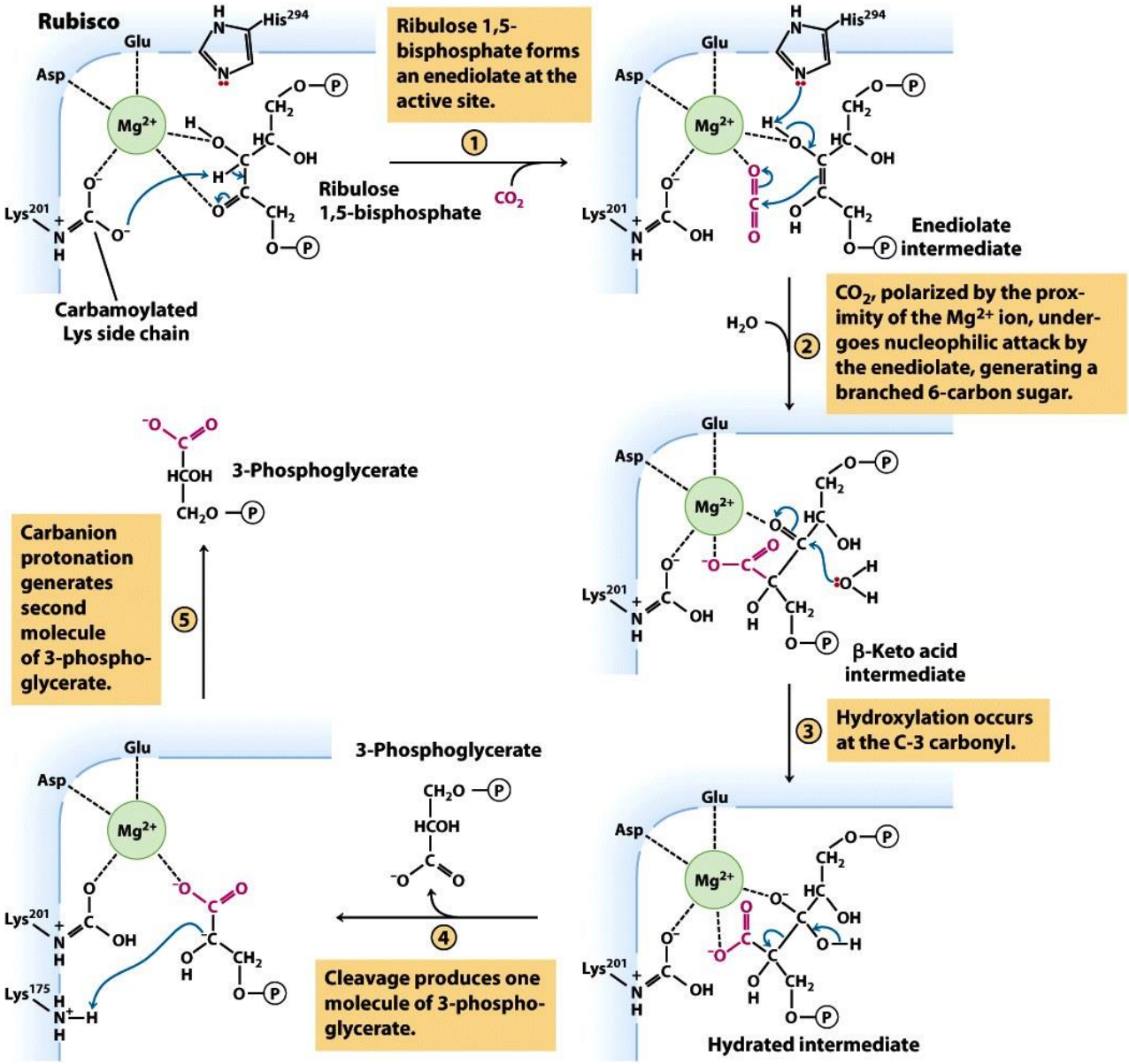
Rubisco is activated by rubisco activase (carbamoylation of Lys^{201} of rubisco)



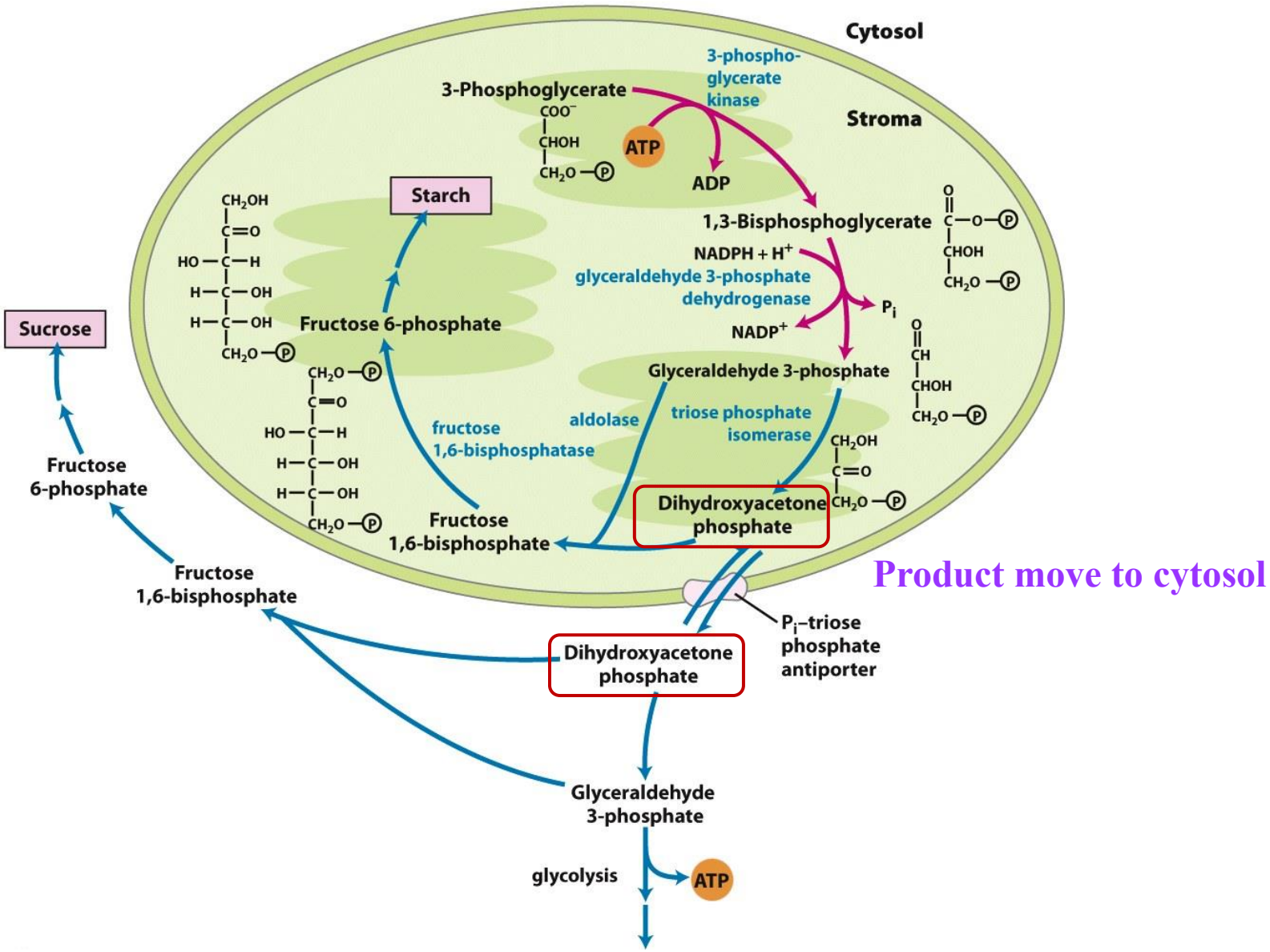
Active site of RuBisCo
(Central role of Mg^{2+} and Carbamoylated Lys)



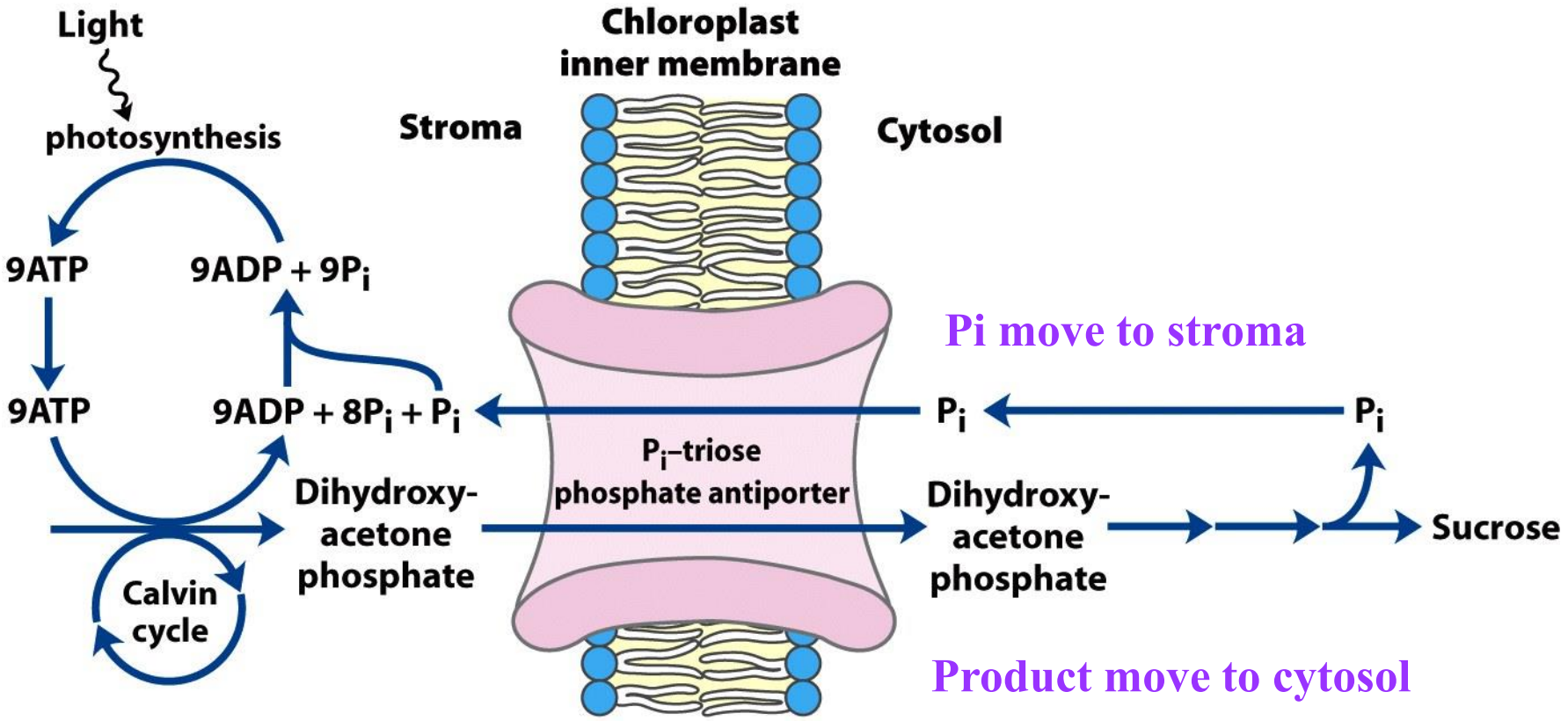
First stage of CO_2 assimilation: 3-PG synthesis



Second stage of CO_2 assimilation (3PG to G3P)

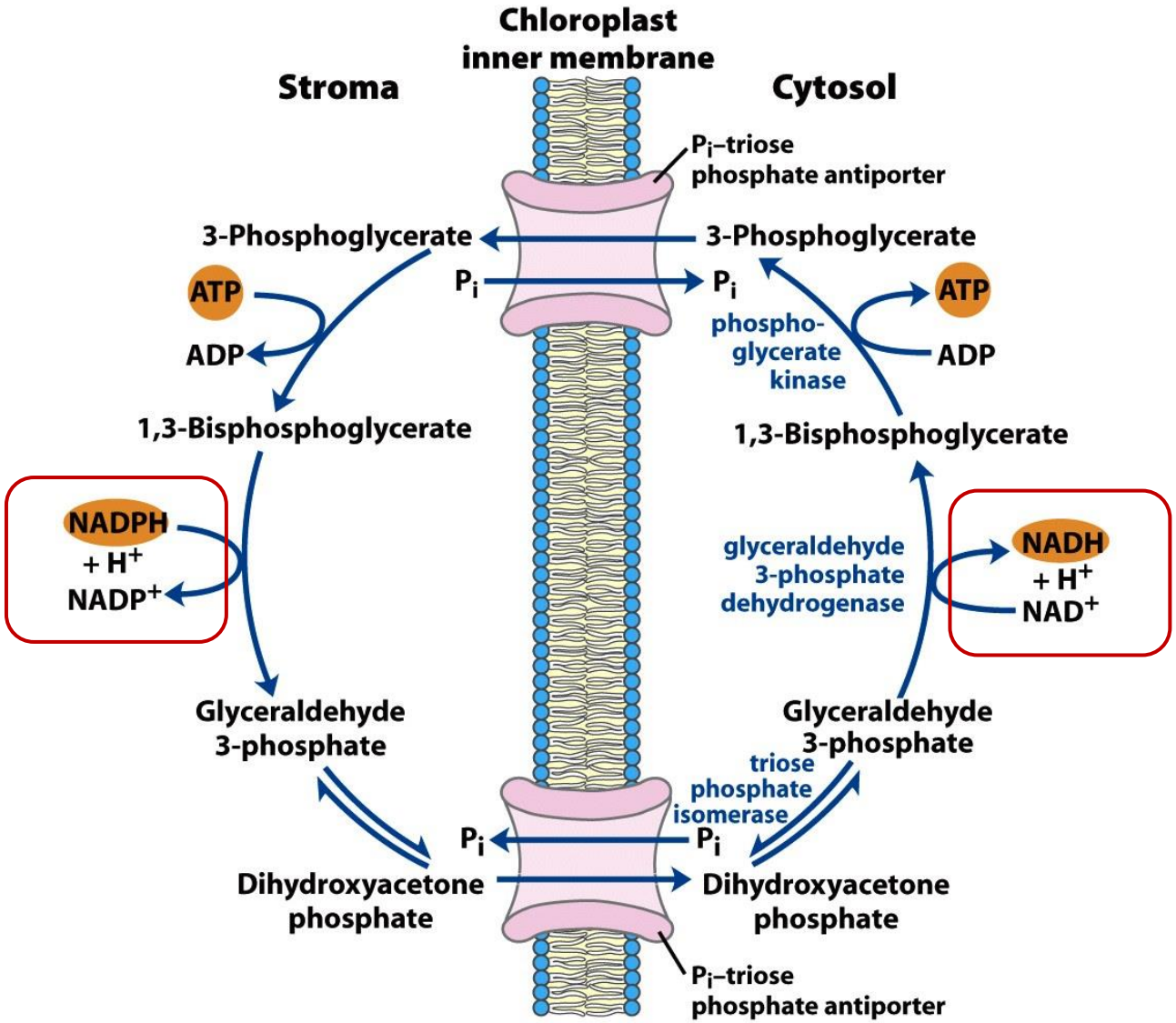


The P_i–triose phosphate antiport system of the inner chloroplast membrane



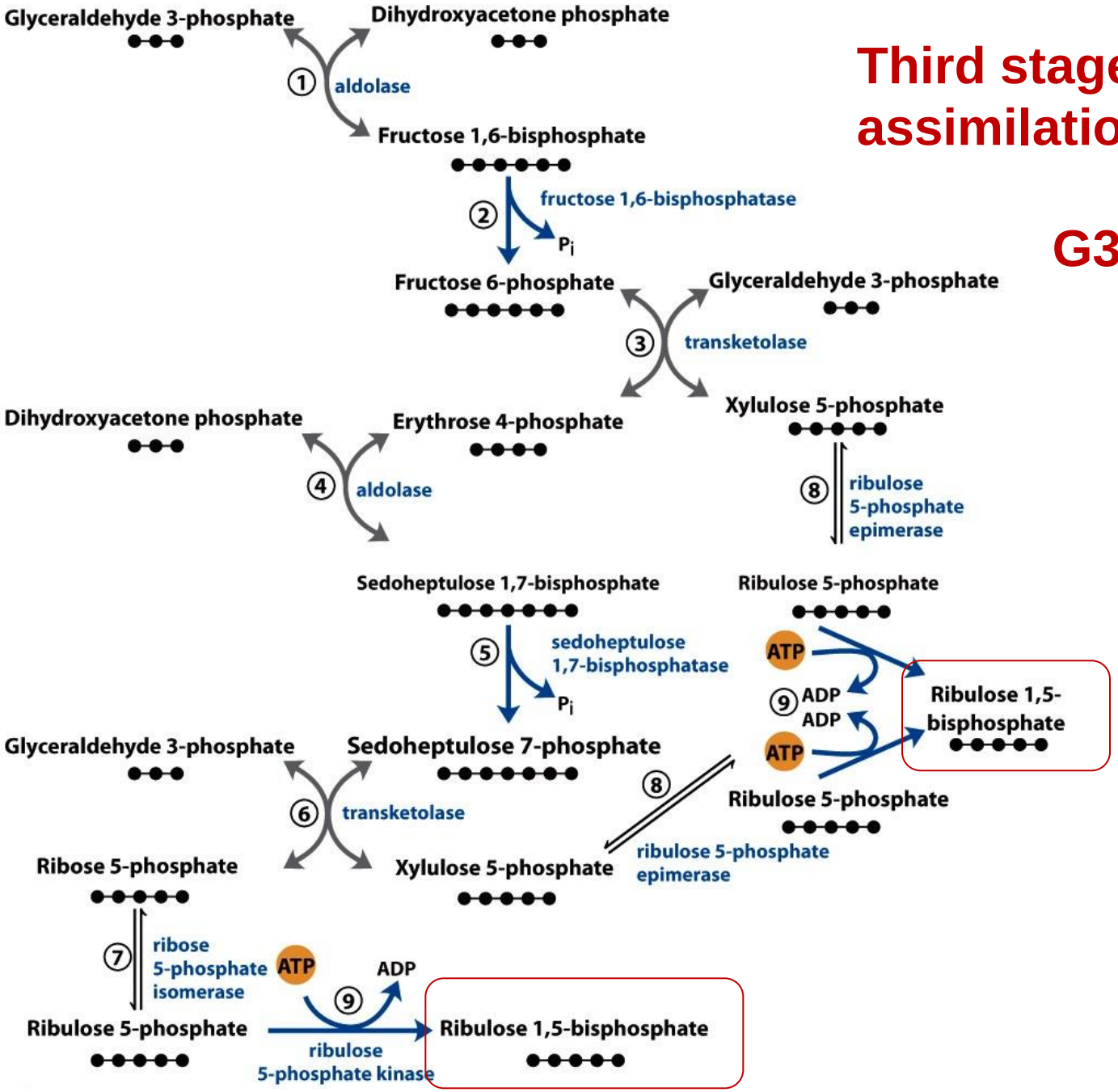
1 Pi is supplied from cytosol by the Pi-triose phosphate antiport system

Role of the P_i-triose phosphate antiporter in the transport of ATP and reducing equivalents (NAD(P)H)

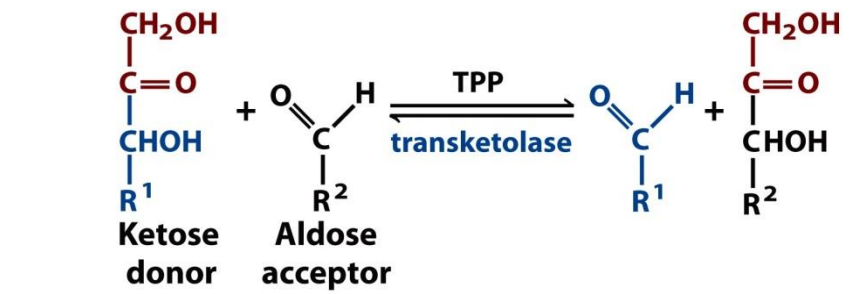


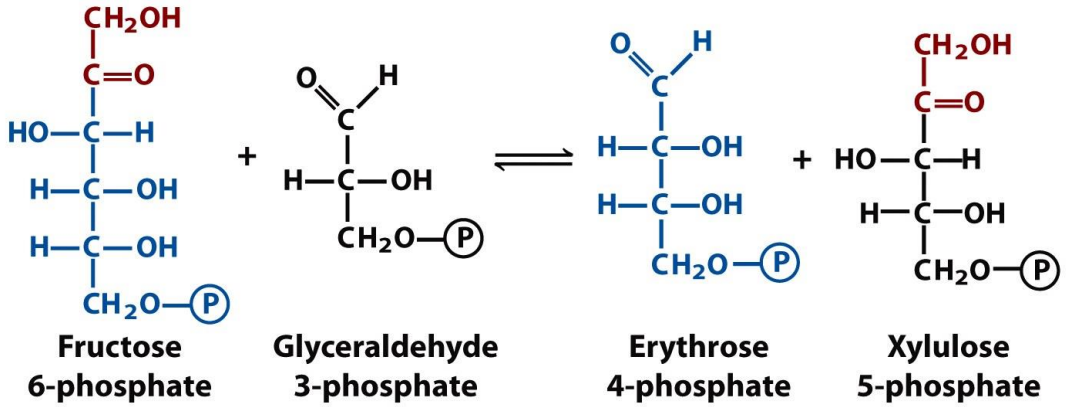
Third stage of CO_2 assimilation

G3P to RuBP

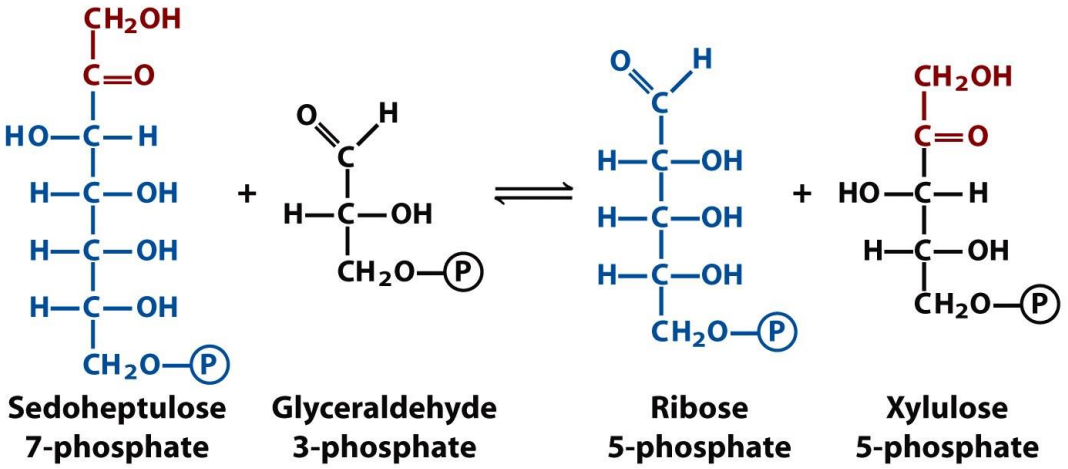


Transketolase-catalyzed reactions of the Calvin cycle



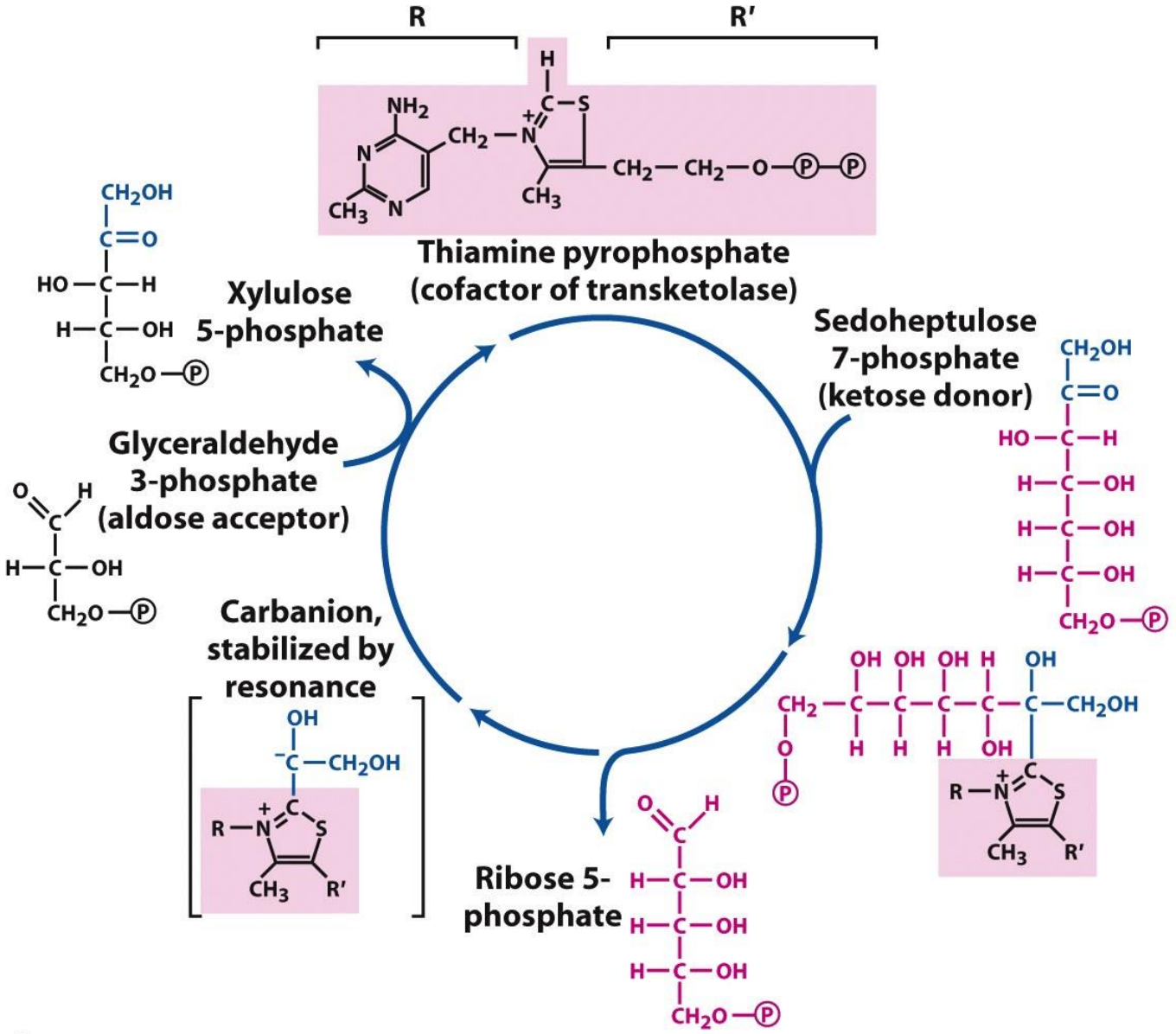


6+3=4+5

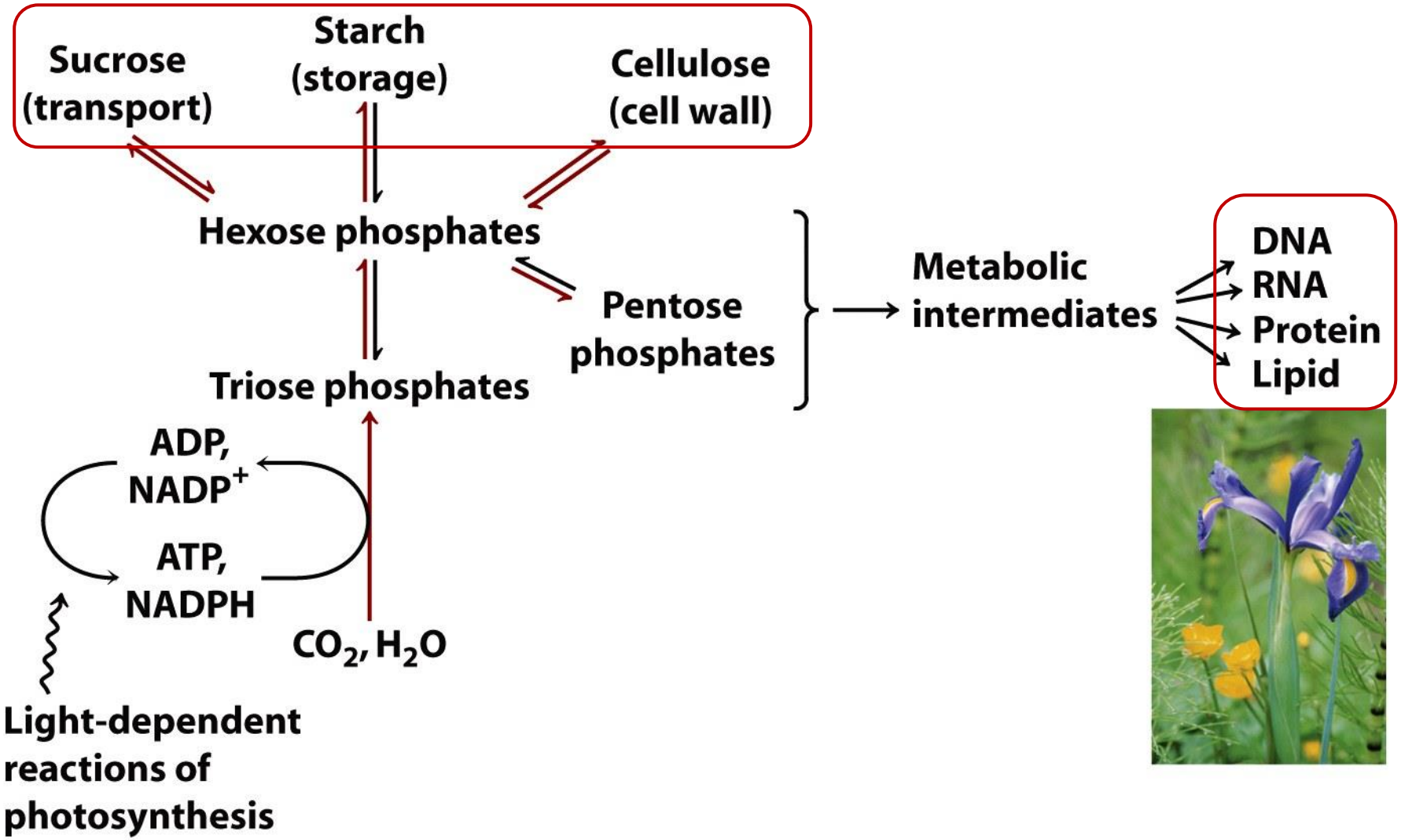


7+3=5+5

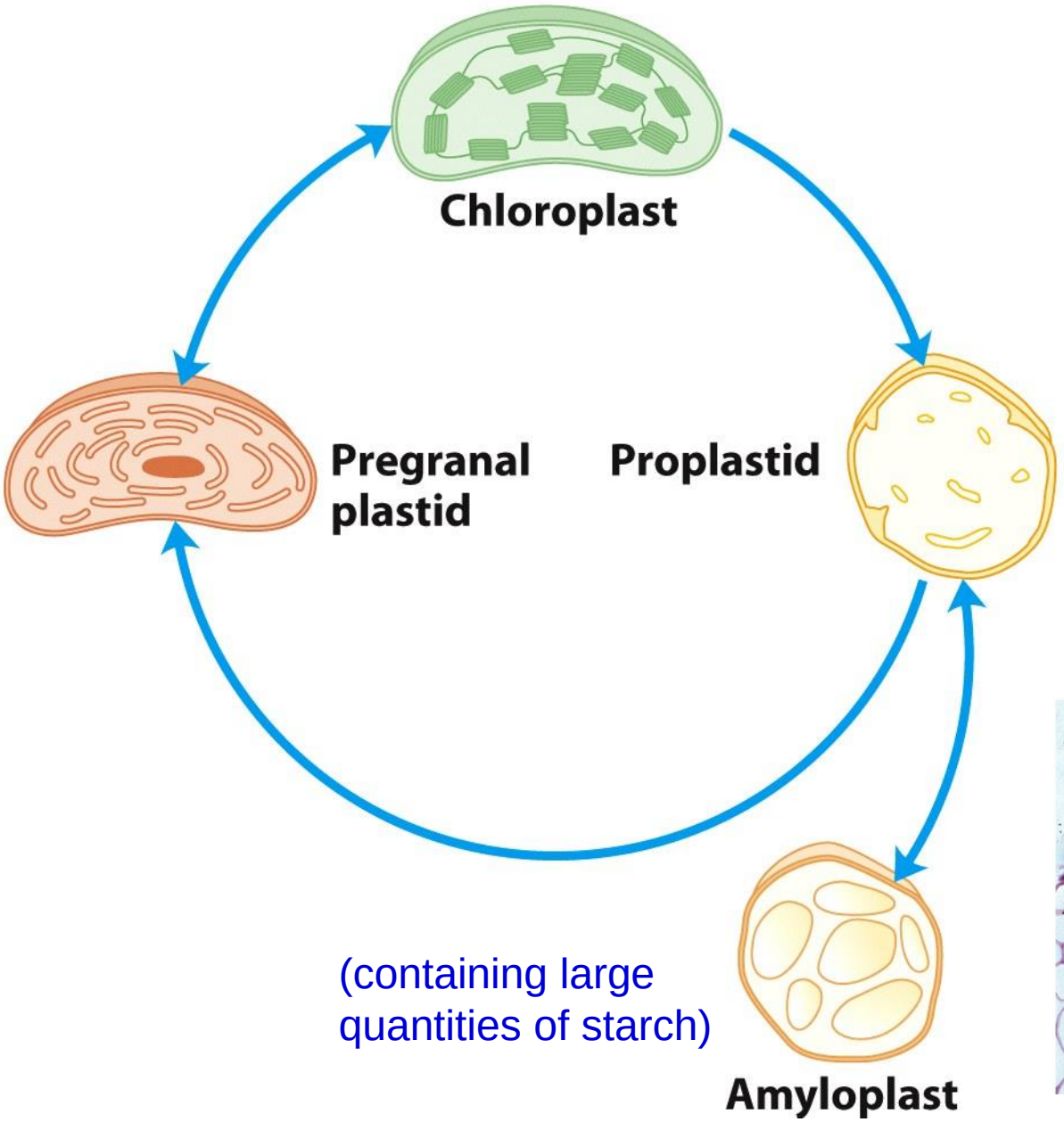
TPP as a cofactor for transketolase

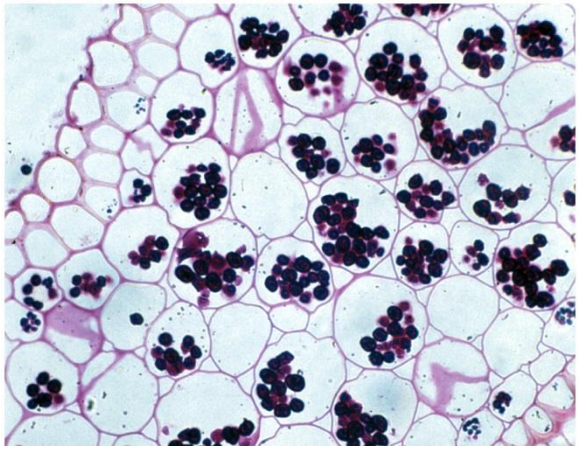


Assimilation of CO_2 into biomass in plants



Plastids: organelles unique to plant cells and algae





The end of chapter 19-20

Oxidative phosphorylation
and
Photophosphorylation